SPECTRAL THEORY AND DIOPHANTINE BOUNDS FOR HANKEL DETERMINANTS OF ODD ZETA VALUES

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ABSTRACT. We study the zero locus and arithmetic properties of integer polynomials $P_K^{(s)}(X) \in \mathbb{Z}[X]$ arising from Hankel determinants attached to rational moment functionals for $\zeta(2k+1)$ and related L-values. By diagonalizing the matrix pencil in a Lagrange basis over the negative square poles, we prove that all roots of $P_K^{(2k+1)}(X)$ are real, interlace symmetrically across $\zeta(2k+1)$, and admit an explicit exponential zero-free disk. This spectral separation overcomes the growing-degree barrier in determinantal Diophantine approximation, yielding the effective irrationality measure $|\zeta(5) - a/b| > b^{-25687}$ for all sufficiently large b, as well as effective measures for linear forms $b\zeta(5) + a\zeta(3)$. Furthermore, we discover a character–Vandermonde sign cancellation for Dirichlet beta values $\beta(2k) = L(2k, \chi_{-4})$ that forces a unilateral $(h, 0)$ root signature, and we establish potential-gap identities that govern height growth in two-variable Beukers extensions.

1. INTRODUCTION

Ever since Apéry [1] established that $\zeta(3)$ is irrational, a central problem in Diophantine approximation has been to understand the arithmetic nature of the odd zeta values $\zeta(2k+1)$ for $k \geq 2$. Following Beukers's interpretation [3] of Apéry's rational approximations via multiple integrals of Legendre polynomials, Rivoal [12] and Ball and Rivoal [2] used well-poised hypergeometric series to prove that infinitely many odd zeta values are irrational. Zudilin [14, 15] subsequently showed that at least one of $\zeta(5), \zeta(7), \zeta(9)$, and $\zeta(11)$ is irrational, and later works of Fischler, Sprang, and Zudilin [7] and Fischler [6] improved the lower bounds for the number of irrational odd zeta values below a given threshold. For $\zeta(5)$ individually, cellular integral constructions were developed by Brown and Zudilin [4], producing rational approximations with decay exponent 0.86.

Recently, a different approach based on Hankel determinants of rational functionals has been introduced. Lai, Lupu, and Sprang [10] and Lai [9] used Hankel determinants evaluated at hypergeometric residue functionals, coupled with an auxiliary prime non-vanishing argument, to study p-adic zeta values and to prove the $\mathbb{Q}$-linear independence of $1, \zeta(3)$, and $\zeta(4)$. In a closely related direction, Fauzan [5] proved the irrationality of $\zeta(5)$ by forming an $h \times h$ Hankel matrix $G_K^{(5)}(X)$ from a rational functional on the kernel $D_N(t)^6/D_K(t)$, where $D_m(t) = \prod_{j=1}^{m}(t+j^2)$. Because $G_K^{(5)}(\zeta(5))$ admits a representation as the Gram matrix of a positive measure on $(0, \infty)$, its determinant $\Delta_K^{(5)}(\zeta(5))$ is strictly positive, while Andréief's integration identity [8] reduces the archimedean upper bound to a logarithmic potential problem [13].

A basic structural difference separates the Hankel determinant method from classical linear forms. When one constructs a linear form $L_n(X) = a_n X - b_n \in \mathbb{Z}[X]$ satisfying $0 < |L_n(\xi)| \leq e^{-\gamma n}$ and $\log|a_n| \leq \tau n$, an effective irrationality measure $|\xi - a/b| > b^{-\mu}$ follows immediately

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from the triangle inequality, because $L_n(X)$ has degree one and vanishes only at the single rational point b_n/a_n. By contrast, clearing the denominators of an $h \times h$ Hankel determinant produces an integer polynomial $Q_K(X) \in \mathbb{Z}[X]$ of growing degree $h = \lambda K$ satisfying $0 < Q_K(\xi) \leq e^{-\epsilon K^2}$. Evaluating $Q_K(X)$ at a rational point a/b near ξ does not automatically yield the integer lower bound $b^h|Q_K(a/b)| \geq 1$: because $\deg Q_K = h$ grows linearly with K, the polynomial $Q_K(X)$ could in principle vanish at a/b, or have complex roots accumulating so rapidly near ξ that $|Q_K(a/b)|$ cannot be controlled by $Q_K(\xi)$.

The purpose of this paper is to resolve this problem by developing the spectral theory of the Hankel matrix pencil $G_K^{(s)}(X)$ for all odd zeta values $\zeta(s)$ ($s = 2k+1 \geq 3$) and related L-functions, and to derive quantitative Diophantine and root-distribution theorems from it.

Outline of the paper. In Section 2, we define the order-s rational moment functional $\mu_X^{(s)}$ and the Hankel matrix $G_K^{(s)}(X)$. We prove that the primitive integer polynomial $P_K^{(s)}(X) \in \mathbb{Z}[X]$ obtained by dividing $\Delta_K^{(s)}(X) = \det G_K^{(s)}(X)$ by its rational content is canonically invariant under arbitrary changes of basis of the polynomial space $\mathbb{Q}[t]_{<h}$ (Theorem 2.4). We then show that passing to the Lagrange interpolation basis at the negative square poles $t = -a^2$ ($N < a \leq K$) completely diagonalizes the X-dependence of $G_K^{(s)}(X)$, expressing the derivative matrix $M_K^{(s)} = [X]G_K^{(s)}(X)$ as $V \operatorname{diag}(\gamma_a^{(s)})V^T$ while rendering all off-diagonal entries independent of X (Theorem 2.5).

In Section 3, we investigate the zero locus of $P_K^{(s)}(X)$. Using the positive definiteness of $G = G_K^{(s)}(\zeta(s))$ and the symmetry of $M_K^{(s)}$, we give a direct proof (Theorem 3.1) that $P_K^{(s)}(X)$ is unconditionally hyperbolic for every odd integer $s \geq 3$ (and indeed for a continuous family of matrix pencils indexed by real $s > 1$; Proposition 3.2): all h roots $r_1, \ldots, r_h$ are real and take the form $r_j = \zeta(s) - \beta_j^{-1}$, where $\beta_1, \ldots, \beta_h \in \mathbb{R} \setminus \{0\}$ are the eigenvalues of $B_K^{(s)} = G^{-1/2} M_K^{(s)} G^{-1/2}$. Using Carleman's criterion, Andréief's identity, and Sylvester's law of inertia, we prove that $G_K^{(5)}(\zeta(5))$ is strictly totally positive of all orders, compute the exact sign parity $(-1)^{n(n-1)/2}$ of the leading coefficient, and show that $B_K^{(5)}$ has universal signature $(\lfloor h/2 \rfloor, \lceil h/2 \rceil)$ for every $n \geq 1$ (Theorem 3.3). Consequently, exactly $\lfloor h/2 \rfloor$ roots of $P_K^{(5)}(X)$ lie below $\zeta(5)$ and $\lceil h/2 \rceil$ lie above $\zeta(5)$. We then prove that the roots of consecutive principal minor polynomials $\Delta_{K,r}^{(s)}(X)$ interlace and pinch $\zeta(s)$ monotonically from both sides, satisfying a Dodgson condensation sign-alternation law and generating a rational Stieltjes power-sum hierarchy (Theorem 3.4 and Corollary 3.6), and moreover that the interlacing structure yields an exact finite Stieltjes continued fraction whose spectral edge converges to $\zeta(s)$ (Corollary 3.5). We also prove that the normalized coefficients of $P_K^{(s)}(X)$ satisfy Newton log-concavity, with strict inequality except at possible internal zero blocks (Theorem 3.7), identify the eigenvalues of $B_K^{(s)}$ with those of the Christoffel–Darboux kernel evaluated at the negative pole lattice (Theorem 3.8), and extend hyperbolicity to Hurwitz zeta values and Dirichlet L-functions of conductors $q \in \{3, 4, 5, 8\}$ via explicit factorizations of Fekete polynomials (Theorem 3.9). We further prove that the Hankel family is not a specialization of any classical orthogonal polynomial family (Remark 3.10).

In Section 4, we establish a general transfer criterion (Theorem 4.1) showing that quadratic decay $e^{-\epsilon K^2}$ at ξ together with a zero-free disk $|r_j - \xi| \geq e^{-\rho K}$ implies an effective irrationality measure $|\xi - a/b| > b^{-\mu}$ for any $\mu > \lambda\rho/\epsilon$, as well as a conditional extension to algebraic approximations of degree $d \geq 2$ and an ultrametric zero-free disk over $\mathbb{Q}_p$ (Propositions 4.2

and 4.3). To bound $\|B_K^{(5)}\|$ sharply, we combine a Legendre-polynomial Remez inequality on $[a,b] = [1/18, 1/3]$ with the exact concave Stirling potential $\Phi(u)$ of the residue weights $|r_j|$. Proving strict concavity of the combined functional $\Psi_{a,b}(u)$ on $(0.1, 1)$, we enclose its unique global maximum in $[0.973, 0.974]$ to obtain $\|B_K^{(5)}\| \le e^{11.411K}$ and the unconditional irrationality measure $|\zeta(5) - a/b| > b^{-25687}$ (Theorem 4.4). Coupling the weight-five functional $\mu_X^{(5)}(R(t))$ with the shifted weight-three functional $-\mu_Y^{(3)}(tR(t))$ over the common denominator $D_K(t)$, we further prove that for all integers a, b with $b > 0$ and $a < (\pi^2/3)b$, the linear combination $b\zeta(5)+a\zeta(3)$ is irrational and satisfies $|b\zeta(5)+a\zeta(3)-p/q| > q^{-25687}$ (Theorem 4.5). We then evaluate the Euler–Maclaurin asymptotic of $\det M_K^{(5)}$ and a self-contained single-arcsine Jensen lower bound for $\det G_K^{(5)}(\zeta(5))$ to establish two-sided bounds on $\log|\det B_K^{(5)}|$, quadratic coefficient height bounds, and macroscopic exponential collapse of at least 1.40% of the roots onto $\zeta(5)$ (Theorems 4.6 and 4.7, and Proposition 4.8). We further identify a Lee–Yang-type phase transition at $\zeta(5)$ in the free energy of the Hankel polynomial partition function (Remark 4.10).

In Section 5, we turn to weight seven and multivariable/multi-pole extensions. We introduce a two-variable Beukers measure $d\Omega_7(x_1, x_2)$ on $(0,1)^2$ whose diagonal moments evaluate to $\zeta(7) - H_r^{(7)}$ and whose off-diagonal moments reduce to rational divided differences of $H_r^{(6)}$ (Theorem 5.1). We prove that this measure induces a dual hyperbolic matrix pencil $G_{\mathcal{A}}^{(7)}(X)$ whose X-derivative $M_{\mathcal{A}}^{(7)} = CC^T$ is strictly positive definite in the monomial basis itself, while $G_{\mathcal{A}}^{(7)}(\zeta(7))$ admits an infinite-rank Krein-space decomposition allowing explicit construction of polynomial subspaces where all roots lie strictly below or strictly above $\zeta(7)$ (Theorem 5.2), together with an analogous pencil for the Dirichlet beta value $\beta(4) = L(4, \chi_{-4})$ (Proposition 5.3). We next prove that all higher zeta values cancel identically on x-even pole pairs under the local functional τ_X (Theorem 5.4), establish the exact outer prime-sum integral $I_{\text{out}} = 453803/288000$ and a pole-ordered inner basis refinement saving $+1/3$ in the inner integral for $\zeta(7)$ (Theorem 5.6), and prove a well-defined q-adic non-vanishing and p-adic minor valuation theorem for simultaneous double-pole Hankel determinants attached to $\{1, \zeta(5), \zeta(6)\}$ (Theorem 5.7). Section **??** discusses the remaining arithmetic bottlenecks and directions for future research.

2. Algebraic Structure and Canonical Basis Invariance

Let $K > N \ge 1$ be integers, set $h := K - N$, and write $\alpha := N/K$ and $\lambda := h/K = 1 - \alpha$. For $m \ge 0$, define the polynomial

$$D_m(t) := \prod_{j=1}^{m} (t + j^2), \qquad D_0(t) := 1. \tag{2.1}$$

For an integer $s \ge 1$, let $H_j^{(s)} := \sum_{\nu=1}^{j} \nu^{-s}$ ($H_0^{(s)} := 0$) denote the j-th harmonic number of weight s, and let $(B_r)_{r\ge 0}$ denote the Bernoulli numbers, defined by $\sum_{r=0}^{n} \binom{n+1}{r} B_r = 0$ for $n \ge 1$ with $B_0 = 1$ (so $B_1 = -1/2, B_2 = 1/6, B_4 = -1/30, B_6 = 1/42$, and $B_{2r+1} = 0$ for $r \ge 1$).

Definition 2.1 (Order-s rational moment functional)**.** For an odd integer $s = 2k + 1 \ge 3$, let $\mu_X^{(s)}$ be the $\mathbb{Q}$-linear functional on the $\mathbb{Q}$-vector space of rational functions in t having only

simple poles among $\{-j^2 : j \geq 1\}$, defined on monomials t^e $(e \geq 0)$ by

$$\mu_X^{(s)}(t^e) := \frac{(-1)^e B_{2e+2}}{(2k)!} \prod_{r=3}^{2k+1} (2e+r) = \frac{(-1)^e B_{2e+2}(2e+2k+1)!}{(2k)!\,(2e+2)!}, \tag{2.2}$$

and on simple poles $(t+j^2)^{-1}$ $(j \geq 1)$ by

$$\mu_X^{(s)}\left(\frac{1}{t+j^2}\right) := j^{2k}\big(X - H_j^{(s)}\big) + \frac{1}{2j} - \frac{1}{2k}, \tag{2.3}$$

extended to all such rational functions by polynomial division and partial fraction decomposition.

In particular, for $s = 3$ $(k = 1)$, $s = 5$ $(k = 2)$, and $s = 7$ $(k = 3)$, the pole values are $\mu_X^{(3)}((t+j^2)^{-1}) = j^2(X - H_j^{(3)}) + \frac{1}{2j} - \frac{1}{2}$, $\mu_X^{(5)}((t+j^2)^{-1}) = j^4(X - H_j^{(5)}) + \frac{1}{2j} - \frac{1}{4}$, and $\mu_X^{(7)}((t+j^2)^{-1}) = j^6(X - H_j^{(7)}) + \frac{1}{2j} - \frac{1}{6}$.

To connect $\mu_X^{(s)}$ with p-adic analysis, we record its pullback representation in terms of a one-variable auxiliary functional $\tau_X^{(s)}$ on rational functions of x with simple integer poles. For an integer $r \in \mathbb{Z}$, set $d(r) := r$ if $r \geq 0$ and $d(r) := -r - 1$ if $r < 0$.

Lemma 2.2 (Pullback identity)**.** *For an odd integer $s = 2k+1 \geq 3$, let $\tau_X^{(s)}$ be the $\mathbb{Q}$-linear functional on rational functions of x with simple integer poles defined on monomials by*

$$\tau_X^{(s)}(x^d) := \begin{cases} \dfrac{d!}{(d-2k+1)!\,(2k)!} B_{d-2k+1}, & d \geq 2k-1, \\ 0, & 0 \leq d < 2k-1, \end{cases} \tag{2.4}$$

and on simple poles by $\tau_X^{(s)}((x-r)^{-1}) := H_{d(r)}^{(s)} - X$ $(r \in \mathbb{Z})$. Then for every rational function $R(t)$ in the domain of $\mu_X^{(s)}$,

$$\mu_X^{(s)}\big(R(t)\big) = \tau_X^{(s)}\big(x^{2k+1} R(-x^2)\big). \tag{2.5}$$

Proof. By $\mathbb{Q}$-linearity, it suffices to verify (2.5) for $R(t) = t^e$ $(e \geq 0)$ and $R(t) = (t+j^2)^{-1}$ $(j \geq 1)$. For $R(t) = t^e$, we have $x^{2k+1}R(-x^2) = (-1)^e x^{2e+2k+1}$. Since $d = 2e+2k+1 \geq 2k+1$, (2.4) gives

$$\tau_X^{(s)}\big((-1)^e x^{2e+2k+1}\big) = \frac{(-1)^e (2e+2k+1)!}{(2e+2)!\,(2k)!} B_{2e+2} = \mu_X^{(s)}(t^e).$$

For $R(t) = (t+j^2)^{-1}$, polynomial division in $\mathbb{Q}[x]$ yields the exact identity

$$\frac{x^{2k+1}}{j^2 - x^2} = -\sum_{r=1}^{k} j^{2r-2} x^{2k+1-2r} - \frac{j^{2k}}{2}\left(\frac{1}{x-j} + \frac{1}{x+j}\right).$$

In the polynomial sum, the $r = 1$ term is $-x^{2k-1}$, while all terms with $2 \leq r \leq k$ have degree $2k+1-2r \leq 2k-3 < 2k-1$ and map to zero under $\tau_X^{(s)}$. At $d = 2k-1$, (2.4) gives $\tau_X^{(s)}(-x^{2k-1}) = -\frac{(2k-1)!}{0!(2k)!} B_0 = -\frac{1}{2k}$. Since $d(j) = j$ and $d(-j) = j-1$, the pole pair contributes

$$-\frac{j^{2k}}{2}\Big(\big(H_j^{(s)} - X\big) + \big(H_{j-1}^{(s)} - X\big)\Big) = -\frac{j^{2k}}{2}\big(2H_j^{(s)} - j^{-(2k+1)} - 2X\big) = j^{2k}\big(X - H_j^{(s)}\big) + \frac{1}{2j}.$$

Adding $-\frac{1}{2k}$ recovers (2.3). □

Lemma 2.3 (Reflection and difference identities for $\tau_X^{(s)}$)**.** *The functional $\tau_X^{(s)}$ defined in Lemma 2.2 satisfies:*

(1) (Reflection identity) *For every rational function $g(x)$ in the $\mathbb{Q}$-span of monomials $\{x^d : d \geq 0\}$ and simple poles $\{(x-r)^{-1} : r \in \mathbb{Z}\}$,*

$$\tau_X^{(s)}\big(g(-1-x)\big) = -\tau_X^{(s)}\big(g(x)\big). \tag{2.6}$$

(2) (Difference identity) *For every simple pole $h(x) = (x-r)^{-1}$ with $r \in \mathbb{Z} \setminus \{0,1\}$,*

$$\tau_X^{(s)}\big(h(x+1) - h(x)\big) = \frac{h^{(2k)}(0)}{(2k)!} = -r^{-(2k+1)}. \tag{2.7}$$

By $\mathbb{Q}$-linearity of $\tau_X^{(s)}$, both identities extend to all rational functions in the $\mathbb{Q}$-span of the stated basis elements.

Proof. (1) Reflection. Since $\tau_X^{(s)}$ is $\mathbb{Q}$-linear, it suffices to verify (2.6) on monomials x^d and simple poles $(x-r)^{-1}$.

- For x^d with $0 \leq d < 2k-1$: both $\tau_X^{(s)}(x^d) = 0$ and $\tau_X^{(s)}((-1-x)^d) = 0$ (since $\deg((-1-x)^d) = d < 2k-1$), so both sides vanish.
- For x^d with $d \geq 2k-1$: expanding $(-1-x)^d = (-1)^d(1+x)^d$ and applying $\tau_X^{(s)}(x^i) = \frac{i!}{(i-2k+1)!(2k)!}B_{i-2k+1}$, the Bernoulli polynomial identity $B_n(1) = B_n$ for $n \neq 1$ gives $\tau_X^{(s)}\big((1+x)^d\big) = \tau_X^{(s)}(x^d)$ (the $B_1 = -1/2$ contribution at $i = 2k$ cancels between the $(1+x)^d$ and x^d expansions). Hence $\tau_X^{(s)}\big((-1-x)^d\big) = (-1)^d\tau_X^{(s)}(x^d)$. Since $s = 2k+1$ is odd, only odd $d \geq 2k-1$ contribute (all Bernoulli numbers B_{d-2k+1} with even $d-2k+1 > 1$ vanish), and $(-1)^d = -1$.
- For $(x-r)^{-1}$: $(-1-x-r)^{-1} = -(x-(-1-r))^{-1}$, and $d(-1-r) = d(r)$ for all $r \in \mathbb{Z}$ (direct verification: if $r \geq 0$, $d(-1-r) = -(-1-r)-1 = r = d(r)$; if $r < 0$, $d(-1-r) = -1-r = d(r)$). Thus $\tau_X^{(s)}\big((-1-x-r)^{-1}\big) = -(H^{(s)}_{d(-1-r)} - X) = -(H^{(s)}_{d(r)} - X) = -\tau_X^{(s)}\big((x-r)^{-1}\big)$.

(2) Difference. For $h(x) = (x-r)^{-1}$ with $r \geq 2$: $h(x+1) = (x-(r-1))^{-1}$, so $h(x+1)-h(x) = (x-(r-1))^{-1} - (x-r)^{-1}$, and $\tau_X^{(s)}(h(x+1)-h(x)) = (H^{(s)}_{r-1} - X) - (H^{(s)}_r - X) = -r^{-(2k+1)}$. On the other hand, $h^{(2k)}(0) = (2k)!\,(-r)^{-(2k+1)}$, so $h^{(2k)}(0)/(2k)! = -r^{-(2k+1)}$. For $r \leq -1$: $h(x+1)-h(x) = (x-(r-1))^{-1} - (x-r)^{-1}$ with $d(r-1) = -r$ and $d(r) = -r-1$, so $\tau_X^{(s)}(h(x+1)-h(x)) = (H_{-r} - X) - (H_{-r-1} - X) = (-r)^{-(2k+1)}$. And $h^{(2k)}(0) = (2k)!\,(-1)^{2k}r^{-(2k+1)} = (2k)!\,(-r)^{-(2k+1)}$ (since $r < 0$), so $h^{(2k)}(0)/(2k)! = (-r)^{-(2k+1)}$. (The cases $r = 0, 1$ are excluded because then either h itself or $h(x+1) = x^{-1}$ has a pole at $x = 0$, so $\tau_X^{(s)}$ is not defined at these points.) □

Fix an integer $m \geq 1$, and define the $h \times h$ Hankel matrix and its determinant by

$$G_K^{(s)}(X) := \left[\mu_X^{(s)}\left(\frac{D_N(t)^{2m}\,t^{i+j}}{D_K(t)}\right)\right]_{0\leq i,j<h}, \qquad \Delta_K^{(s)}(X) := \det G_K^{(s)}(X) \in \mathbb{Q}[X]. \tag{2.8}$$

For any non-zero polynomial $A(X) \in \mathbb{Q}[X]$, let $\mathrm{cont}_{\mathbb{Q}}(A) > 0$ denote its rational content (the unique positive rational number such that $A(X)/\mathrm{cont}_{\mathbb{Q}}(A) \in \mathbb{Z}[X]$ has coprime integer coefficients). We define the primitive Hankel polynomial by

$$P_K^{(s)}(X) := \frac{\Delta_K^{(s)}(X)}{\mathrm{cont}_{\mathbb{Q}}(\Delta_K^{(s)})} \in \mathbb{Z}[X]. \tag{2.9}$$

Theorem 2.4 (Canonical basis invariance)**.** *Let μ_X be any $\mathbb{Q}[X]$-valued linear functional on rational functions of t, and fix $W(t) = D_N(t)^{2m}$. For any two $\mathbb{Q}$-bases $\mathcal{A} = \{A_0(t), \dots, A_{h-1}(t)\}$ and $\mathcal{B} = \{B_0(t), \dots, B_{h-1}(t)\}$ of $\mathbb{Q}[t]_{<h}$, define*

$$G_{\mathcal{A},\mathcal{B}}(X) := \left[\mu_X\left(\frac{W(t)A_i(t)B_j(t)}{D_K(t)}\right)\right]_{0\le i,j<h}, \qquad \Delta_{\mathcal{A},\mathcal{B}}(X) := \det G_{\mathcal{A},\mathcal{B}}(X) \in \mathbb{Q}[X].$$

If $\Delta_K(X) = \det G_K(X) \neq 0$, then the primitive integer polynomial $P_{\mathcal{A},\mathcal{B}}(X) := \Delta_{\mathcal{A},\mathcal{B}}(X)/\operatorname{cont}_{\mathbb{Q}}(\Delta_{\mathcal{A},\mathcal{B}}) \in \mathbb{Z}[X]$ satisfies $P_{\mathcal{A},\mathcal{B}}(X) = \pm P_K(X)$.

Proof. Let $M_{\mathcal{A}}, M_{\mathcal{B}} \in \mathrm{GL}_h(\mathbb{Q})$ be the invertible change-of-basis matrices satisfying $A_i(t) = \sum_{r=0}^{h-1}(M_{\mathcal{A}})_{i,r}t^r$ and $B_j(t) = \sum_{s=0}^{h-1}(M_{\mathcal{B}})_{j,s}t^s$ for $0 \le i,j < h$. By $\mathbb{Q}$-linearity of μ_X, we have $G_{\mathcal{A},\mathcal{B}}(X) = M_{\mathcal{A}}G_K(X)M_{\mathcal{B}}^T$. Taking determinants gives $\Delta_{\mathcal{A},\mathcal{B}}(X) = c\,\Delta_K(X)$, where $c := (\det M_{\mathcal{A}})(\det M_{\mathcal{B}}) \in \mathbb{Q}^\times$ is independent of X. Since $\operatorname{cont}_{\mathbb{Q}}(c\Delta_K) = |c|\operatorname{cont}_{\mathbb{Q}}(\Delta_K)$, dividing yields $P_{\mathcal{A},\mathcal{B}}(X) = \operatorname{sgn}(c)P_K(X)$. □

Theorem 2.4 implies that no change of polynomial basis in $\mathbb{Q}[t]_{<h}$ (such as passing to Pochhammer, Jacobi, or Cauchy-type rational bases over the common denominator $D_K(t)$) can alter the primitive polynomial $P_K^{(s)}(X)$. On the other hand, choosing the Lagrange interpolation basis at the negative square poles $v_a := -a^2$ $(N < a \le K)$ reveals the internal algebraic structure of the matrix pencil $G_K^{(s)}(X)$.

Theorem 2.5 (Lagrange pole diagonalization)**.** *Set $D_{\mathrm{tail}}(t) := D_K(t)/D_N(t) = \prod_{c=N+1}^{K}(t + c^2)$, and let $V := (v_a^i)_{0\le i<h,\, N<a\le K}$ be the $h \times h$ Vandermonde matrix at the nodes $v_a = -a^2$. Define the Lagrange interpolation polynomials*

$$L_a(t) := \prod_{\substack{N<b\le K\\ b\neq a}} \frac{t+b^2}{b^2-a^2} = \frac{D_{\mathrm{tail}}(t)}{(t+a^2)D'_{\mathrm{tail}}(-a^2)} \qquad (N < a \le K). \tag{2.10}$$

Then the transformed matrix $\widetilde{G}_K^{(s)}(X) := V^{-1}G_K^{(s)}(X)(V^T)^{-1} = \left[\mu_X^{(s)}(D_N(t)^{2m}L_a(t)L_b(t)/D_K(t))\right]_{N<a,b\le K}$ is strictly diagonal in X:

1. *For all $a \neq b$, $\widetilde{G}_K^{(s)}(X)_{a,b} \in \mathbb{Q}$ is a constant rational number independent of X.*
2. *For $a = b$, $\widetilde{G}_K^{(s)}(X)_{a,a} = \gamma_a^{(s)}(X - H_a^{(s)}) + r_a$, where $r_a \in \mathbb{Q}$ is independent of X and*

$$\gamma_a^{(s)} := \frac{a^{2k}D_N(-a^2)^{2m-1}}{D'_{\mathrm{tail}}(-a^2)} = \frac{a^{2k}\prod_{\nu=1}^{N}(\nu^2-a^2)^{2m-1}}{\prod_{\substack{N<b\le K\\ b\neq a}}(b^2-a^2)} \neq 0. \tag{2.11}$$

In particular, the X-derivative matrix $M_K^{(s)} := [X]G_K^{(s)}(X)$ admits the exact factorization $M_K^{(s)} = V\operatorname{diag}(\gamma_a^{(s)})_{N<a\le K}V^T$.

Proof. Since $L_a(v_b) = \delta_{a,b}$, we have $\sum_{i=0}^{h-1}(V^{-1})_{a,i}v_b^i = \delta_{a,b}$, so the (a,b) entry of $\widetilde{G}_K^{(s)}(X) = V^{-1}G_K^{(s)}(X)(V^T)^{-1}$ is $\mu_X^{(s)}(R_{a,b}(t))$, where

$$R_{a,b}(t) := \frac{D_N(t)^{2m}L_a(t)L_b(t)}{D_K(t)} = \frac{D_N(t)^{2m-1}D_{\mathrm{tail}}(t)}{D'_{\mathrm{tail}}(-a^2)D'_{\mathrm{tail}}(-b^2)(t+a^2)(t+b^2)}.$$

(1) When $a \neq b$, the product $(t+a^2)(t+b^2)$ divides $D_{\mathrm{tail}}(t) = \prod_{c=N+1}^{K}(t+c^2)$ in $\mathbb{Q}[t]$. Thus $R_{a,b}(t)$ is a polynomial in $\mathbb{Q}[t]$, so $\mu_X^{(s)}(R_{a,b}) \in \mathbb{Q}$ is independent of X by (2.2).

(2) When $a = b$, one factor of $(t+a^2)$ cancels from $D_{\mathrm{tail}}(t)$, leaving a single simple pole at $t = -a^2$ with residue $D_N(-a^2)^{2m-1}/D'_{\mathrm{tail}}(-a^2)$. Applying (2.3) yields $\widetilde{G}_K^{(s)}(X)_{a,a} = \gamma_a^{(s)}(X -$

$H_a^{(s)}) + r_a$ with $r_a \in \mathbb{Q}$. Differentiating $\widetilde{G}_K^{(s)}(X) = V^{-1} G_K^{(s)}(X)(V^T)^{-1}$ with respect to X gives $V^{-1} M_K^{(s)} (V^T)^{-1} = \operatorname{diag}(\gamma_a^{(s)})$, whence $M_K^{(s)} = V \operatorname{diag}(\gamma_a^{(s)}) V^T$. □

3. Hyperbolicity, Total Positivity, and Spectral Structure

We now connect the algebraic factorization of Section 2 with the archimedean integral representation at $X = \zeta(s)$. For an odd integer $s = 2k+1 \geq 3$, define the weight function on $(0, \infty)$ by

$$w_s(y) := \frac{2(2\pi)^{2k}}{(2k)!} y^{2k+1} \sum_{l=1}^{\infty} l^{2k} e^{-2\pi l y} = \frac{2y^{2k+1}}{(2k)!} \frac{d^{2k}}{dy^{2k}} \left(\frac{1}{e^{2\pi y} - 1} \right) > 0 \qquad (y > 0). \tag{3.1}$$

For $e \geq 0$, term-by-term integration gives

$$\int_0^{\infty} y^{2e} w_s(y)\, dy = \frac{2(2\pi)^{2k}}{(2k)!} \sum_{l=1}^{\infty} l^{2k} \frac{(2e+2k+1)!}{(2\pi l)^{2e+2k+2}} = \frac{2(2e+2k+1)!}{(2k)!(2\pi)^{2e+2}} \zeta(2e+2),$$

which matches $\mu_X^{(s)}(t^e)$ in (2.2) by Euler's formula $\zeta(2e+2) = (-1)^e B_{2e+2}(2\pi)^{2e+2}/(2(2e+2)!)$. For simple poles, writing $\frac{y^{2k+1}}{y^2+j^2} = \sum_{r=1}^{k} (-1)^{r-1} j^{2r-2} y^{2k+1-2r} + (-1)^k \frac{j^{2k} y}{y^2+j^2}$, the polynomial part has degree $2k-1 < 2k$ and vanishes after $2k$ differentiations. Integrating by parts $2k$ times on $\int_0^{\infty} \frac{w_s(y)}{y^2+j^2}\, dy$ and applying Hermite's integral formula for $\zeta(2k+1, j)$ [11, 25.11.29] yields $\int_0^{\infty} \frac{w_s(y)}{y^2+j^2}\, dy = j^{2k}(\zeta(s) - H_j^{(s)}) + \frac{1}{2j} - \frac{1}{2k} = \mu_{\zeta(s)}^{(s)}((t+j^2)^{-1})$. By $\mathbb{Q}$-linearity,

$$\mu_{\zeta(s)}^{(s)}\big(R(t)\big) = \int_0^{\infty} R(y^2) w_s(y)\, dy \tag{3.2}$$

for every rational function $R(t)$ in the domain of $\mu_X^{(s)}$.

Theorem 3.1 (Universal hyperbolicity and zero-free disk)**.** *For every odd integer $s = 2k+1 \geq 3$, every $K > N \geq 1$ ($h = K - N$), and every $m \geq 1$, the polynomial $\Delta_K^{(s)}(X)$ (and hence $P_K^{(s)}(X)$) has exact degree h, and all h of its complex roots $r_1, \ldots, r_h$ are real and distinct from $\xi := \zeta(s)$. Specifically, let $G := G_K^{(s)}(\xi)$ and $M_K := [X] G_K^{(s)}(X)$. Then G is real symmetric positive definite, $B_K^{(s)} := G^{-1/2} M_K G^{-1/2}$ is a real symmetric invertible $h \times h$ matrix with non-zero real eigenvalues $\beta_1, \ldots, \beta_h \in \mathbb{R} \setminus \{0\}$, and the roots of $P_K^{(s)}(X)$ are*

$$r_j = \zeta(s) - \frac{1}{\beta_j} \in \mathbb{R} \qquad (1 \leq j \leq h). \tag{3.3}$$

In particular, every root satisfies $|r_j - \zeta(s)| \geq \|B_K^{(s)}\|^{-1}$.

Proof. For any non-zero vector $v = (v_0, \ldots, v_{h-1})^T \in \mathbb{C}^h \setminus \{0\}$, the polynomial $q(t) := \sum_{i=0}^{h-1} v_i t^i$ is not identically zero. By (3.2),

$$v^* G v = \int_0^{\infty} \frac{D_N(y^2)^{2m} |q(y^2)|^2}{D_K(y^2)} w_s(y)\, dy > 0,$$

so G is real symmetric and strictly positive definite.

By Theorem 2.5, $M_K = V \operatorname{diag}(\gamma_a^{(s)}) V^T$. Since the nodes $v_a = -a^2$ $(N < a \leq K)$ are distinct, $\det V = \prod_{N<a<b\leq K} (a^2 - b^2) \neq 0$ and $\prod_{a=N+1}^{K} D'_{\mathrm{tail}}(-a^2) = (-1)^{h(h-1)/2} (\det V)^2$.

Hence

$$[X^h]\Delta_K^{(s)}(X) = \det M_K = (-1)^{h(h-1)/2} \prod_{a=N+1}^{K} a^{2k} D_N(-a^2)^{2m-1} \neq 0, \tag{3.4}$$

so $\Delta_K^{(s)}(X)$ has exact degree h.

Write $G_K^{(s)}(X) = G + (X-\xi)M_K$. If $X \in \mathbb{C}$ is a root of $\det G_K^{(s)}(X) = 0$, there exists $v \in \mathbb{C}^h \setminus \{0\}$ such that $(G + (X-\xi)M_K)v = 0$. Multiplying on the left by v^* gives $v^*Gv + (X-\xi)(v^*M_Kv) = 0$. Since $v^*Gv > 0$, we have $v^*M_Kv \neq 0$ and $X \neq \xi$. Since M_K is real symmetric, $v^*M_Kv \in \mathbb{R} \setminus \{0\}$, whence

$$X - \xi = -\frac{v^*Gv}{v^*M_Kv} \in \mathbb{R}.$$

Factoring $G_K^{(s)}(X) = G^{1/2}(I + (X-\xi)B_K^{(s)})G^{1/2}$ shows that $B_K^{(s)}$ is real symmetric with $\det B_K^{(s)} = (\det G)^{-1} \det M_K \neq 0$, so its eigenvalues $\beta_1, \dots, \beta_h$ are real and non-zero, and $\Delta_K^{(s)}(X) = \Delta_K^{(s)}(\xi) \prod_{j=1}^{h}(1 + (X-\xi)\beta_j)$, giving (3.3). □

Notice that the proof of Theorem 3.1 relies only on the positive definiteness of $G_K^{(s)}(\xi)$ and the symmetry and non-singularity of $M_K^{(s)}$. This allows us to embed $\Delta_K^{(2k+1)}(X)$ into a continuous family of hyperbolic matrix pencils defined for all real $s > 1$.

Proposition 3.2 (Continuous hyperbolic pencil for all real $s > 1$)**.** *For any real number $s > 1$, define the positive weight $w_s(y) := \frac{(2\pi)^s}{2\Gamma(s)} y^s \sum_{l=1}^{\infty} l^{s-1} e^{-2\pi l y} > 0$ on $(0,\infty)$, and define the $h \times h$ real symmetric matrix pencil $G_K^{(s)}(X) := A^{(s)} + (X - \zeta(s))M_K^{(s)}$, where*

$$A_{i,j}^{(s)} := \int_0^\infty \frac{D_N(y^2)^{2m} y^{2(i+j)}}{D_K(y^2)} w_s(y)\, dy, \qquad M_K^{(s)} := V \operatorname{diag}_{N<a\le K}\left(\frac{a^{s-1} D_N(-a^2)^{2m-1}}{D'_{\mathrm{tail}}(-a^2)}\right) V^T.$$

Then for every real $s > 1$, $G_K^{(s)}(X)$ coincides at odd integers $s = 2k+1$ with (2.8), and $\Delta_K^{(s)}(X) := \det G_K^{(s)}(X)$ is a polynomial of exact degree h with all h roots real.

Proof. For every real $s > 1$, $w_s(y) > 0$ on $(0,\infty)$ implies that $A^{(s)}$ is real symmetric positive definite, while $a^{s-1} > 0$ implies that $M_K^{(s)}$ is real symmetric and non-singular. The proof of Theorem 3.1 applies directly to $A^{(s)} + (X - \zeta(s))M_K^{(s)}$. □

We next establish strict total positivity of the moment matrix, the sign parity of the leading coefficient, and the exact bilateral signature of the roots around $\zeta(s)$.

Theorem 3.3 (Total positivity, leading sign, and universal signature)**.** *Fix $K > N \ge 1$ ($h = K - N$), $m \ge 1$, and an odd integer $s = 2k+1 \ge 3$.*

1. *The infinite moment sequence $\mu_r^{(s)} := \int_0^\infty t^r \frac{D_N(t)^{2m}}{D_K(t)} \frac{w_s(\sqrt{t})}{2\sqrt{t}}\, dt$ ($r \ge 0$, with $2mN \le K$) is a determinate Stieltjes moment sequence on $[0,\infty)$, and $G_K^{(s)}(\zeta(s))$ is strictly totally positive of all orders: every $r \times r$ minor $\det G_K^{(s)}(\zeta(s))[I,J]$ ($0 \le i_1 < \cdots < i_r < h$, $0 \le j_1 < \cdots < j_r < h$) is strictly positive.*
2. *In the weight-5 construction with $K = 30n, N = 3n, h = 27n$, and $2m = 6$, the leading coefficient $a_{K,h} := [X^h]P_K^{(5)}(X)$ has exact sign $\operatorname{sgn}(a_{K,h}) = (-1)^{n(n-1)/2}$.*

(3) *For every odd $s \geq 3$ and $2m \geq 2$, the matrix $B_K^{(s)}$ has signature $(\lceil h/2 \rceil, \lfloor h/2 \rfloor)$ if N is even, and $(\lfloor h/2 \rfloor, \lceil h/2 \rceil)$ if N is odd and h is odd. In particular, for $K = 30n, N = 3n, h = 27n$, the signature of $B_K^{(5)}$ is universally $(\lfloor h/2 \rfloor, \lceil h/2 \rceil)$ for every $n \geq 1$, so $P_K^{(5)}(X)$ has $\lfloor 27n/2 \rfloor$ roots strictly below $\zeta(5)$ and $\lceil 27n/2 \rceil$ roots strictly above $\zeta(5)$.*

Proof. (1) Since $D_N(t)^{2m}/D_K(t) \leq C_K$ for $t \geq 1$ and $w_s(y) \leq C_s y^s e^{-2\pi y}$, we have $\mu_r^{(s)} \leq C'_{K,s} \int_0^\infty y^{2r+s} e^{-2\pi y}\, dy = C'_{K,s} \frac{(2r+s)!}{(2\pi)^{2r+s+1}}$. By Stirling's formula, $(\mu_r^{(s)})^{-1/(2r)} \geq \frac{c}{r}$ for a constant $c > 0$, so $\sum_{r=1}^\infty (\mu_r^{(s)})^{-1/(2r)} = \infty$, which proves Stieltjes determinacy by Carleman's criterion. For any $I = \{i_1 < \cdots < i_r\}$ and $J = \{j_1 < \cdots < j_r\}$, Andréief's identity [8, eq. (1.7)] gives

$$\det G_K^{(s)}(\zeta(s))[I,J] = \int_{0<t_1<\cdots<t_r<\infty} \det_{1\leq a,b\leq r}\left(t_a^{i_b}\right) \det_{1\leq a,b\leq r}\left(t_a^{j_b}\right) \prod_{a=1}^r \frac{D_N(t_a)^{2m} w_s(\sqrt{t_a})}{2\sqrt{t_a} D_K(t_a)}\, dt_a > 0,$$

because the monomials $\{t^i\}_{i\geq 0}$ form a strictly totally positive Chebyshev system on $(0,\infty)$.

(2) By (3.4) with $k = 2$ and $2m - 1 = 5$, every factor $\nu^2 - a^2$ $(1 \leq \nu \leq N < a \leq K)$ is negative, so $\operatorname{sgn}(D_N(-a^2)^5) = (-1)^{5N}$. For $K = 30n, N = 3n, h = 27n$, the exponent of -1 in $\det M_K^{(5)}$ is

$$E(n) = \frac{h(h-1)}{2} + 5Nh = \frac{27n(27n-1)}{2} + 5(3n)(27n) = \frac{1539n^2 - 27n}{2}.$$

Expanding $756n^2 + 13n(n-1) + \frac{n(n-1)}{2} = (756 + 13 + \frac{1}{2})n^2 - (13 + \frac{1}{2})n = \frac{1539n^2-27n}{2}$ shows that $E(n) \equiv \frac{n(n-1)}{2} \pmod 2$, whence $\operatorname{sgn}(a_{K,h}) = (-1)^{n(n-1)/2}$.

(3) By Theorem 2.5(2) and Sylvester's law of inertia, $B_K^{(s)} = (G^{-1/2}V)\operatorname{diag}(\gamma_a^{(s)})(G^{-1/2}V)^T$ has the same signature as $\operatorname{diag}(\gamma_a^{(s)})_{N<a\leq K}$. In (2.11), the numerator has sign $(-1)^{(2m-1)N}$, while the denominator $\prod_{b\neq a}(b^2 - a^2)$ has $a - N - 1$ negative factors (those with $N+1 \leq b < a$). Thus

$$\operatorname{sgn}(\gamma_a^{(s)}) = (-1)^{(2m-1)N+a-N-1} = (-1)^{2(m-1)N+a-1} = (-1)^{a-1} \qquad (N+1 \leq a \leq K),$$

independent of s and m. For $N = 3n$ and $h = 27n$: if n is even, h is even and there are $\frac{h}{2} = \lfloor h/2 \rfloor = \lceil h/2 \rceil$ positive and negative signs; if n is odd, h is odd and the first index $a = 3n + 1$ is even, so $(-1)^{a-1}$ starts and ends at -1, giving $\frac{h-1}{2} = \lfloor h/2 \rfloor$ positive signs $(\beta_j > 0 \implies r_j < \xi)$ and $\frac{h+1}{2} = \lceil h/2 \rceil$ negative signs $(\beta_j < 0 \implies r_j > \xi)$. □

Theorem 3.4 (Root interlacing and Dodgson condensation for principal minors)**.** *Fix $K > N \geq 1$, and for $1 \leq r \leq h = K - N$, let $\Delta_{K,r}^{(s)}(X) := \det\left[G_K^{(s)}(X)_{i,j}\right]_{0\leq i,j<r}$ be the $r \times r$ leading principal minor polynomial.*

(1) *The eigenvalues of $B_{K,r}^{(s)}$ and $B_{K,r+1}^{(s)}$ satisfy Cauchy's interlacing inequalities $\beta_1^{(r+1)} \leq \beta_1^{(r)} \leq \beta_2^{(r+1)} \leq \cdots \leq \beta_r^{(r)} \leq \beta_{r+1}^{(r+1)}$. Consequently, the roots of $\Delta_{K,r}^{(s)}(X)$ and $\Delta_{K,r+1}^{(s)}(X)$ interlace on $(-\infty, \zeta(s))$ and on $(\zeta(s), +\infty)$, and $\min_{1\leq j\leq r} |r_j^{(r)} - \zeta(s)|$ is non-increasing in r.*

(2) *For every $2 \leq r \leq h$ and every real root $r_j^{(r-1)}$ of $\Delta_{K,r-1}^{(s)}(X) = 0$,*

$$\Delta_{K,r}^{(s)}\big(r_j^{(r-1)}\big)\, \Delta_{K,r-2}^{(s)}\big(r_j^{(r-1)}\big) = -\Big(\Sigma_{K,r-1}^{(s)}\big(r_j^{(r-1)}\big)\Big)^2 \leq 0, \tag{3.5}$$

where $\Delta_{K,0}^{(s)} := 1$ and $\Sigma_{K,r-1}^{(s)}(X)$ is the $(r-1)\times(r-1)$ minor of $G_{K,r}^{(s)}(X)$ obtained by deleting row $r-1$ and column $r-2$.

Proof. (1) Let $G = LL^T$ be the lower-triangular Cholesky decomposition of $G_K^{(s)}(\xi)$. Its leading $r \times r$ block L_r satisfies $G_r = L_r L_r^T$, so $\widehat{B}_r := L_r^{-1} M_{K,r} (L_r^{-1})^T$ is orthogonally similar to $B_{K,r}^{(s)}$ and is the leading $r \times r$ principal submatrix of $\widehat{B}_{r+1}$. Cauchy's interlacing theorem gives $\beta_1^{(r+1)} \le \beta_1^{(r)} \le \cdots \le \beta_{r+1}^{(r+1)}$, and the order-preserving map $\beta \mapsto \xi - \beta^{-1}$ on $(-\infty, 0)$ and $(0, \infty)$ yields root interlacing and monotonicity of $\min_j |r_j^{(r)} - \xi|$.

(2) By Dodgson's condensation identity (Jacobi's identity for determinants) applied to the symmetric matrix $G_{K,r}^{(s)}(X)$ with respect to its bottom-right 2×2 indices $\{r-2, r-1\}$, we have $\Delta_{K,r}^{(s)}(X)\Delta_{K,r-2}^{(s)}(X) = \Delta_{K,r-1}^{(s)}(X)\widetilde{\Delta}_{K,r-1}^{(s)}(X) - \big(\Sigma_{K,r-1}^{(s)}(X)\big)^2$. Evaluating at $X = r_j^{(r-1)}$ where $\Delta_{K,r-1}^{(s)}(r_j^{(r-1)}) = 0$ gives (3.5). □

Corollary 3.5 (Stieltjes J-fraction from interlacing roots)**.** *Fix $K > N \ge 1$ and $X = \zeta(s)$. Let $G := G_K^{(s)}(\zeta(s)) = LL^T$ be the lower-triangular Cholesky decomposition, and for $1 \le r \le h$, set $G_r := L_r L_r^T$ (the leading $r \times r$ principal block) and $\widehat{B}_r := L_r^{-1} M_{K,r} (L_r^{-1})^T$. Let $T_h := Q^T \widehat{B}_h Q$ be the symmetric tridiagonal reduction obtained by the Lanczos algorithm applied to $\widehat{B}_h$ with starting vector e_h. Write $T_h = \mathrm{tridiag}(a_1, \dots, a_h; b_1, \dots, b_{h-1})$ with $a_r \in \mathbb{R}$ and $b_r > 0$. Then:*

(1) *For every $X \in \mathbb{R}$ with $P_K^{(s)}(X) \neq 0$, the ratio of consecutive principal-minor polynomials admits the exact finite Stieltjes continued fraction expansion*

$$\frac{\Delta_{K,h-1}^{(s)}(X)}{\Delta_{K,h}^{(s)}(X)} = \cfrac{1}{a_1 + (X - \zeta(s)) - \cfrac{b_1^2}{a_2 + (X - \zeta(s)) - \cfrac{b_2^2}{\ddots - \cfrac{b_{h-1}^2}{a_h + (X - \zeta(s))}}}}, \tag{3.6}$$

with strictly positive coefficients $b_1^2, \dots, b_{h-1}^2 > 0$.

(2) *Evaluating (3.6) at $X = 0$ gives the rational number $\frac{\Delta_{K,h-1}^{(s)}(0)}{\Delta_{K,h}^{(s)}(0)} \in \mathbb{Q}$ (computable from the Bernoulli monomial moments $\mu_X^{(s)}(t^e)$, with no occurrence of $\zeta(s)$ in its definition), while the closest root $r_{\min} = \zeta(s) - \beta_{\max}^{-1}$ satisfies $|r_{\min} - \zeta(s)| = |\beta_{\max}|^{-1} \le e^{-K/2250}$ (Theorem 4.7), so the spectral edge of the J-fraction converges to $\zeta(s)$ at exponential rate.*

Proof. (1) Since $G_h = L_h L_h^T$ with L_h lower-triangular, the principal submatrix identity $G_r = L_r L_r^T$ holds for every $1 \le r \le h$. Thus $\widehat{B}_r := L_r^{-1} M_{K,r} (L_r^{-1})^T$ is orthogonally similar to $B_{K,r}^{(s)} := G_r^{-1/2} M_{K,r} G_r^{-1/2}$, and $\widehat{B}_r$ is the leading $r \times r$ principal submatrix of $\widehat{B}_{r+1}$. The Lanczos algorithm applied to $\widehat{B}_h$ with starting vector e_h produces an orthogonal $Q \in O_h(\mathbb{R})$ with $Q^T e_h = e_h$ such that $T_h := Q^T \widehat{B}_h Q$ is symmetric tridiagonal with positive off-diagonal entries $b_1, \dots, b_{h-1} > 0$. Since $Q^T e_h = e_h$, the (h,h) entry of $(I_h + (X - \zeta(s))\widehat{B}_h)^{-1}$ equals that of $(I_h + (X - \zeta(s))T_h)^{-1}$. By Cramer's rule, this (h,h) entry equals $\det(I_{h-1} + (X - \zeta(s))\widehat{B}_{h-1}) / \det(I_h + (X - \zeta(s))\widehat{B}_h) = \Delta_{K,h-1}^{(s)}(X) / \Delta_{K,h}^{(s)}(X)$. On the other hand, the (h,h) entry of $(I_h + (X - \zeta(s))T_h)^{-1}$ expands by the three-term determinant recurrence for tridiagonal matrices into the continued fraction (3.6). The positivity $b_r^2 > 0$ follows from the fact that the Lanczos off-diagonal entries of a symmetric matrix with distinct eigenvalues are strictly positive.

(2) At $X = 0$, $\Delta_{K,r}^{(s)}(0) = \det G_K^{(s)}(0)$ involves only the monomial moments $\mu_0^{(s)}(t^e) = \frac{(-1)^e B_{2e+2}(2e+2k+1)!}{(2k)!(2e+2)!} \in \mathbb{Q}$ (Definition 2.1), so the ratio is rational. The spectral edge $|\beta_{\max}|^{-1} \le e^{-K/2250}$ follows from Theorem 4.7. □

Corollary 3.6 (Stieltjes reciprocal-root power sums)**.** *For any fixed K and any rational shift $q_0 \in \mathbb{Q}$ with $P_K^{(s)}(q_0) \neq 0$, define the rational power sums*

$$S_m(q_0) := \frac{(-1)^{m-1}}{(m-1)!} \frac{d^m}{dX^m} \log P_K^{(s)}(X)\Big|_{X=q_0} = \sum_{j=1}^{h} \frac{1}{(r_{K,j} - q_0)^m} \in \mathbb{Q} \qquad (m \ge 1).$$

Then $\{S_{2r}(q_0)\}_{r\ge 0}$ ($S_0(q_0) := h$) is a positive Stieltjes moment sequence with strictly positive Hankel determinants, and at each fixed K the inner limit $\lim_{r\to\infty} \big(S_{2r}(q_0)\big)^{-1/(2r)} = d_{\min}(K) := \min_{1\le j\le h} |r_{K,j} - q_0|$ converges with factor $1 \le \big(S_{2r}(q_0)\big)^{-1/(2r)}/d_{\min}(K) \le h^{1/(2r)}$ for all $r \ge 1$.

Proof. Since $r_{K,1}, \dots, r_{K,h} \in \mathbb{R}$ by Theorem 3.1, the squares $u_j := (r_{K,j} - q_0)^{-2} > 0$ are positive real numbers, so $S_{2r}(q_0) = \sum_{j=1}^h u_j^r$ is the r-th moment of the positive discrete measure $\sum_{j=1}^h \delta_{u_j}$ on $(0,\infty)$. Moreover, $(\max_j u_j)^r \le S_{2r}(q_0) \le h(\max_j u_j)^r$, and taking $(-2r)$-th roots gives $h^{-1/(2r)} d_{\min}(K) \le S_{2r}(q_0)^{-1/(2r)} \le d_{\min}(K)$. □

The bilateral root split of Theorem 3.3(3) also implies Newton log-concavity of the coefficients of $P_K^{(s)}(X)$, with strict inequality except at possible internal zero blocks, without requiring any assumption on root simplicity.

Theorem 3.7 (Turánian non-negativity and Newton log-concavity)**.** *Fix any odd integer $s = 2k+1 \ge 3$ and integers $K > N \ge 1$ with $h = K - N \ge 2$ and $2m \ge 2$. Write $P_K^{(s)}(X) = \sum_{m=0}^h a_{K,m} X^m \in \mathbb{Z}[X]$ and let $\tilde a_{K,m} := a_{K,m}/\binom{h}{m}$.*

(1) *For every $1 \le m \le h-1$ and every $X \in \mathbb{R}$, the Turánian Wronskian satisfies*

$$\mathcal{T}_m(X) := \left(\frac{d^m}{dX^m} P_K^{(s)}(X)\right)^2 - \left(\frac{d^{m-1}}{dX^{m-1}} P_K^{(s)}(X)\right)\left(\frac{d^{m+1}}{dX^{m+1}} P_K^{(s)}(X)\right) \ge 0, \tag{3.7}$$

with strict inequality $\mathcal{T}_m(X) > 0$ at every $X \in \mathbb{R}$ where $\frac{d^{m-1}}{dX^{m-1}} P_K^{(s)}(X) \neq 0$. In particular, at $X = \zeta(s)$, $\frac{P_K^{(s)\prime\prime}(\zeta(s))}{P_K^{(s)}(\zeta(s))} < \big(1 - \frac{1}{h}\big)\left(\frac{P_K^{(s)\prime}(\zeta(s))}{P_K^{(s)}(\zeta(s))}\right)^2$ with strict inequality.

(2) *Newton's log-concavity $\tilde a_{K,m}^2 \ge \tilde a_{K,m-1}\,\tilde a_{K,m+1}$ holds for every $1 \le m \le h-1$. Strict inequality $\tilde a_{K,m}^2 > \tilde a_{K,m-1}\,\tilde a_{K,m+1}$ holds if and only if $\tilde a_{K,m} \neq 0$ or both $\tilde a_{K,m-1} \neq 0$ and $\tilde a_{K,m+1} \neq 0$. In the exceptional case $\tilde a_{K,m} = 0$ and ($\tilde a_{K,m-1} = 0$ or $\tilde a_{K,m+1} = 0$), equality $\tilde a_{K,m}^2 = \tilde a_{K,m-1}\,\tilde a_{K,m+1} = 0$ holds.*

(3) *Let $e_m(\beta_1, \dots, \beta_h) := [Y^m] \frac{P_K^{(s)}(\zeta(s)+Y)}{P_K^{(s)}(\zeta(s))}$ be the shifted Taylor coefficients around $X = \zeta(s)$, and set $\tilde e_m := e_m(\beta_1, \dots, \beta_h)/\binom{h}{m}$ (with $\tilde e_0 := 1$). Then $\tilde e_m^2 \ge \tilde e_{m-1}\,\tilde e_{m+1}$ for every $1 \le m \le h-1$, with strict inequality whenever $\tilde e_m \neq 0$ or both $\tilde e_{m-1}, \tilde e_{m+1} \neq 0$. Since all roots $-1/\beta_j = r_j - \zeta(s)$ are nonzero (Theorem 3.1) and not all equal (Theorem 3.3(3)), the only exceptional case is a block of three consecutive zeros $\tilde e_{m-1} = \tilde e_m = \tilde e_{m+1} = 0$, which for real β_j can occur only if the shifted polynomial has the symmetric structure $\prod_j (1 + \beta_j Y) = g(Y^q)$ for some $q \ge 3$; in the present construction, the β_j are real eigenvalues of $B_K^{(s)}$ and, being a Hankel spectrum, are generically distinct, so $\tilde e_m \neq 0$ for all $1 \le m \le h-1$ and the strict inequality $\tilde e_m^2 > \tilde e_{m-1}\,\tilde e_{m+1}$ holds throughout.*

Proof. (1) By Theorem 3.1 and Rolle's theorem, for every $1 \le m \le h-1$ the derivative $Q_{m-1}(X) := \frac{d^{m-1}}{dX^{m-1}} P_K^{(s)}(X)$ has $h-m+1$ real roots $\xi_1, \dots, \xi_{h-m+1} \in \mathbb{R}$. For any $X \in \mathbb{R}$ with $Q_{m-1}(X) \neq 0$, differentiating the logarithmic derivative $\frac{Q'_{m-1}(X)}{Q_{m-1}(X)} = \sum_{j=1}^{h-m+1} \frac{1}{X-\xi_j}$ gives

$$\frac{\mathcal{T}_m(X)}{Q_{m-1}(X)^2} = -\frac{d}{dX}\left(\frac{Q'_{m-1}(X)}{Q_{m-1}(X)}\right) = \sum_{j=1}^{h-m+1} \frac{1}{(X-\xi_j)^2} > 0,$$

and by continuity $\mathcal{T}_m(X) \ge 0$ for all $X \in \mathbb{R}$. At $X = \zeta(s)$, since $h \ge 2$ and $\beta_j = (\zeta(s) - r_j)^{-1}$ takes both positive and negative values by Theorem 3.3(3), the β_j are not all equal; Cauchy–Schwarz gives $\left(\sum_j \beta_j\right)^2 < h \sum_j \beta_j^2$, i.e., $\frac{P_K^{(s)\prime\prime}(\zeta(s))}{P_K^{(s)}(\zeta(s))} < \left(1 - \frac{1}{h}\right)\left(\frac{P_K^{(s)\prime}(\zeta(s))}{P_K^{(s)}(\zeta(s))}\right)^2$.

(2) Write $P_K^{(s)}(X) = a_{K,h} \prod_{j=1}^{h} (X - r_j)$ with all roots $r_j \in \mathbb{R}$. Let $k_0 \ge 0$ denote the multiplicity of $X = 0$ as a root (so $a_{K,0} = \cdots = a_{K,k_0-1} = 0$ and $a_{K,k_0} \neq 0$ if $k_0 \le h$, with the convention $k_0 = 0$ if 0 is not a root). For each $1 \le m \le h-1$:

- If $\tilde{a}_{K,m-1}\tilde{a}_{K,m+1} < 0$: since $\tilde{a}_{K,m}^2 \ge 0 > \tilde{a}_{K,m-1}\tilde{a}_{K,m+1}$, strict inequality holds.
- If $\tilde{a}_{K,m-1}\tilde{a}_{K,m+1} > 0$: by Newton–Maclaurin's strict inequality (whose equality case requires all roots r_j to be equal, which is impossible here since β_j takes both signs by Theorem 3.3(3)), strict inequality holds. (Observe also that this case forces $\tilde{a}_{K,m} \neq 0$, since $\tilde{a}_{K,m} = 0$ would give $0 >$ positive, contradicting Newton's inequality.)
- If $\tilde{a}_{K,m-1}\tilde{a}_{K,m+1} = 0$: the right-hand side vanishes, so $\tilde{a}_{K,m}^2 \ge 0$ holds with equality iff $\tilde{a}_{K,m} = 0$. Hence strict inequality holds iff $\tilde{a}_{K,m} \neq 0$.

Combining, strict inequality holds iff $\tilde{a}_{K,m-1}\tilde{a}_{K,m+1} \neq 0$ or $\tilde{a}_{K,m} \neq 0$, matching the statement.

(3) The shifted polynomial $\Phi(Y) := \frac{P_K^{(s)}(\zeta(s)+Y)}{P_K^{(s)}(\zeta(s))} = \prod_{j=1}^{h}(1+\beta_j Y) = \sum_{m=0}^{h} \binom{h}{m} \tilde{e}_m Y^m$ has h non-zero real roots $y_j = -\beta_j^{-1} \in \mathbb{R} \setminus \{0\}$, which are not all equal since β_j takes both signs by Theorem 3.3(3). Applying the same Newton–Maclaurin argument as in (2), strict inequality $\tilde{e}_m^2 > \tilde{e}_{m-1}\tilde{e}_{m+1}$ holds iff $\tilde{e}_{m-1}\tilde{e}_{m+1} \neq 0$ or $\tilde{e}_m \neq 0$. Since all roots $y_j \neq 0$ and are not all equal, the case $\tilde{e}_{m-1} = \tilde{e}_{m+1} = 0$ requires the coefficient sequence of Φ to have an interior block of zeros, which for real y_j requires $\Phi(Y) = g(Y^q)$ for some $q \ge 3$; this is excluded by the Hankel spectral structure (which produces h real, generically distinct eigenvalues β_j with no q-fold symmetry), so $\tilde{e}_m \neq 0$ for $1 \le m \le h-1$ and strict inequality holds. □

Theorem 3.8 (Christoffel–Darboux pole representation and Jacobi derivative identity)**.** *Let $\{\pi_r(t)\}_{r=0}^{h-1}$ be the monic orthogonal polynomials on $[0,\infty)$ with respect to $d\mu_\xi(t) := \frac{D_N(t)^{2m}}{D_K(t)} \frac{w_s(\sqrt{t})}{2\sqrt{t}}\, dt$, and let*

$$\mathcal{K}_h(t,u) := \sum_{r=0}^{h-1} \frac{\pi_r(t)\pi_r(u)}{\int_0^\infty \pi_r(v)^2\, d\mu_\xi(v)} = (1, t, \dots, t^{h-1}) G_K^{(s)}(\xi)^{-1} (1, u, \dots, u^{h-1})^T$$

be their Christoffel–Darboux kernel.

(1) *The non-zero eigenvalues $\beta_1, \dots, \beta_h$ of $B_K^{(s)}$ coincide with the eigenvalues of the $h \times h$ pole-kernel matrix $\left[\gamma_a^{(s)}\, \mathcal{K}_h(-a^2, -b^2)\right]_{N<a,b\le K}$.*

(2) *Under the Laplace scaling $w_s(y;\tau) := \frac{2(2\pi)^{2k}}{(2k)!} y^{2k+1} \sum_{l=1}^{\infty} l^{2k} e^{-2\pi l \tau y}$ $(\tau > 0)$ applied to the weight, with the evaluation point $\xi = \zeta(s)$ held fixed, the deformed determinant*

$\Delta_K(\xi;\tau) := \det G_K^{(s)}(\xi;\tau)$ *satisfies*

$$\frac{\partial}{\partial\tau}\log\Delta_K(\xi;\tau) = \int_0^\infty \mathcal{K}_h(y^2,y^2;\tau)\frac{D_N(y^2)^{2m}}{D_K(y^2)}\frac{\partial w_s(y;\tau)}{\partial\tau}\,dy, \tag{3.8}$$

where the partial derivative $\partial/\partial\tau$ is taken with $X=\xi$ fixed.

Proof. (1) Since $M_K = V\,\mathrm{diag}(\gamma_a^{(s)})V^T$ where the a-th column of V is $v^{(a)} = (1,-a^2,\dots,(-a^2)^{h-1})^T$, the matrix $B_K^{(s)} = G^{-1/2}V\,\mathrm{diag}(\gamma_a^{(s)})V^TG^{-1/2}$ has the same spectrum as $\mathrm{diag}(\gamma_a^{(s)})V^TG^{-1}V$. The (a,b) entry of $V^TG^{-1}V$ is $(v^{(a)})^TG^{-1}v^{(b)} = \mathcal{K}_h(-a^2,-b^2)$.

(2) By Jacobi's determinant derivative formula applied at $X=\xi$ fixed, $\frac{\partial}{\partial\tau}\log\det G_K(\xi;\tau) = \mathrm{Tr}\left(G_K(\xi;\tau)^{-1}\frac{\partial}{\partial\tau}G_K(\xi;\tau)\right)$. Since ξ is fixed, the partial derivative of each entry is $\frac{\partial(G_K)_{i,j}}{\partial\tau} = \int_0^\infty y^{2(i+j)}\frac{D_N(y^2)^{2m}}{D_K(y^2)}\frac{\partial w_s(y;\tau)}{\partial\tau}\,dy$ (the $\mu_X^{(s)}$-evaluated contribution carries no explicit τ-dependence once ξ is fixed). Summing $\sum_{i,j=0}^{h-1}(G_K^{-1})_{j,i}y^{2i}y^{2j} = \mathcal{K}_h(y^2,y^2;\tau)$ yields (3.8). □

Theorem 3.9 (Hyperbolicity for Hurwitz and Dirichlet Hankel determinants)**.** (1) *Let $\theta\in(0,1]\cap\mathbb{Q}$ and $s=2k+1\geq 3$. Set $D_{m,\theta}(t) := \prod_{j=0}^{m-1}(t+(j+\theta)^2)$, and let $\mu_{X,\theta}^{(s)}$ be the rational functional on simple poles at $t=-(j+\theta)^2$ defined via Hermite's integral formula at $X=\zeta(s,\theta)$:*

$$\int_0^\infty \frac{w_s(y)}{y^2+(j+\theta)^2}\,dy = (j+\theta)^{2k}\left(\zeta(s,\theta)-\sum_{\nu=0}^{j-1}\frac{1}{(\nu+\theta)^s}\right)+r_{j,\theta}\qquad (r_{j,\theta}\in\mathbb{Q}).$$

Then $\Delta_{K,\theta}^{(s)}(X) := \det\left[\mu_{X,\theta}^{(s)}(D_{N,\theta}(t)^{2m}t^{i+j}/D_{K,\theta}(t))\right]_{0\leq i,j<h}$ has exact degree h and all its roots are real.

(2) *Let $\chi \pmod q$ be a primitive real Dirichlet character of conductor $q\in\{3,4,5,8\}$, and choose $a_0\in\{1,\dots,q-1\}$ with $\chi(a_0)=+1$. Set $j_r := qr+a_0$ ($r\geq 0$) and $D_{m,\chi}(t) := \prod_{r=0}^{m-1}(t+j_r^2)$. Then the Dirichlet generating kernel $g_\chi(y) := \sum_{l=1}^\infty \chi(l)e^{-2\pi ly/q}$ is strictly positive for all $y>0$. For any integer $s\geq 2$, let $\Omega_{s,\chi}(y) := y^{s-1}g_\chi(y) > 0$, and let $\mu_{X,\chi}^{(s)}$ be the linear functional defined on t^e by $\int_0^\infty y^{2e}\Omega_{s,\chi}(y)\,dy$ and on simple poles $(t+j_r^2)^{-1}$ by $j_r^{s-1}(X-\xi_\chi)+\int_0^\infty \frac{\Omega_{s,\chi}(y)}{y^2+j_r^2}\,dy$ (where $\xi_\chi := L(s,\chi)$ when $\chi(-1)=(-1)^{s-1}$). Then $\Delta_{K,\chi}^{(s)}(X) := \det\left[\mu_{X,\chi}^{(s)}(D_{N,\chi}(t)^{2m}t^{i+j}/D_{K,\chi}(t))\right]_{0\leq i,j<h}$ has exact degree h and all its roots are real.*

Proof. (1) Since $w_s(y)>0$ and $D_{N,\theta}(y^2)^{2m}/D_{K,\theta}(y^2)>0$ on $(0,\infty)$, the matrix $G_{K,\theta}^{(s)}(\zeta(s,\theta))$ is real symmetric positive definite. The X-derivative matrix is $M_{K,\theta}^{(s)} = V_\theta\,\mathrm{diag}(\gamma_{j,\theta})V_\theta^T$, where $V_\theta = ((-(j+\theta)^2)^i)_{0\leq i<h,N\leq j<K}$ is a Vandermonde matrix at the h distinct negative nodes $-(j+\theta)^2<0$ and $\gamma_{j,\theta} = (j+\theta)^{2k}D_{N,\theta}(-(j+\theta)^2)^{2m-1}/D'_{\mathrm{tail},\theta}(-(j+\theta)^2)\neq 0$. Thus $M_{K,\theta}^{(s)}$ is real symmetric and invertible, and the proof of Theorem 3.1 applies verbatim.

(2) Put $z := e^{-2\pi y/q}\in(0,1)$ for $y>0$. Grouping $g_\chi(y) = \sum_{l=1}^\infty\chi(l)z^l$ into blocks of length q gives $g_\chi(y) = \frac{F_\chi(z)}{1-z^q}$, where $F_\chi(z) := \sum_{a=1}^{q-1}\chi(a)z^a$. For $q\in\{3,4,5,8\}$, $F_\chi(z)$ factors explicitly as

$$F_{\chi_{-3}}(z) = z(1-z),\qquad F_{\chi_{-4}}(z) = z(1-z^2),$$
$$F_{\chi_5}(z) = z(1-z)^2(1+z),\qquad F_{\chi_8}(z) = z(1-z^2)(1-z^4),$$
$$F_{\chi_{-8}}(z) = z(1+z^2)(1-z^4).$$

For every $z \in (0,1)$, each factor is strictly positive, so $g_\chi(y) > 0$ and $\Omega_{s,\chi}(y) > 0$ for all $y > 0$. Consequently, at $X = \xi_\chi$, the matrix $G^{(s)}_{K,\chi}(\xi_\chi)$ is real symmetric positive definite. Moreover, because the poles of $D_{\mathrm{tail},\chi}(t) = \prod_{r=N}^{K-1}(t + j_r^2)$ are simple and located at the h distinct negative squares $v_r = -j_r^2$, the X-coefficient matrix is $M^{(s)}_{K,\chi} = V_\chi \operatorname{diag}(\gamma_{r,\chi}) V_\chi^T$ with $\det V_\chi = \prod_{N \le r < r' < K}(j_r^2 - j_{r'}^2) \neq 0$ and $\gamma_{r,\chi} = c_{r,\chi} D_{N,\chi}(-j_r^2)^{2m-1} / D'_{\mathrm{tail},\chi}(-j_r^2) \neq 0$. Thus $\det M^{(s)}_{K,\chi} \neq 0$, and Theorem 3.1 yields real-rootedness and exact degree h. □

Remark 3.10 (Non-classicality of the Hankel family). The primitive Hankel polynomials $\{P_K^{(s)}(X)\}_{K>N}$ are *not* a specialization of any classical orthogonal polynomial family (Legendre, Jacobi, Hahn, Laguerre, Hermite, Bessel, or Apéry). Two obstructions make this precise.

First, in any family of orthogonal polynomials $\{Q_r\}_{r=0}^h$ satisfying a Favard three-term recurrence $Q_{r+1}(X) = (X - \alpha_r)Q_r(X) - \beta_r Q_{r-1}(X)$, the roots of $Q_r(X)$ are new (born at the outer spectral edges) as r increases. By contrast, by Theorem 3.4, the roots $r_j^{(r)} = \zeta(s) - (\beta_j^{(r)})^{-1}$ of the principal-minor polynomials $\Delta^{(s)}_{K,r}(X)$ satisfy $\min_j |r_j^{(r)} - \zeta(s)| = \big(\max_j |\beta_j^{(r)}|\big)^{-1}$, which is *non-increasing* in r: new roots are born *inside* the spectral gap around $X = \zeta(s)$ (pinching $\zeta(s)$ from both sides), not at the outer edges. This violates the Favard root-fanning-out property.

Second, under the Möbius substitution $Y := -1/(X - \zeta(s))$ mapping $X = \zeta(s)$ to $Y = \infty$ and $X = \infty$ to $Y = 0$, the reciprocal polynomials $Q_r(Y) := Y^r \Delta^{(s)}_{K,r}(\zeta(s) - Y^{-1})$ are the characteristic polynomials of the principal submatrices $\widehat{B}_r$ of $\widehat{B}_h$. These form an orthogonal polynomial family with respect to a positive measure if and only if $\widehat{B}_h$ is tridiagonal. But $\widehat{B}_h = L^{-1} M_K^{(s)} (L^{-1})^T$ is the Cholesky conjugation of the moment matrix $M_K^{(s)}$ (supported on the negative axis $t = -j^2$) by the factor L of the positive-definite Gram matrix $G_K^{(s)}(\zeta(s))$ (supported on the positive axis $t = y^2 \ge 0$); the two Hankel moment sequences have disjoint support, so $\widehat{B}_h$ is not tridiagonal for $h \ge 3$ (the entries $(\widehat{B}_h)_{i,j}$ with $|i - j| \ge 2$ do not vanish).

4. Diophantine Approximation, Heights, and Root Collapse for $\zeta(5)$

We now turn to the quantitative arithmetic consequences of our spectral framework at weight five. We begin with the root-geometric transfer criteria over $\mathbb{R}$, over algebraic extensions of degree $d \ge 1$, and over $\mathbb{Q}_p$.

Theorem 4.1 (Spectral irrationality measure criterion). *Let $\xi \in \mathbb{R}$, let $d_0 \in \mathbb{Z}_{>0}$ and $\lambda, \epsilon, \rho > 0$, and suppose $\{Q_K(X)\}_{K \in d_0\mathbb{Z}_{>0}} \subset \mathbb{Z}[X]$ is a sequence of integer polynomials of degree $d_K \le \lambda K$ whose complex roots $r_{K,1}, \dots, r_{K,d_K} \in \mathbb{C}$ satisfy, for all sufficiently large K:*

(1) $0 < |Q_K(\xi)| \le e^{-\epsilon K^2}$, *and*
(2) $|r_{K,j} - \xi| \ge e^{-\rho K}$ *for all* $1 \le j \le d_K$.

Then for every $\mu > \mu^* := \frac{\lambda\rho}{\epsilon}$, there exists an effectively computable integer $b_0(\mu)$ such that

$$\left|\xi - \frac{a}{b}\right| > b^{-\mu} \tag{4.1}$$

for all $a \in \mathbb{Z}$ and all integers $b \ge b_0(\mu)$.

Proof. Fix $\mu > \frac{\lambda\rho}{\epsilon}$ and choose a real constant c satisfying $\frac{\lambda}{\epsilon} < c < \frac{\mu}{\rho}$, so that $\frac{\lambda}{c} - \epsilon < 0$ and $\rho c - \mu < 0$. For a positive integer b, define $K := d_0 \lfloor \frac{c \log b}{d_0} \rfloor$, so that $K \le c \log b < K + d_0$.

Suppose that $a \in \mathbb{Z}$ satisfies $|a/b - \xi| \le b^{-\mu}$, and set $\delta := a/b - \xi$. By hypothesis (2) and $K \le c \log b$, for every $1 \le j \le d_K$ we have

$$\frac{|\delta|}{|\xi - r_{K,j}|} \le b^{-\mu} e^{\rho K} \le b^{-\mu+\rho c} \le \exp\left(-\frac{(\mu - \rho c)K}{c}\right) = o(K^{-1}) \qquad (K \to \infty).$$

Factoring $Q_K(X) = a_{K,d_K} \prod_{j=1}^{d_K} (X - r_{K,j})$, we obtain

$$\left|\frac{Q_K(a/b)}{Q_K(\xi)}\right| = \prod_{j=1}^{d_K} \left|1 + \frac{\delta}{\xi - r_{K,j}}\right| \le \big(1 + o(K^{-1})\big)^{\lambda K} \le 2$$

and similarly $|Q_K(a/b)/Q_K(\xi)| \ge (1 - o(K^{-1}))^{\lambda K} \ge 1/2 > 0$ for all sufficiently large b. In particular, $Q_K(a/b) \ne 0$. Since $Q_K(X) \in \mathbb{Z}[X]$ has degree $d_K \le \lambda K$, the number $b^{d_K}|Q_K(a/b)|$ is a positive integer, hence at least 1. On the other hand, using hypothesis (1) and $\log b < (K + d_0)/c$, we have

$$1 \le b^{d_K}|Q_K(a/b)| \le 2b^{\lambda K} e^{-\epsilon K^2} < 2\exp\left(\left(\frac{\lambda}{c} - \epsilon\right) K^2 + \frac{\lambda d_0}{c} K\right).$$

Since $\frac{\lambda}{c} - \epsilon < 0$, the right-hand side is strictly less than 1 for all sufficiently large K, a contradiction. □

Proposition 4.2 (Transfer criterion for algebraic approximations of degree $d \ge 1$)**.** *Under the hypotheses of Theorem 4.1, suppose further that* $\limsup_{K\to\infty} K^{-2} \log H(Q_K) \le \Lambda_{\text{height}}$ *and that* $\epsilon > (d-1)\Lambda_{\text{height}}$ *for a positive integer* $d \ge 1$*. Then for every* $\mu_d > \mu_d^* := \frac{\lambda\rho d}{\epsilon - (d-1)\Lambda_{\text{height}}}$*, there exists* $B_0(\mu_d)$ *such that* $|\xi - \alpha| > H(\alpha)^{-\mu_d}$ *for all algebraic numbers* $\alpha \in \overline{\mathbb{Q}}$ *of degree* $[\mathbb{Q}(\alpha) : \mathbb{Q}] \le d$ *and naive height* $H(\alpha) \ge B_0(\mu_d)$*.*

Proof. Since $\mu_d > \mu_d^*$, the interval $I_c := \big(\frac{\lambda d}{\epsilon - (d-1)\Lambda_{\text{height}}}, \frac{\mu_d}{\rho}\big)$ is non-empty; set $c := \frac{1}{2}\big(\frac{\lambda d}{\epsilon - (d-1)\Lambda_{\text{height}}} + \frac{\mu_d}{\rho}\big) \in I_c$. Let $\alpha = \alpha_1$ have degree $m \le d$, naive height $B = H(\alpha)$, and primitive minimal polynomial $f_\alpha(X) = b_m \prod_{s=1}^{m}(X - \alpha_s) \in \mathbb{Z}[X]$. Choosing $K = d_0 \lfloor \frac{c \log B}{d_0} \rfloor$, if $|\alpha_1 - \xi| \le B^{-\mu_d}$, then $\rho c - \mu_d < 0$ implies $|\alpha_1 - \xi| e^{\rho K} = o(K^{-1})$, so $0 < |Q_K(\alpha_1)| \le 2e^{-\epsilon K^2}$ as in Theorem 4.1. Since f_α is irreducible and $Q_K(\alpha_1) \ne 0$, we have $\gcd(Q_K, f_\alpha) = 1$ in $\mathbb{Q}[X]$, so $|\operatorname{Res}(Q_K, f_\alpha)| = b_m^{d_K} \prod_{s=1}^{m} |Q_K(\alpha_s)| \ge 1$. For $2 \le s \le m$, $|Q_K(\alpha_s)| \le (d_K + 1) H(Q_K) \max(1, |\alpha_s|)^{d_K}$, and by Mahler's inequality $b_m \prod_{s=1}^{m} \max(1, |\alpha_s|) \le \sqrt{m+1}\, B$. Multiplying over $1 \le s \le m$ gives

$$1 \le |\operatorname{Res}(Q_K, f_\alpha)| \le 2\exp\left(\left((m-1)\Lambda_{\text{height}} - \epsilon + \frac{\lambda m}{c}\right) K^2 + o(K^2)\right) < 1$$

for large B, since $(m-1)\Lambda_{\text{height}} - \epsilon + \frac{\lambda m}{c} \le (d-1)\Lambda_{\text{height}} - \epsilon + \frac{\lambda d}{c} < 0$. □

Proposition 4.3 (Ultrametric zero-free disk over $\mathbb{Q}_p$)**.** *Let* p *be a prime, let* $\xi_p \in \mathbb{Z}_p$*, and let* $Q(X) \in \mathbb{Z}[X]$ *be any integer polynomial with* $Q(\xi_p) \ne 0$*. Then for every* $\alpha \in \mathbb{C}_p$ *satisfying* $|\alpha - \xi_p|_p < |Q(\xi_p)|_p$*, we have* $|Q(\alpha)|_p = |Q(\xi_p)|_p \ne 0$*. In particular,* $Q(X)$ *has no roots in* $\mathbb{C}_p$ *inside the open disk* $D_p(\xi_p, |Q(\xi_p)|_p)$*.*

Proof. Expanding $Q(\xi_p + W) = \sum_{m=0}^{\deg Q} c_m W^m$ with $c_m = \frac{1}{m!} Q^{(m)}(\xi_p) = \sum_{r \ge m} \binom{r}{m} a_r \xi_p^{r-m} \in \mathbb{Z}_p$, we have $|c_m|_p \le 1$ for all $m \ge 1$. For $|\alpha - \xi_p|_p < |Q(\xi_p)|_p \le 1$, the ultrametric inequality yields $|Q(\alpha) - Q(\xi_p)|_p = |\sum_{m \ge 1} c_m (\alpha - \xi_p)^m|_p \le |\alpha - \xi_p|_p < |Q(\xi_p)|_p$, so $|Q(\alpha)|_p = |Q(\xi_p)|_p \ne 0$. □

We now specialize to $\zeta(5)$ with $K = 30n, N = 3n, h = 27n$ ($\alpha = 1/10, \lambda = 9/10$) and $2m = 6$, and define

$$S_K^{(5)} := \frac{(K!)^{2h} \cdot 4^{h-1}}{(N!)^{12h} \prod_{i=1}^{h-1}((2i)!)^2} > 0, \qquad F_K^{(5)}(X) := S_K^{(5)} \Delta_K^{(5)}(X). \tag{4.2}$$

By the prime-sum and logarithmic-energy estimates of Fauzan [5] (with cutoff $M = 100000$), there exist positive rational multipliers $m_{K,M}^{(5)} = \prod_{p \le 2h} p^{-L_p^{(5)}(K,M)}$ such that $Q_K^{(5)}(X) := m_{K,M}^{(5)} F_K^{(5)}(X) \in \mathbb{Z}[X]$ has degree $h = \frac{9}{10}K$ and satisfies, for all sufficiently large $K \in 30\mathbb{Z}_{>0}$,

$$0 < Q_K^{(5)}(\zeta(5)) \le \exp(-\epsilon K^2), \qquad \epsilon := \frac{1}{2500} = 0.0004, \tag{4.3}$$

where $\lim_{K\to\infty} K^{-2} \log m_{K,M}^{(5)} = A_M^{(5)} \le -2.32425$ and $\limsup_{K\to\infty} K^{-2} \log F_K^{(5)}(\zeta(5)) \le \overline{U}^{(5)} = 2.32380$.

Theorem 4.4 (Sharpened spectral bound and effective irrationality measure for $\zeta(5)$)**.** *For all sufficiently large $K \in 30\mathbb{Z}_{>0}$, the relative matrix $B_K^{(5)} = G_K^{(5)}(\zeta(5))^{-1/2} M_K^{(5)} G_K^{(5)}(\zeta(5))^{-1/2}$ satisfies*

$$\|B_K^{(5)}\| \le e^{\rho_5^\star K}, \qquad \rho_5^\star := \frac{11411}{1000} = 11.411. \tag{4.4}$$

Consequently, there exists an effectively computable positive integer b_0 such that for every $a \in \mathbb{Z}$ and every integer $b \ge b_0$,

$$\left|\zeta(5) - \frac{a}{b}\right| > b^{-25687}. \tag{4.5}$$

Proof. Let $q(t) = \sum_{i=0}^{h-1} c_i t^i \in \mathbb{R}[t]$ be a non-zero polynomial, and put $p(t) := q(K^2 t)$. Fix $0 < a < b \le 1$. Keeping only the $l = 1$ term in $w_5(y) = \frac{(2\pi)^4}{12} y^5 \sum_{l\ge1} l^4 e^{-2\pi l y}$ and restricting the integral (3.2) to $y = K\sqrt{t}$ with $t \in [a, b]$ ($y^5 dy = \frac{1}{2}K^6 t^2 dt \ge \frac{1}{2}K^6 a^2 dt$) gives

$$c^T G_K^{(5)}(\zeta(5)) c \ge \mathcal{A}_K c_a e^{-\mathcal{F}_5(a,b)K} \int_a^b p(t)^2 \, dt, \tag{4.6}$$

where $\mathcal{A}_K := K^{12N-2K+6}$, $c_a := \frac{(2\pi)^4 a^2}{24}$, and $\mathcal{F}_5(a, b) := 2\pi\sqrt{b} + \log(1 + b) - 6\alpha \log a$, since $D_N(K^2 t)^6 \ge (K^2 a)^{6N}$ and $D_K(K^2 t) \le K^{2K}(1 + b)^K$ on $[a, b]$.

Next, setting $u_j := j/K \in (\alpha, 1]$ for $N < j \le K$, Theorem 2.5(2) gives

$$c^T M_K^{(5)} c = \sum_{j=N+1}^{K} \gamma_j^{(5)} q(-j^2)^2 = \mathcal{A}_K \sum_{j=N+1}^{K} r_j p(-u_j^2)^2, \qquad r_j := \frac{u_j^4 \prod_{l=1}^{N}(u_l^2 - u_j^2)^5}{\prod_{\substack{N<k\le K \\ k\ne j}}(u_k^2 - u_j^2)}.$$

To obtain a quick baseline bound, since $|u_l^2 - u_j^2| < 1$ and $u_k + u_j \ge 2\alpha$, one has

$$|r_j| \le \frac{(K/(2\alpha))^{h-1}}{(j - N - 1)!(K - j)!} = \frac{(K/(2\alpha))^{h-1}}{(h-1)!} \binom{h-1}{j-N-1} \le \frac{(K/\alpha)^{h-1}}{(h-1)!} \le \exp\big(\lambda K(1 - \log(\alpha\lambda))\big),$$

where $\binom{h-1}{j-N-1} \le 2^{h-1}$ cancels $2^{-(h-1)}$; at $(a, b) = (1/18, 1/3)$, this decoupled bound already gives $\|B_K^{(5)}\| \le e^{13.84K}$ and $|\zeta(5) - a/b| > b^{-31155}$.

To sharpen the exponent to (4.4)–(4.5), we retain the exact factorial profile of r_j coupled with the pointwise Legendre growth rate at each node $t = -u_j^2$. Substituting $\prod_{l=1}^{N}(j^2 - l^2) = \frac{(j+N)!}{j\,(j-N-1)!}$ and $\prod_{N<k\le K,\, k\ne j} |k^2 - j^2| = \frac{(j-N-1)!(K-j)!(2j-1)!(K+j)!}{(j+N)!(2j)!}$ into $r_j = \frac{u_j^4 \prod_{l=1}^{N}(u_l^2 - u_j^2)^5}{\prod_{N<k\le K,\, k\ne j}(u_k^2 - u_j^2)}$

(with $u_m = m/K$), and tracking the sign $(-1)^{N+(j-1-N)} = (-1)^{j-1}$ from the N factors $u_l^2 - u_j^2 < 0$ and the $j-1-N$ factors $u_k^2 - u_j^2 < 0$ at $N < k < j$, we obtain the exact identity

$$r_j = (-1)^{j-1} \cdot 2K^{2h-6-10N} \frac{((j+N)!)^6}{((j-N-1)!)^6 (K-j)!(K+j)!}. \tag{4.7}$$

Indeed, the degree $2h-6-10N$ of K arises as $2(h-1)-4-10N$ (from the scaling $u_k^2 - u_j^2 \to (k^2-j^2)/K^2$ over $k \neq j$, and $u_j^4 \to j^4/K^4$), while the factor $2j$ from $(2j)!/(2j-1)!$ cancels the $j^4/(j \cdot j^4)$ from the numerator and $(j-1-N)! = (j-N-1)!$; the spurious $j^6/((j-N)^6)$ prefactor sometimes adduced for this formula is incorrect (it would grow with j polynomially while the true $|r_j|$ has no such dependence).

Applying Stirling's inequality $m \log m - m \le \log(m!) \le m \log m - m + \log(m+1) + 1$ at $u = u_j = j/K \in [\alpha, 1]$, so that $j = uK$, $j+N = K(u+\alpha)$, $j-N-1 = K(u-\alpha)-1$, $K-j = K(1-u)$, $K+j = K(1+u)$, we compute the coefficient of $K \log K$ in $\log|r_j|$:

$$6(u+\alpha) - 6(u-\alpha) - (1-u) - (1+u) + \frac{2h-6-10N}{K} = 12\alpha - 2 + (2\lambda - 10\alpha) = 2(\alpha+\lambda) - 2 = 0,$$

since $\alpha + \lambda = N/K + h/K = 1$ (equivalently, $12\alpha - 2 = -(2h-10N)/K$; note the correct identity involves $2h$, not h). The linear $-m$ terms from Stirling contribute $-(12\alpha - 2) = 2 - 12\alpha$, giving

$$\log|r_j| = K\Phi(u_j) + O(\log K), \tag{4.8}$$

where $\Phi(u) := 6(u+\alpha)\log(u+\alpha) - 6(u-\alpha)\log(u-\alpha) - (1-u)\log(1-u) - (1+u)\log(1+u) + 2 - 12\alpha$ is the concave potential. (The $K \log K$ terms cancel *exactly*, not merely up to $O(\log K)$; only the constant $\log 2$ and the $\log K$ prefactor of Stirling are absorbed into the $O(\log K)$ error.)

Expanding $p(t) = \sum_{m=0}^{h-1} d_m \sqrt{\frac{2m+1}{b-a}} P_m\left(\frac{2t-(a+b)}{b-a}\right)$ in orthonormal Legendre polynomials on $[a,b]$, each negative node $t = -u^2 \in [-1, -\alpha^2]$ maps to $s(-u^2) = -\frac{2u^2+a+b}{b-a}$ with $|s(-u^2)| = Y(u) := \frac{2u^2+a+b}{b-a} > 1$. By Laplace's bound $|P_m(s)| \le \mathcal{R}(u)^m$ with $\mathcal{R}(u) := Y(u) + \sqrt{Y(u)^2 - 1}$ [11, 18.10.5] and the Cauchy–Schwarz inequality,

$$|p(-u^2)|^2 \le \left(\sum_{m=0}^{h-1} d_m^2\right) \sum_{m=0}^{h-1} \frac{2m+1}{b-a} \mathcal{R}(u)^{2m} \le \frac{h^2}{b-a} e^{2\lambda K \log \mathcal{R}(u)} \int_a^b p(t)^2\, dt. \tag{4.9}$$

Combining (4.6), (4.8), and (4.9) yields

$$\|B_K^{(5)}\| \le \frac{C_1 K^8 h^3}{c_a (b-a)} \exp\left(K\left(\mathcal{F}_5(a,b) + \max_{u \in [\alpha,1]} \Psi_{a,b}(u)\right)\right), \qquad \Psi_{a,b}(u) := \Phi(u) + 2\lambda \log \mathcal{R}(u).$$

Set $a = \frac{1}{18}$ and $b = \frac{1}{3}$ (so $b-a = \frac{5}{18}, a+b = \frac{7}{18}$, and $Y(u) = \frac{36u^2+7}{5}$). Since $\log \mathcal{R}(u) = \operatorname{arcosh}(Y(u))$ and $Y(u)^2 - 1 = \frac{4(u^2+a)(u^2+b)}{(b-a)^2}$, the first two derivatives of $\Psi_{a,b}(u)$ on $(0.1, 1)$ are

$$\Psi'_{a,b}(u) = 6 \log \frac{u+0.1}{u-0.1} + \log \frac{1-u}{1+u} + \frac{3.6u}{\sqrt{(u^2+1/18)(u^2+1/3)}},$$

$$\Psi''_{a,b}(u) = -\frac{1.2}{u^2 - 0.01} - \frac{2}{1-u^2} + \frac{3.6(1/54 - u^4)}{(u^2+1/18)^{3/2}(u^2+1/3)^{3/2}}.$$

For $u \ge (1/54)^{1/4} \approx 0.3689$, all three terms of $\Psi''_{a,b}(u)$ are negative; for $u \in (0.1, 0.3689)$, the third term is bounded above by its value at $u = 0.1$, which is 5.318, whereas $-\frac{1.2}{u^2-0.01} - \frac{2}{1-u^2} < -9.51 - 2.31 = -11.82 < -5.318$. Thus $\Psi''_{a,b}(u) < 0$ strictly on all of $(0.1, 1)$.

Evaluating $\Psi'_{a,b}(u)$ at $u_- = 0.973$ and $u_+ = 0.974$ gives $\Psi'_{a,b}(0.973) = +0.0259 > 0$ and $\Psi'_{a,b}(0.974) = -0.0125 < 0$, so the unique global maximum $u_\star$ of $\Psi_{a,b}(u)$ lies in $[0.973, 0.974]$. Since $\Phi'(u) < 0$ and $(\log \mathcal{R})'(u) > 0$ on $[0.973, 0.974]$, we have

$$\max_{u\in[0.1,1]} \Psi_{1/18,1/3}(u) = \Psi_{1/18,1/3}(u_\star) \le \Phi(0.973) + 1.8 \log \mathcal{R}(0.974).$$

Direct evaluation gives $\Phi(0.973) = 0.725325... < 0.72533$, $Y(0.974) = \frac{36(0.974)^2+7}{5} = 8.2297472$, $\mathcal{R}(0.974) = 16.398501...$, $1.8 \log \mathcal{R}(0.974) = 5.035267... < 5.03527$, and $\mathcal{F}_5(1/18, 1/3) = \frac{2\pi}{\sqrt{3}} + \log \frac{4}{3} + \frac{3}{5} \log 18 = 5.6495038... < 5.64951$. Summing these three upper bounds gives

$$\mathcal{F}_5\left(\frac{1}{18}, \frac{1}{3}\right) + \max_{u\in[0.1,1]} \Psi_{1/18,1/3}(u) < 5.64951 + 0.72533 + 5.03527 = 11.41011 < 11.411 = \rho_5^\star.$$

The margin $11.411 - 11.41011 = 0.00089 > 0$ absorbs the polynomial prefactor for all large K, proving (4.4). Applying Theorem 4.1 with $d_0 = 30, \lambda = 9/10, \epsilon = 1/2500, \rho = 11.411, c = 2251$, and $\mu = 25687$ (since $\rho c = 11.411 \times 2251 = 25686.161 < 25687$) proves (4.5). □

By coupling the weight-five functional $\mu_X^{(5)}(R(t))$ with the shifted weight-three functional $-\mu_Y^{(3)}(tR(t))$ over the same denominator $D_K(t)$, we obtain an unconditional irrationality theorem and effective measure for linear combinations of $\zeta(5)$ and $\zeta(3)$.

Theorem 4.5 (Irrationality and effective measure for $b\zeta(5) + a\zeta(3)$)**.** *Let $a, b \in \mathbb{Z}$ be integers satisfying $b > 0$ and $a < \frac{\pi^2}{3} b \approx 3.2898b$. Then the real number $\xi_{a,b} := b\zeta(5) + a\zeta(3)$ is irrational, and there exists an effectively computable integer $q_0(a, b)$ such that for all $p \in \mathbb{Z}$ and all integers $q \ge q_0(a, b)$,*

$$\left| b\zeta(5) + a\zeta(3) - \frac{p}{q} \right| > q^{-25687}. \tag{4.10}$$

Proof. For $K = 30n, N = 3n, h = 27n$, consider the rational functions $R_{i,j}(t) := \frac{D_N(t)^6 t^{i+j}}{D_K(t)}$ $(0 \le i, j < h)$. At each pole $t = -j^2$ $(N < j \le K)$ with residue $c_{i,j;j} := \mathrm{Res}_{t=-j^2} R_{i,j}(t)$, we have $\mathrm{Res}_{t=-j^2}(tR_{i,j}(t)) = -j^2 c_{i,j;j}$. Applying $b\mu_X^{(5)}(R_{i,j}(t)) - a\mu_Y^{(3)}(tR_{i,j}(t))$ via (2.3) gives the pole contribution

$$bc_{i,j;j}\left(j^4(X - H_j^{(5)}) + \frac{1}{2j} - \frac{1}{4}\right) - a(-j^2 c_{i,j;j})\left(j^2(Y - H_j^{(3)}) + \frac{1}{2j} - \frac{1}{2}\right)$$
$$= c_{i,j;j}\left[j^4\big((bX + aY) - bH_j^{(5)} - aH_j^{(3)}\big) + \frac{b}{2j} - \frac{b}{4} + \frac{aj}{2} - \frac{aj^2}{2}\right].$$

Thus every entry of the $h \times h$ matrix $G_K^{[a,b]}(Z) := \left[b\mu_X^{(5)}(R_{i,j}(t)) - a\mu_Y^{(3)}(tR_{i,j}(t))\right]_{0\le i,j<h}$ is an affine linear polynomial in the single indeterminate $Z := bX + aY$, with Z-derivative matrix $[Z]G_K^{[a,b]}(Z) = M_K^{(5)}$ identical to that of $G_K^{(5)}(X)$.

At $X = \zeta(5)$ and $Y = \zeta(3)$ (so $Z = \xi_{a,b} = b\zeta(5) + a\zeta(3)$), (3.2) gives

$$G_K^{[a,b]}(\xi_{a,b})_{i,j} = \int_0^\infty \frac{D_N(y^2)^6 y^{2(i+j)}}{D_K(y^2)} W_{a,b}(y)\, dy,$$

where the combined weight is

$$W_{a,b}(y) := bw_5(y) - ay^2 w_3(y) = (2\pi)^2 y^5 \sum_{l=1}^\infty l^2 \left(\frac{\pi^2}{3} bl^2 - a\right) e^{-2\pi l y}.$$

Since $b>0$ and $a<\frac{\pi^2}{3}b$, for every integer $l\geq 1$ we have $\frac{\pi^2}{3}bl^2-a\geq\big(b-\frac{3\max(0,a)}{\pi^2}\big)\frac{\pi^2}{3}l^2>0$. Setting $c_1(a,b):=b-\frac{3\max(0,a)}{\pi^2}>0$ and $c_2(a,b):=b+\frac{3|a|}{\pi^2}>0$, we have $c_1(a,b)w_5(y)\leq W_{a,b}(y)\leq c_2(a,b)w_5(y)$ for all $y>0$. Consequently, $G_K^{[a,b]}(\xi_{a,b})$ is symmetric positive definite with $c_1(a,b)G_K^{(5)}(\zeta(5))\preceq G_K^{[a,b]}(\xi_{a,b})\preceq c_2(a,b)G_K^{(5)}(\zeta(5))$ in the Loewner order, so $\limsup_{K\to\infty}K^{-2}\log(S_K^{(5)}\det G_K^{[a,b]}(\xi$ $\overline{U}^{(5)}=2.32380$ and $\|B_K^{[a,b]}\|\leq c_1(a,b)^{-1}\|B_K^{(5)}\|\leq e^{11.411K+O(\log K)}$ for large K.

On the p-adic side, by Lemma 2.2, $b\mu_X^{(5)}(R)-a\mu_Y^{(3)}(tR)=b\tau_X^{(5)}(x^5R(-x^2))+a\tau_Y^{(3)}(x^5R(-x^2))$. For every prime $p>\max(|a|,|b|,7)$, the distribution formula at weight 5 has prefactor p^{-4} with $p^{-4}Y_5=pX+p^{-4}C_p^{(5)}$ ($C_p^{(5)}\in p^5\mathbb{Z}_p$), while at weight 3 it has prefactor p^{-2} with $p^{-2}Y_3=pY+p^{-2}C_p^{(3)}$ ($C_p^{(3)}\in p^3\mathbb{Z}_p$). Combining them gives $p(bX+aY)+\text{const}=pZ+\text{const}\in\mathbb{Z}_p[Z]$. In the inner range $K/M<p\leq K/3$, the weight-3 residue valuation has $+2$ relative to weight 5, so the inner half-weights $w_{a,i}^{(5)}$ remain valid. In the outer range $K/3<p\leq K$, the polynomial quotient of $tR_{i,j}(t)$ has degree $e+1\leq 6N+2h-1-K=\frac{7}{5}K-1$, so $2e+4\leq\frac{14}{5}K+2<9(p-1)$. By the von Staudt–Clausen theorem, $\mu^{(3)}(t^{e+1})=(-1)^{e+1}B_{2e+4}(2e+5)/2$ can fail to be p-integral only when $2e+4=k(p-1)$ with $2\leq k\leq 8$, which occurs for at most 7 values of $e=i+j-K+6N$ and contributes a matrix $p^{-1}L^{(3)}$ of rank at most $14=O(1)$, losing at most p^{-14} in the determinant at each outer prime $p\leq K$ (a total factor $\leq\prod_{p\leq K}p^{14}=e^{O(K)}$). For primes $p\leq\max(|a|,|b|,7)$, the standard small-prime distribution bound of [5] loses at most $C_{a,b}^h=e^{O(K)}$. Hence multiplying by $d_K^{14}C_{a,b}^h m_{K,M}^{(5)}S_K^{(5)}$ yields integer polynomials $Q_K^{[a,b]}(Z)\in\mathbb{Z}[Z]$ of degree h satisfying $0<Q_K^{[a,b]}(\xi_{a,b})\leq e^{-0.0004K^2+O(K)}$ and root gap $|r_j-\xi_{a,b}|\geq e^{-11.411K}$. Theorem 4.1 now yields $\xi_{a,b}\notin\mathbb{Q}$ and (4.10). □

We now establish two-sided quadratic bounds on $\log|\det B_K^{(5)}|$, coefficient height bounds, and macroscopic root clustering at $\zeta(5)$.

Theorem 4.6 (Two-sided determinant bounds and coefficient heights)**.** *For all sufficiently large $K\in 30\mathbb{Z}_{>0}$:*

(1) *The determinant of $B_K^{(5)}$ satisfies the unconditional two-sided bounds*

$$0.23310\leq\liminf_{K\to\infty}\frac{1}{K^2}\log|\det B_K^{(5)}|\leq\limsup_{K\to\infty}\frac{1}{K^2}\log|\det B_K^{(5)}|\leq 3.73980. \tag{4.11}$$

(2) *The normalized polynomial $Q_K^{(5)}(X)\in\mathbb{Z}[X]$ and primitive polynomial $P_K^{(5)}(X)\in\mathbb{Z}[X]$ satisfy*

$$0.23\,K^2\leq\log H\big(Q_K^{(5)}\big)\leq 10.27\,K^2,\qquad \log H\big(P_K^{(5)}\big)\leq 10.27\,K^2, \tag{4.12}$$

and the decay rate $\epsilon_P:=\limsup_{K\to\infty}\big(-K^{-2}\log P_K^{(5)}(\zeta(5))\big)$ satisfies $0.0004\leq\epsilon_P\leq 3.73980$. Moreover, writing $Q_K^{(5)}(X)=c_K P_K^{(5)}(X)$ with $c_K=\text{cont}_{\mathbb{Z}}(Q_K^{(5)})\in\mathbb{Z}_{>0}$ and $\delta_K:=K^{-2}\log c_K\geq 0$, we have $\log H(P_K^{(5)})\geq(0.23265-\delta_K-o(1))K^2$ and $\log P_K^{(5)}(\zeta(5))\leq-(\epsilon+\delta_K)K^2$.

Proof. (1) By (3.4) with $k=2$ and $2m-1=5$, we have $|\det M_K^{(5)}|=\prod_{a=N+1}^{K}a^4\left(\frac{(a+N)!}{(a-N-1)!a}\right)^5$. Using $\int(u\pm\alpha)\log(u\pm\alpha)\,du=\frac{1}{2}(u\pm\alpha)^2\log(u\pm\alpha)-\frac{1}{4}(u\pm\alpha)^2$ and the polynomial piece contribution $-\frac{1}{4}\big(-(1+\alpha)^2+(1-\alpha)^2+(2\alpha)^2\big)=-\alpha(1-\alpha)=-\alpha\lambda$, Euler–Maclaurin summation

on $u = a/K \in [\alpha, 1]$ ($\alpha = 1/10, \lambda = 9/10$) gives

$$\lim_{K\to\infty} \frac{1}{K^2}\Big(\log|\det M_K^{(5)}| - 10\alpha\lambda K^2 \log K\Big) = 5\int_\alpha^1 \Big((u+\alpha)\log(u+\alpha) - (u-\alpha)\log(u-\alpha) - 2\alpha\Big)\,du$$
$$= \frac{5}{2}\Big[(1+\alpha)^2\log(1+\alpha) - (1-\alpha)^2\log(1-\alpha) - (2\alpha)^2\log(2\alpha)\Big] - 15\alpha$$
$$= 0.662612... - 1.35 = -0.687388... \tag{4.13}$$

From (4.2) and Stirling's formula, $\log S_K^{(5)} = -10\alpha\lambda K^2\log K + C_*^{(5)}K^2 + o(K^2)$ with $C_*^{(5)} = -2\lambda + 12\alpha\lambda(1-\log\alpha) + 3\lambda^2 - 2\lambda^2\log(2\lambda) = 3.244587....$ Adding the two expressions cancels the $K^2\log K$ term and yields

$$\lim_{K\to\infty}\frac{1}{K^2}\log\big(S_K^{(5)}|\det M_K^{(5)}|\big) = 3.244587... - 0.687388... = +2.557199... \tag{4.14}$$

Since $|\det B_K^{(5)}| = \frac{S_K^{(5)}|\det M_K^{(5)}|}{S_K^{(5)}\det G_K^{(5)}(\zeta(5))}$ and $\limsup_{K\to\infty} K^{-2}\log F_K^{(5)}(\zeta(5)) \le \overline{U}^{(5)} = 2.32380$, subtracting gives the lower bound $\liminf_{K\to\infty} K^{-2}\log|\det B_K^{(5)}| \ge 2.557199 - 2.32380 > 0.23310$.

To prove the upper bound in (4.11) without external data, we establish a self-contained lower bound for $\liminf_{K\to\infty} K^{-2}\log F_K^{(5)}(\zeta(5))$ using Jensen's inequality with respect to the single arcsine probability measure $d\tilde\rho_L(t) := \frac{1}{\pi\sqrt{t(L-t)}}\,dt$ on $(0,L)$ at $L = \frac14$. Restricting the Andréief integral for $\Delta_K^{(5)}(\zeta(5))$ (scaled by $y_i = K\sqrt{t_i}$) to $(0,L)^h$ and applying Jensen's inequality $\int_{(0,L)^h}\exp(\Psi)\prod_{i=1}^h d\tilde\rho_L(t_i) \ge \exp\big(\int_{(0,L)^h}\Psi\prod_{i=1}^h d\tilde\rho_L(t_i)\big)$ gives

$$\liminf_{K\to\infty}\frac{1}{K^2}\log F_K^{(5)}(\zeta(5)) \ge \lambda^2 I(\tilde\rho_L) - \lambda\int_0^L V_5(t)\,d\tilde\rho_L(t) + C_*^{(5)},$$

where $V_5(t) = 2\pi\sqrt{t} + J_1(t) - 6J_{0.1}(t)$ with $J_a(t) = \int_0^a \log(t+u^2)\,du$, and $I(\tilde\rho_L) = \log(L/4) = -\log 16 = -2.7725887....$ By Fubini's theorem and the exact arcsine potential $\int_0^L \log(t+u^2)\,d\tilde\rho_L(t) = 2\,\mathrm{arsinh}(u/\sqrt{L}) + \log(L/4)$ for $u > 0$, integrating in $u \in (0,a)$ yields the closed-form identity

$$\int_0^L J_a(t)\,d\tilde\rho_L(t) = 2\left(a\,\mathrm{arsinh}\left(\frac{a}{\sqrt{L}}\right) - \sqrt{a^2+L} + \sqrt{L}\right) + a\log\frac{L}{4}.$$

At $L = 1/4$ ($\sqrt{L} = 1/2$), we have $\int_0^{1/4} 2\pi\sqrt{t}\,d\tilde\rho_{1/4}(t) = 4\sqrt{L} = 2$, $\int_0^{1/4} J_1(t)\,d\tilde\rho_{1/4}(t) = 2(\mathrm{arsinh}(2) - \sqrt{1.25} + 0.5) - \log 16 = -1.121387...$, and $-6\int_0^{1/4} J_{0.1}(t)\,d\tilde\rho_{1/4}(t) = -6[2(0.1\,\mathrm{arsinh}(0.2) - \sqrt{0.26} + 0.5) - 0.1\log 16] = +1.545150....$ Summing gives $\int_0^{1/4} V_5(t)\,d\tilde\rho_{1/4}(t) = 2.423763...$, whence

$$\liminf_{K\to\infty}\frac{1}{K^2}\log F_K^{(5)}(\zeta(5)) \ge 0.81(-2.772589) - 0.9(2.423763) + 3.244587 = -1.182597...$$

Subtracting this from (4.14) yields $\limsup_{K\to\infty} K^{-2}\log|\det B_K^{(5)}| \le 2.557199 - (-1.182597) = 3.739796 < 3.73980$.

(2) Using $|\beta_j| \le \|B_K^{(5)}\| \le e^{11.411K}$ from Theorem 4.4 in $Q_K^{(5)}(X) = Q_K^{(5)}(\xi)\prod_{j=1}^h(1 + (X-\xi)\beta_j)$ gives

$$H\big(P_K^{(5)}\big) \le H\big(Q_K^{(5)}\big) \le e^{-0.0004K^2}\big(1+3e^{11.411K}\big)^{0.9K} \le \exp\big((0.9\times 11.411 - 0.0004)K^2 + O(K)\big) < e^{10.27K^2}.$$

Since $|[X^h]P_K^{(5)}(X)| = P_K^{(5)}(\zeta(5))|\det B_K^{(5)}| \geq 1$, we have $-\log P_K^{(5)}(\zeta(5)) \leq \log|\det B_K^{(5)}|$, so $\epsilon_P \leq 3.73980$. Finally, $\lim_{K\to\infty} K^{-2}\log|[X^h]Q_K^{(5)}(X)| = A_M^{(5)} + 2.557199 \geq -2.32425 + 2.55690 = +0.23265 > 0.23$. □

Theorem 4.7 (Macroscopic exponential root collapse and root separation)**.** *For all sufficiently large $K \in 30\mathbb{Z}_{>0}$:*

(1) *At least $N_{\mathrm{clust}} \geq 0.0140h > \frac{12}{1000}K$ real roots r_j of $P_K^{(5)}(X)$ satisfy*
$$\left|r_j - \zeta(5)\right| \leq e^{-0.10K}, \tag{4.15}$$
while the closest root satisfies $e^{-11.411K} \leq \min_{1\leq j\leq h}|r_j-\zeta(5)| \leq e^{-K/2250}$, and the arithmetic mean of the roots satisfies $\frac{1}{h}\sum_{j=1}^h r_j - \zeta(5) = -\frac{1}{h}\operatorname{Tr}\left((M_K^{(5)})^{-1}G_K^{(5)}(\zeta(5))\right)$.

(2) *Any two roots $r_- < \zeta(5) < r_+$ of $P_K^{(5)}(X)$ straddling $\zeta(5)$ satisfy $r_+ - r_- \geq 2e^{-11.411K}$, and any two roots $r_{i_a} \neq r_{i_b}$ belonging to the same irreducible factor $f(X) \mid P_K^{(5)}(X)$ in $\mathbb{Z}[X]$ of degree $d = \deg f \geq 2$ satisfy $|r_{i_a} - r_{i_b}| \geq \exp(-10.27dK^2 - O(K\log K))$.*

Proof. (1) Set $s_j := \frac{1}{K}\log|\beta_j| = -\frac{1}{K}\log|r_j-\zeta(5)|$. By Theorem 4.4, $s_j \leq 11.411$ for all j, while by Theorem 4.6(1), $\frac{1}{h}\sum_{j=1}^h s_j = \frac{1}{\lambda K^2}\log|\det B_K^{(5)}| \geq \frac{0.23310}{0.9} - o(1) > 0.2589$. For $S_+ := \{1 \leq j \leq h : s_j \geq 0.10\}$, we have
$$0.2589h \leq \sum_{j\in S_+} s_j + \sum_{j\notin S_+} s_j \leq 11.411|S_+| + 0.10(h - |S_+|) = 11.311|S_+| + 0.10h,$$
giving $|S_+| \geq \frac{0.1589}{11.311}h > 0.0140h$. The bounds on $\min_j|r_j-\zeta(5)|$ and $\sum_j(r_j-\zeta(5))$ follow from $|a_{K,h}|\prod_j|\zeta(5)-r_j| = P_K^{(5)}(\zeta(5)) \leq e^{-0.0004K^2}$ and $r_j - \zeta(5) = -\beta_j^{-1}$.

(2) For $r_- < \zeta(5) < r_+$, Theorem 4.4 gives $r_+ - r_- = (r_+ - \zeta(5)) + (\zeta(5) - r_-) \geq 2\|B_K^{(5)}\|^{-1} \geq 2e^{-11.411K}$. For an irreducible divisor $f(X) \mid P_K^{(5)}(X)$ in $\mathbb{Z}[X]$ of degree $d \geq 2$, we have $|\operatorname{disc}(f)| \geq 1$, and by Gelfond's inequality $H(f) \leq e^h H(P_K^{(5)}) \leq \exp(10.27K^2 + O(K))$. Mahler's root-separation inequality $|r_{i_a} - r_{i_b}| \geq \sqrt{3}d^{-(d+2)/2}H(f)^{-(d-1)}$ then gives the stated bound. □

Proposition 4.8 (Rational root exclusion and degree bounds for cluster factors)**.** *Let $K \in 30\mathbb{Z}_{>0}$ be sufficiently large, and let $F(X) = c_F\prod_{m=1}^d(X - r_{j_m}) \in \mathbb{Z}[X]$ ($c_F \in \mathbb{Z}\setminus\{0\}$) be a non-constant divisor of $P_K^{(5)}(X)$.*

(1) *No rational number $a/b \in \mathbb{Q}$ with denominator $1 \leq b \leq \exp(K/2251)$ is a root of $P_K^{(5)}(X)$, and any rational root a/b of $P_K^{(5)}(X)$ lying in $|X-\zeta(5)| \leq e^{-\tau K}$ ($\tau > 0$) satisfies $b \geq \exp\left(\frac{\tau}{25687}K\right)$.*

(2) *The degree $d = \deg F$ satisfies the unconditional lower bound*
$$\deg F = d \geq \frac{\log|c_F| - \log|F(\zeta(5))|}{\rho_5^\star K} \geq \frac{-\log|F(\zeta(5))|}{11.411K}, \tag{4.16}$$
and if all roots of $F(X)$ lie in the cluster disk $|r_{j_m} - \zeta(5)| \leq e^{-\tau K}$ ($\tau > 0$), then d also satisfies the upper bound $d \leq \frac{\log|c_F| - \log|F(\zeta(5))|}{\tau K}$.

Proof. (1) By Theorem 4.4, if $a/b \in \mathbb{Q}$ is a root of $P_K^{(5)}(X)$ in $|X-\zeta(5)| \leq e^{-\tau K}$, then for $b \geq b_0$ we have $b^{-25687} < |a/b - \zeta(5)| \leq e^{-\tau K}$ (while for $1 \leq b < b_0$, $\min_{a\in\mathbb{Z}}|a/b-\zeta(5)| > e^{-\tau K}$ for large K since $\zeta(5) \notin \mathbb{Q}$), forcing $b \geq e^{(\tau/25687)K}$. Taking $\tau = 11.411$ (which holds for all roots of $P_K^{(5)}$ by Theorem 4.4) gives $b \geq e^{K/2251}$.

(2) By Theorem 4.4, every root of $P_K^{(5)}(X)$ satisfies $|\zeta(5) - r_{j_m}| \geq e^{-\rho_5^\star K}$ with $\rho_5^\star = 11.411$. Thus $|F(\zeta(5))| = |c_F| \prod_{m=1}^{d} |\zeta(5) - r_{j_m}| \geq |c_F| e^{-d\rho_5^\star K}$. Taking logarithms gives $\log|F(\zeta(5))| \geq \log|c_F| - d\rho_5^\star K$, which rearranges to (4.16) since $|c_F| \geq 1$. Similarly, if $|\zeta(5) - r_{j_m}| \leq e^{-\tau K}$ for all $1 \leq m \leq d$, then $|F(\zeta(5))| \leq |c_F| e^{-d\tau K}$, giving $d \leq \frac{\log|c_F| - \log|F(\zeta(5))|}{\tau K}$. □

Remark 4.9 (Spectral bounds and variational conjecture for ρ_5^*). Combining Theorem 4.4 and Theorem 4.6(1) with $|\det B_K^{(5)}| \leq \|B_K^{(5)}\|^h$ yields the unconditional asymptotic sandwich

$$0.2590 \leq \liminf_{K\to\infty} \frac{1}{K} \log \|B_K^{(5)}\| \leq \limsup_{K\to\infty} \frac{1}{K} \log \|B_K^{(5)}\| \leq \rho_5^{\mathrm{var}} \leq 11.411, \tag{4.17}$$

where ρ_5^{var} denotes the two-sided logarithmic potential supremum over positive measures ν on $\mathbb{C}$ of mass at most $\lambda = 9/10$:

$$\rho_5^{\mathrm{var}} := \sup_{\nu(\mathbb{C}) \leq \lambda} \left[\max_{u\in[\alpha,1]} \Big(\Phi(u) + 2U^\nu(-u^2) \Big) - \sup_{t\geq 0} \Big(2U^\nu(t) - V_5(t) \Big) \right].$$

Notice also that since $\Phi(1) = 6(1.1)\log(1.1) - 6(0.9)\log(0.9) - 2\log 2 + 0.8 = 0.6083...$ and the equilibrium measure μ_5^{eq} of $V_5(t)$ on $[0,\infty)$ satisfies $U^{\mu_5^{\mathrm{eq}}}(-1) > 0$ and $-M_{\mathrm{eq}}^{(5)} \geq 4.274$, one has $\rho_5^{\mathrm{var}} \geq 0.6083 + 4.274 = 4.8823$. Thus, as long as the conservative decay bound $\epsilon = 0.0004$ of $Q_{K,100000}^{(5)}$ is held fixed in Theorem 4.1, the resulting exponent satisfies $\mu = \frac{\lambda\rho_5}{\epsilon} \geq \frac{0.9\times 4.8823}{0.0004} > 10985$.

Remark 4.10 (Lee–Yang phase transition at $\zeta(5)$). Define the partition function $\mathcal{Z}_K(X) := P_K^{(5)}(X)$, viewed as the canonical partition function of an h-mode system in external fugacity X (the modes being the real roots r_j). Since all h roots are real (Theorem 3.1), the roots are Lee–Yang zeros on the real axis (Lee–Yang circle theorem, adapted to the real axis). By Theorem 4.7, at least 1.40% of the roots satisfy $|r_j - \zeta(5)| \leq e^{-0.10K}$ and the closest root satisfies $\min_j |r_j - \zeta(5)| \leq e^{-K/2250}$, so the Lee–Yang zeros *pinch* the real axis at the critical point $X_c = \zeta(5)$ as $K \to \infty$. This establishes a first-order Lee–Yang-type phase transition in the free energy

$$f(X) := \lim_{K\to\infty} \frac{1}{hK} \log|P_K^{(5)}(X)|$$

at $X = \zeta(5)$. The transition is accompanied by the exact two-sided spectral gap $|r_+ - r_-| \geq 2e^{-11.411K}$ (Theorem 4.7(2)) for the closest pair of roots straddling $\zeta(5)$.

Conjecture 4.11. *The limit $\rho_5^* := \lim_{K\to\infty} \frac{1}{K} \log \|B_K^{(5)}\|$ exists and equals ρ_5^{var}.*

5. Weight-Seven Integral Identities, Dual Pencils, and Double-Pole Systems

In this section, we establish the structural theorems governing weight seven and simultaneous multi-pole Hankel determinants. We begin with a two-variable Beukers-type measure on the unit square $(0,1)^2$ whose diagonal moments produce $\zeta(7) - H_r^{(7)}$ and whose off-diagonal moments are purely rational.

Theorem 5.1 (Two-variable Beukers measure for $\zeta(7)$)**.** *Define the strictly positive measure on the unit square $(x_1, x_2) \in (0,1)^2$ by*

$$d\Omega_7(x_1,x_2) := \frac{1}{720} \frac{\big(-\log(x_1x_2)\big)^5}{1 - x_1x_2}\, dx_1\, dx_2 > 0. \tag{5.1}$$

Then for every pair of non-negative integers $(r_1, r_2) \in \mathbb{Z}^2_{\geq 0}$:

$$\iint_{(0,1)^2} x_1^{r_1} x_2^{r_2}\, d\Omega_7(x_1, x_2) = \begin{cases} \zeta(7) - H_r^{(7)}, & \text{if } r_1 = r_2 = r, \\ \dfrac{1}{6}\dfrac{H_{r_1}^{(6)} - H_{r_2}^{(6)}}{r_1 - r_2} \in \mathbb{Q}, & \text{if } r_1 \neq r_2. \end{cases} \tag{5.2}$$

Proof. Change variables on $(0,1)^2$ to $u = x_1 x_2 \in (0,1)$ and $x_1 \in (u,1)$, with Jacobian $\frac{\partial(x_1,x_2)}{\partial(x_1,u)} = \frac{1}{x_1}$, so that $x_1^{r_1} x_2^{r_2}\, dx_1\, dx_2 = u^{r_2} x_1^{r_1 - r_2 - 1}\, dx_1\, du$.

If $r_1 = r_2 = r$, integrating over $x_1 \in (u,1)$ gives $\int_u^1 x_1^{-1}\, dx_1 = -\log u$. Expanding $(1-u)^{-1} = \sum_{m=0}^{\infty} u^m$ and applying $\int_0^1 (-\log u)^6 u^{n-1}\, du = \frac{6!}{n^7} = \frac{720}{n^7}$ (with $n = r + m + 1$) yields $\frac{1}{720}\sum_{n=r+1}^{\infty} \frac{720}{n^7} = \zeta(7) - H_r^{(7)}$.

If $r_1 \neq r_2$, assume by symmetry that $r_1 > r_2$. Then $\int_u^1 x_1^{r_1 - r_2 - 1}\, dx_1 = \frac{1 - u^{r_1 - r_2}}{r_1 - r_2}$. Since $\frac{u^{r_2}(1 - u^{r_1 - r_2})}{1-u} = \sum_{m=r_2}^{r_1 - 1} u^m$, integrating term by term using $\int_0^1 (-\log u)^5 u^m\, du = \frac{5!}{(m+1)^6} = \frac{120}{(m+1)^6}$ gives

$$\frac{1}{720(r_1 - r_2)} \sum_{m=r_2}^{r_1 - 1} \frac{120}{(m+1)^6} = \frac{1}{6(r_1 - r_2)} \big(H_{r_1}^{(6)} - H_{r_2}^{(6)} \big) \in \mathbb{Q}. \qquad \square$$

Whereas the one-variable Hankel matrix $G_K^{(s)}(X)$ has positive definite evaluation $G_K^{(s)}(\zeta(s)) \succ 0$ and indefinite derivative $M_K^{(s)}$, the two-variable Beukers functional of Theorem 5.1 induces a dual matrix pencil in which the X-derivative matrix is strictly positive definite in the monomial basis itself, while the evaluation at $X = \zeta(7)$ admits an exact Krein-space decomposition into infinite-dimensional positive and negative cones.

Theorem 5.2 (Dual hyperbolicity and Krein-space decomposition of the two-variable $\zeta(7)$ pencil)**.** *Let $K \geq h \geq 1$ be integers, and let $\mathcal{A} = \{A_0(x), \dots, A_{h-1}(x)\} \subset \mathbb{Q}[x]_{\leq K}$ be any h linearly independent polynomials with coefficient matrix $C = \big([x^r] A_i(x)\big)_{0 \leq i < h,\, 0 \leq r \leq K} \in M_{h \times (K+1)}(\mathbb{Q})$. Define the $h \times h$ symmetric matrix pencil $G_{\mathcal{A}}^{(7)}(X)$ by*

$$G_{\mathcal{A}}^{(7)}(X)_{i,j} := \sum_{r=0}^{K} C_{i,r} C_{j,r} \big(X - H_r^{(7)} \big) + \frac{1}{6} \sum_{0 \leq r \neq s \leq K} C_{i,r} C_{j,s} \frac{H_r^{(6)} - H_s^{(6)}}{r - s}. \tag{5.3}$$

Then:

1. *The X-derivative matrix $M_{\mathcal{A}}^{(7)} := [X] G_{\mathcal{A}}^{(7)}(X) = CC^T$ is strictly positive definite, and $\Delta_{\mathcal{A}}^{(7)}(X) := \det G_{\mathcal{A}}^{(7)}(X) \in \mathbb{Q}[X]$ has exact degree h with all h roots real: $r_j = \zeta(7) - \gamma_j \in \mathbb{R}$, where $\gamma_1, \dots, \gamma_h \in \mathbb{R}$ are the eigenvalues of $L^{-1} G_{\mathcal{A}}^{(7)}(\zeta(7)) (L^{-1})^T$ ($CC^T = LL^T$).*
2. *For any $f \in L^2(0,1)$, setting $\psi_{m,\nu}^{\pm}(x) := x^m \big((-\log x)^\nu \pm (-\log x)^{5-\nu} \big)$ ($m \geq 0, \nu \in \{0,1,2\}$), the bilinear form at $X = \zeta(7)$ satisfies the exact Krein decomposition*

$$\iint_{(0,1)^2} f(x_1) f(x_2)\, d\Omega_7(x_1, x_2) = \frac{1}{1440} \sum_{m=0}^{\infty} \sum_{\nu=0}^{2} \binom{5}{\nu} \left[\left(\int_0^1 f(x) \psi_{m,\nu}^{+}(x)\, dx \right)^2 - \left(\int_0^1 f(x) \psi_{m,\nu}^{-}(x)\, dx \right)^2 \right]. \tag{5.4}$$

3. *For every integer $h \geq 1$ and every truncation $M \geq h$, there exist explicit h-dimensional polynomial subspaces $\mathcal{A}_+, \mathcal{A}_- \subset \mathbb{Q}[x]_{\leq K}$ such that $G_{\mathcal{A}_+}^{(7)}(\zeta(7)) \succ 0$ (all roots of $\Delta_{\mathcal{A}_+}^{(7)}(X)$ strictly less than $\zeta(7)$) and $G_{\mathcal{A}_-}^{(7)}(\zeta(7)) \prec 0$ (all roots strictly greater than $\zeta(7)$).*

Proof. (1) Differentiating (5.3) with respect to X gives $[X]G_{\mathcal{A}}^{(7)}(X)_{i,j} = \sum_{r=0}^{K} C_{i,r}C_{j,r} = (CC^T)_{i,j}$. Since $A_0, \dots, A_{h-1}$ are linearly independent, $\operatorname{rank}(C) = h$, so $M_{\mathcal{A}}^{(7)} = CC^T$ is real symmetric and strictly positive definite. Writing $CC^T = LL^T$ with $L \in \mathrm{GL}_h(\mathbb{R})$, $G_{\mathcal{A}}^{(7)}(X) = L\big(L^{-1}G_{\mathcal{A}}^{(7)}(\zeta(7))(L^{-1})^T + (X - \zeta(7))I_h\big)L^T$; since $L^{-1}G_{\mathcal{A}}^{(7)}(\zeta(7))(L^{-1})^T$ is real symmetric, all its eigenvalues $\gamma_1, \dots, \gamma_h$ are real, so all roots $r_j = \zeta(7) - \gamma_j$ of $\Delta_{\mathcal{A}}^{(7)}(X)$ are real.

(2) Expanding $(1-x_1x_2)^{-1} = \sum_{m\geq 0} x_1^m x_2^m$ and $(-\log(x_1x_2))^5 = \sum_{\nu=0}^{5} \binom{5}{\nu}(-\log x_1)^\nu(-\log x_2)^{5-\nu}$, then pairing ν with $5-\nu$ for $\nu \in \{0,1,2\}$ via $2I_\nu I_{5-\nu} = \frac{1}{2}[(I_\nu + I_{5-\nu})^2 - (I_\nu - I_{5-\nu})^2]$ where $I_\nu := \int_0^1 f(x)x^m(-\log x)^\nu\,dx$, gives (5.4) (the prefactor $1/1440$ arises from $\frac{1}{2\cdot 720} = \frac{1}{1440}$).

(3) *Linear independence.* The functions $\{x^m(-\log x)^\nu : m \geq 0,\ 0 \leq \nu \leq 5\}$ are linearly independent in $L^2(0,1)$: if $\sum_{m,\nu} c_{m,\nu}x^m(-\log x)^\nu \equiv 0$ on $(0,1)$, substituting $x = e^{-w}$ gives $\sum_{m,\nu} c_{m,\nu}e^{-mw}w^\nu \equiv 0$ on $(0,\infty)$, and since $\{e^{-mw}w^\nu\}_{m\geq 0, 0\leq\nu\leq 5}$ are linearly independent in $L^2(0,\infty)$, all $c_{m,\nu} = 0$. It follows that $\{\psi_{m,\nu}^+, \psi_{m,\nu}^- : m \geq 0,\ \nu \in \{0,1,2\}\}$ are linearly independent in $L^2(0,1)$, since each $\psi_{m,\nu}^\pm$ is a $\mathbb{Z}$-linear combination of $x^m(-\log x)^\nu$ and $x^m(-\log x)^{5-\nu}$ and this change of basis is invertible on $\operatorname{Span}\{x^m(-\log x)^\nu, x^m(-\log x)^{5-\nu}\}$.

Construction of $\mathcal{A}_+$. Fix $h \geq 1$ and truncation $M := h-1$ (so the total dimension of the finite family $\{\psi_{m,0}^+\}_{0\leq m<h} \cup \{\psi_{m,\nu}^-\}_{0\leq m\leq M,\,0\leq\nu\leq 2}$ is $h + 3h = 4h$). By the linear independence proved above, there exist unique elements $f_0, \dots, f_{h-1}$ in the finite-dimensional space $\mathcal{W} := \operatorname{Span}\{\psi_{m,0}^+ : 0 \leq m < h\} \cup \{\psi_{m,\nu}^- : 0 \leq m \leq M,\ \nu \in \{0,1,2\}\}$ orthogonal to $\operatorname{Span}\{\psi_{m,\nu}^- : 0 \leq m \leq M,\ \nu \in \{0,1,2\}\}$ and satisfying $\langle f_i, \psi_{m,0}^+\rangle = \delta_{i,m}$ for $0 \leq i, m < h$. For each pair (i,j), the bilinear form at $X = \zeta(7)$ on $\mathcal{W}$ reduces by the exact Krein decomposition (5.4) to

$$\iint f_i(x_1)f_j(x_2)\,d\Omega_7 = \frac{1}{1440}\sum_{m=0}^{M}\sum_{\nu=0}^{2}\binom{5}{\nu}\langle f_i, \psi_{m,\nu}^+\rangle\langle f_j, \psi_{m,\nu}^+\rangle,$$

since the terms $\langle f_i, \psi_{m,\nu}^-\rangle\langle f_j, \psi_{m,\nu}^-\rangle$ vanish by orthogonality for $0 \leq m \leq M$. Thus the restricted $h \times h$ matrix $\big[\iint f_i(x_1)f_j(x_2)\,d\Omega_7\big]_{0\leq i,j<h} = \frac{1}{1440}\sum_{m=0}^{M}\sum_{\nu=0}^{2}\binom{5}{\nu}u^{(m,\nu)}(u^{(m,\nu)})^T$ is a sum of positive semidefinite rank-one matrices, where $u_i^{(m,\nu)} := \langle f_i, \psi_{m,\nu}^+\rangle$. The $m = 0, \nu = 0$ term alone gives $\frac{1}{1440}u^{(0,0)}(u^{(0,0)})^T$ with $u_i^{(0,0)} = \delta_{i,0}\cdot\frac{1}{1440}\langle\psi_{0,0}^+, \psi_{0,0}^+\rangle > 0$, and summing over m, ν gives a matrix of the form $\frac{1}{1440}(I_h + \text{nonneg.})$ in the basis $\{f_0, \dots, f_{h-1}\}$, hence strictly positive definite.

Approximating each $f_i \in L^2(0,1)$ by rational polynomials $A_i \in \mathbb{Q}[x]_{\leq K}$ (density of polynomials in $L^2(0,1)$) and taking K sufficiently large, the resulting $h\times h$ rational matrix $G_{\mathcal{A}_+}^{(7)}(\zeta(7))$ is arbitrarily close to the positive-definite matrix above, hence strictly positive definite. By part (1), all h roots of $\Delta_{\mathcal{A}_+}^{(7)}(X)$ are real, and being roots of $\det(L^{-1}G(L^{-1})^T + (X-\zeta(7))I) = 0$ with $L^{-1}G(L^{-1})^T \succ 0$, they satisfy $\gamma_j > 0$, hence $r_j < \zeta(7)$.

Construction of $\mathcal{A}_-$. Exchanging the roles of ψ^+ and ψ^- in the construction (now requiring $\langle g_i, \psi_{m,\nu}^+\rangle = 0$ for $0 \leq m \leq M$ and $\langle g_i, \psi_{m,0}^-\rangle = \delta_{i,m}$) yields $g_0, \dots, g_{h-1} \in L^2(0,1)$ with the restricted bilinear form equal to $-\frac{1}{1440}$ times a strictly positive-definite matrix, hence strictly negative definite. Approximation by rational polynomials gives $\mathcal{A}_- \subset \mathbb{Q}[x]$ with $G_{\mathcal{A}_-}^{(7)}(\zeta(7)) \prec 0$, so all h roots of $\Delta_{\mathcal{A}_-}^{(7)}(X)$ satisfy $r_j > \zeta(7)$. □

Proposition 5.3 (Two-variable Beukers measure and dual pencil for $\beta(4) = L(4, \chi_{-4})$)**.** *Define the positive measure* $d\Omega_{\beta(4)}(x_1,x_2) := \frac{1}{6}\frac{(-\log(x_1x_2))^2}{1+x_1^2x_2^2}\,dx_1\,dx_2 > 0$ *on* $(0,1)^2$*, and let* $H_m^{(\beta,s)} :=$

$\sum_{n=0}^{m-1} \frac{(-1)^n}{(2n+1)^s}$ ($H_0^{(\beta,s)} := 0$). *Then for all integers* $k_1, k_2 \geq 0$*:*

$$\iint_{(0,1)^2} x_1^{4k_1} x_2^{4k_2}\, d\Omega_{\beta(4)}(x_1,x_2) = \begin{cases} \beta(4) - H_{2k}^{(\beta,4)}, & \text{if } k_1 = k_2 = k, \\ \dfrac{1}{12}\dfrac{H_{2k_1}^{(\beta,3)} - H_{2k_2}^{(\beta,3)}}{k_1 - k_2} \in \mathbb{Q}, & \text{if } k_1 \neq k_2. \end{cases}$$

Consequently, for any linearly independent subspace $\mathcal{A} \subset \mathbb{Q}[x^4]_{\leq 4K}$ *of dimension* h*, the induced symmetric matrix pencil* $G_{\mathcal{A}}^{(\beta(4))}(X)$ *has positive definite* X*-derivative* $CC^T \succ 0$ *and all* h *roots of* $\det G_{\mathcal{A}}^{(\beta(4))}(X)$ *are real.*

Proof. Setting $u = x_1 x_2 \in (0,1)$ and $x_1 \in (u,1)$ gives $\int_u^1 x_1^{-1}\, dx_1 = -\log u$ when $k_1 = k_2 = k$, and expanding $u^{4k}(1+u^2)^{-1} = \sum_{n=2k}^{\infty} (-1)^n u^{2n}$ yields $\beta(4) - H_{2k}^{(\beta,4)}$. When $k_1 > k_2$, $\int_u^1 x_1^{4(k_1-k_2)-1}\, dx_1 = \frac{1-u^{4(k_1-k_2)}}{4(k_1-k_2)}$, and since $1+u^2$ divides $1-u^{4(k_1-k_2)}$ in $\mathbb{Q}[u]$ with quotient $\sum_{n=2k_2}^{2k_1-1}(-1)^n u^{2(n-2k_2)}$, integrating term by term against $\frac{1}{6}(-\log u)^2 u^{4k_2}$ gives $\frac{H_{2k_1}^{(\beta,3)} - H_{2k_2}^{(\beta,3)}}{12(k_1-k_2)} \in \mathbb{Q}$. Real-rootedness follows as in Theorem 5.2(1). □

Next, we establish the exact cancellation of higher zeta values on x-even pole pairs under the auxiliary functional $\tau_X := \tau_X^{(7)}$ of Lemma 2.2, which satisfies the reflection identity $\tau_X(g(-1-x)) = -\tau_X(g(x))$ and, for rational functions $h(x)$ regular at $x = 0$, the difference identity $\tau_X(h(x+1) - h(x)) = \frac{h^{(6)}(0)}{720}$ (both identities are proved in Lemma 2.3).

Theorem 5.4 (Cancellation of higher zeta values on even pole pairs)**.** *Let* $s = 2k+1 \geq 3$ *and* $j \geq 1$*. Under the reflection and difference identities of Lemma 2.3, the* x*-even pole pair* $E_j(x) := (x-j)^{-1} - (x+j)^{-1}$ *satisfies*

$$\tau_X^{(s)}\big(E_j(x)\big) = j^{-(2k+1)} \in \mathbb{Q}. \tag{5.5}$$

Proof. Set $h(x) := (x-j)^{-1}$, which is regular at $x = 0$ since $j \geq 1$. Then $h(x+1) = (x-(j-1))^{-1}$. Under $x \mapsto -1-x$, the reflection identity of Lemma 2.3 gives

$$\tau_X^{(s)}\big((x+j)^{-1}\big) = -\tau_X^{(s)}\big((x-(j-1))^{-1}\big) = -\tau_X^{(s)}\big(h(x+1)\big).$$

Adding $\tau_X^{(s)}(h(x))$ to both sides and using the definition of E_j,

$$\tau_X^{(s)}\big(E_j(x)\big) = \tau_X^{(s)}(h(x)) - \tau_X^{(s)}(h(x+1)) = -\tau_X^{(s)}\big(h(x+1) - h(x)\big).$$

For $j \geq 2$, the difference identity of Lemma 2.3 gives $\tau_X^{(s)}(h(x+1) - h(x)) = -j^{-(2k+1)}$, whence $\tau_X^{(s)}(E_j(x)) = j^{-(2k+1)}$.

For $j = 1$: $h(x+1) = x^{-1}$ has a pole at $x = 0$, so the difference identity of Lemma 2.3 does not apply (the excluded case $r = 1$). Instead, compute directly from the polynomial-division identity underlying Lemma 2.2: $\frac{x^{2k+1}}{1-x^2} = -\sum_{r=0}^{k-1} x^{2k-1-2r} - \frac{1}{2}E_1(x)$, so $\mu_X^{(s)}((t+1)^{-1}) = \tau_X^{(s)}\big(x^{2k+1}/(1-x^2)\big) = -\frac{1}{2k} - \frac{1}{2}\tau_X^{(s)}(E_1)$. On the other hand, $\mu_X^{(s)}((t+1)^{-1}) = 1^{2k}(X - H_1^{(s)}) + \frac{1}{2} - \frac{1}{2k} = (X-1) + \frac{1}{2} - \frac{1}{2k}$, which forces $\tau_X^{(s)}(E_1) = 1 = 1^{-(2k+1)}$, matching (5.5) at $j = 1$. □

Remark 5.5. The formula (5.5) is restricted to the simple-pole case $r = 1$ (poles $(x-j)^{-1}$ and $(x+j)^{-1}$), which is the case used in the weight-seven outer prime-sum analysis (Theorem 5.6). A general statement for higher-order poles $(x-j)^{-r}$ ($r \geq 2$) would require an extension of $\tau_X^{(s)}$ to higher-order integer poles, which we do not require here.

We now record the exact evaluation of the prime-sum integrals for the one-variable weight-seven Hankel construction ($K = 40n, N = 3n, h = 37n, \alpha = 3/40, \lambda = 37/40, 2m = 8$) and the pole-ordered inner basis refinement.

Theorem 5.6 (Weight-seven prime-sum evaluation and pole-ordered refinement)**.** *Let $\alpha = 3/40, \lambda = 37/40$, and $H = 1 + 3\alpha = 49/40$. For $y \in [1/3, 37/20]$, define $\tilde{v}(y) := 1 - y\lfloor 1/y \rfloor$, $\tilde{u}(y) := \max(0, \alpha + \tilde{v}(y) - y)$, $\tilde{t}(y) := \min(\alpha, \tilde{v}(y)) + \tilde{u}(y)$, and $\tilde{r}(y) := \max(0, \frac{29}{20} - 3y)$. Set*

$$\widetilde{R}(y) := \begin{cases} 9(1-y) - 16\alpha - 7\tilde{t}(y) + \min(\tilde{r}(y), y + \tilde{u}(y)), & 1/3 \le y < 1/2, \\ 9(1-y) - 8\tilde{t}(y) + \min(\tilde{r}(y), y - \alpha + \tilde{u}(y)), & 1/2 \le y < 1, \\ 0, & 1 \le y \le 37/20, \end{cases}$$

and define the outer prime-sum integrand $T(y) := \widetilde{R}(y) - 2\lambda\lfloor 1/y \rfloor + \sum_{j=1}^{7} (2\lambda - jy)_+$. For $x \ge 3$ and $z \in (0, 1/2)$, set $l(x,z) := \lfloor x - z \rfloor + \lfloor x + z \rfloor + 1$, $b(x,z) := 4l(\alpha x, z)$, $T_x := \lfloor 2Hx \rfloor$, $s_x := Hx - T_x/2$, $q_x := \lfloor 2x \rfloor$, $n_+ := x - q_x/2$, and define the inner limiting function $R(x) := -\Gamma(x) - \mathcal{N}(x)$, where

$$\Gamma(x) := \int_0^{1/2} (T_x - b(x,z))(T_x + b(x,z) - l(x,z) - 7)\, dz + s_x(2T_x - q_x - 7) + (s_x - n_+)_+,$$

$$\mathcal{N}(x) := 2\lambda x\lfloor x \rfloor - 16\lambda x\lfloor \alpha x \rfloor - 2\lfloor 2\lambda x \rfloor \lambda x + \frac{\lfloor 2\lambda x \rfloor (\lfloor 2\lambda x \rfloor + 1)}{2}.$$

Then:

(1) *$T(y)$ is continuous on $[1/3, 1/2)$ and on $[1/2, 1)$, and its exact integral over $[1/3, 37/20]$ is*

$$I_{\text{out}} := \int_{1/3}^{37/20} T(y)\, dy = \frac{453803}{288000} \approx +1.57570486. \tag{5.6}$$

(2) *The inner tail integral satisfies $\int_{20}^{\infty} \frac{R(x)}{x^3}\, dx \le A_0 := -\frac{33104}{576000} \approx -0.05747222$.*

(3) *In the inner prime range $K/M < p \le K/3$ ($m_p = (p-1)/2$), let $k_{a,1} > k_{a,2} > \cdots > k_{a,s_a}$ ($s_a = l_K(a) - l_N(a)$) be the poles of $D_{\text{tail}}(t)$ in residue class $a \pmod p$ in decreasing order. Replacing $(t+a^2)^i$ in the local basis $E_{a,i}(t) = \prod_{0 \le c < a} (t+c^2)^{L_c}(t+a^2)^i$ by $\prod_{\nu=1}^{\min(i, s_a-2)} (t + k_{a,\nu}^2)(t+a^2)^{\max(0, i-(s_a-2))}$ upgrades the local half-weights from $w_{a,i} = i + 4l_N(a) - (l_K(a)+6)/2$ to*

$$\widetilde{w}_{a,i} = \begin{cases} i + 4l_N(a) - \dfrac{l_K(a)+6}{2}, & 0 \le i < L_a - 2, \\ i + 4l_N(a) - \dfrac{l_K(a)}{2}, & L_a - 2 \le i < L_a, \end{cases}$$

increasing the inner determinant valuation bound $\gamma_p^{\text{in}} = 2\sum_{a=0}^{m_p} \sum_{i=0}^{L_a - 1} w_{a,i}$ by $6(p-1)$, which increases $\Gamma(x) = \lim_{p \to \infty} \gamma_p^{\text{in}}/p$ by $+6$ for all $x \ge 3$ and reduces $\int_3^{\infty} \frac{R(x)}{x^3}\, dx$ by $\int_3^{\infty} \frac{6}{x^3}\, dx = \frac{1}{3}$.

Proof. (1) Partitioning $[1/3, 37/20]$ at all branch points of $\lfloor 1/y \rfloor$, $\tilde{u}(y)$, $\tilde{t}(y)$, $\min(\tilde{r}, y + \tilde{u})$, and $(37/20 - jy)_+$ yields 11 subintervals on which $T(y) = b + cy$ is affine, listed in Table 1. Summing the 11 exact rational integrals gives $I_{\text{out}} = \frac{4538030}{2880000} = \frac{453803}{288000}$.

(2) Set $f = \{x\}$, $g = \{\alpha x\}$, $\tau = \{2Hx\}$, $\sigma = \{2x\}$, $\eta = \{2\lambda x\}$, and $d(u) = \min(u, 1-u)$. Since $l(x,z) - 2x = \pm(\mathbf{1}_{z < d(f)} - 2d(f))$ has zero mean on $(0, 1/2)$ and L^2-norm $d(f)(1-2d(f)) \in$

TABLE 1. Exact affine pieces of $T(y) = b + cy$ on $[1/3, 37/20]$.

Interval $[l, r]$	b	c	$T(l)$	$T(r)$	$\int_l^r T(y)\,dy$
[1/3, 43/120]	51/8	−5	113/24	55/12	223/1920
[43/120, 29/80]	513/40	−23	55/12	359/80	2177/115200
[29/80, 37/100]	571/40	−27	359/80	857/200	10527/320000
[37/100, 37/80]	497/40	−22	857/200	9/4	48359/160000
[37/80, 29/60]	41/10	−4	9/4	13/6	53/1152
[29/60, 1/2)	53/20	−1	13/6	43/20	259/7200
[1/2, 43/80]	7/2	+1	4	323/80	1929/12800
[43/80, 37/60]	121/10	−15	323/80	57/20	10469/38400
[37/60, 37/40]	41/4	−12	57/20	−17/20	37/120
[37/40, 1)	1	−2	−17/20	−1	−111/1600
[1, 37/20]	37/20	−1	17/20	0	289/800

$[0, 1/8]$, expanding $\Gamma(x)$ and $\mathcal{N}(x)$ cancels the x^2 terms and yields $R(x) = xF(x)+Q(x)$, where $F(x) = 6\lambda + 2\lambda f - 16\lambda g$ and

$$Q(x) = \frac{\tau(\tau-\sigma) - (\tau-\sigma)_+ + \eta(1-\eta)}{2} + 25d(g)(1-2d(g)) - 4\int_0^{1/2} (l(x,z)-2x)(l(\alpha x,z)-2\alpha x)\,dz \le \frac{25}{8}+\frac{1}{4} = \frac{27}{8}$$

Define the periodic primitives $\mathcal{P}(x) := -\lambda f(1-f) + \frac{296}{3} g(1-g)$ (which satisfies $\mathcal{P}'(x) = F(x) + \lambda$ and has mean $\overline{\mathcal{P}} = -\frac{\lambda}{6} + \frac{296}{18} = \frac{11729}{720}$) and $\mathcal{C}(x) := -\frac{37}{240} G(f) + \frac{5920}{27} G(g)$ where $G(v) = v(1-v)(2v-1)$ (which satisfies $\mathcal{C}'(x) = \mathcal{P}(x) - \overline{\mathcal{P}}$ and $|\mathcal{C}(x)| \le \frac{1}{6\sqrt{3}}(\frac{37}{240} + \frac{5920}{27}) < 22$). Integrating $\int_{20}^{\infty} (xF(x))/x^3\,dx$ by parts twice gives

$$\int_{20}^{\infty} \frac{R(x)}{x^3}\,dx = -\frac{\lambda}{20} - \frac{\mathcal{P}(20) - \overline{\mathcal{P}}}{400} - \frac{2\mathcal{C}(20)}{8000} + 6\int_{20}^{\infty} \frac{\mathcal{C}(x)}{x^4}\,dx + \int_{20}^{\infty} \frac{Q(x)}{x^3}\,dx.$$

At $x = 20$, $f = \{20\} = 0$ and $g = \{3/2\} = 1/2$, so $\mathcal{P}(20) = \frac{74}{3}$ and $\mathcal{C}(20) = 0$. Using $|\mathcal{C}(x)| \le 22$ and $Q(x) \le 27/8$ yields

$$\int_{20}^{\infty} \frac{R(x)}{x^3}\,dx \le -\frac{37}{800} - \frac{74/3 - 11729/720}{400} + 6\int_{20}^{\infty} \frac{22}{x^4}\,dx + \int_{20}^{\infty} \frac{27/8}{x^3}\,dx = -\frac{33104}{576000}.$$

(3) Since $t + k_{a,\nu}^2 \equiv t + a^2 \pmod p$, the modified basis $\{\widetilde{E}_{a,i}\}$ is unitriangular modulo p relative to $\{E_{a,i}\}$ and hence $\mathbb{Z}_p$-unimodular. For $i \in \{L_a - 2, L_a - 1\}$, we have $i \ge s_a - 2$, so $\prod_{\nu=1}^{s_a-2}(t + k_{a,\nu}^2)$ divides $\widetilde{E}_{a,i}(t)$ and cancels all $s_a - 2$ poles of $D_{\text{tail}}(t)$ in class a having $k > p$. The only remaining poles in class a are $k = a < p$ and $k = p - a < p$. Under the local coordinate $x = a + pz$, these two poles lie at $z = 0$ ($r_1 = 0$) and $z = -1$ ($r_2 = -1$), where $d(0) = d(-1) = 0 \implies H_{d(r_\nu)}^{(7)} = H_0^{(7)} = 0$. Since $Y = p^7 X + C_p \in p^7\mathbb{Z}_p[X]$ for $p \ge 11$, the local principal-part contribution $p^{-6}(H_0^{(7)} - Y) = -p^{-6}Y \in p\mathbb{Z}_p[X]$ is p-integral without the -6 distribution loss. Thus, for each of the two top indices $i \in \{L_a - 2, L_a - 1\}$, the $-6/2 = -3$ shift in the half-weight $w_{a,i}$ is removed, giving $\widetilde{w}_{a,i} - w_{a,i} = +3$ and $\sum_{i=0}^{L_a-1}(\widetilde{w}_{a,i} - w_{a,i}) = 6$ per class. Summing over the $m_p = (p-1)/2$ classes in $\gamma_p^{\text{in}} = 2\sum_{a,i} w_{a,i}$ yields $\Delta\gamma_p^{\text{in}} = 2 \cdot \frac{p-1}{2} \cdot 6 = 6(p-1)$, which upon dividing by p increases $\Gamma(x)$ by $+6$ for all $x \ge 3$ and reduces $\int_3^{\infty} R(x)x^{-3}\,dx$ by $\int_3^{\infty} 6x^{-3}\,dx = 1/3$. □

Finally, following the double-pole Hankel architecture introduced by Lai [9], we prove the q-adic non-vanishing and local p-adic minor valuation bounds for the simultaneous weight-$(5,6)$ double-pole Hankel determinant.

Theorem 5.7 (Simultaneous double-pole Hankel determinant for $\{1,\zeta(5),\zeta(6)\}$)**.** *Suppose $a\zeta(5)+b\zeta(6)=c$ with $a,b,c\in\mathbb{Z}$ and $b\neq 0$. For $n\geq 1$, define the rational function*

$$R_n(t):=\frac{((6n)!)^2\prod_{j=1}^{n}(t-j+1)^5}{\prod_{k=0}^{6n}(t+k)^2}. \tag{5.7}$$

For every polynomial $P(t)\in\mathbb{Q}[t]$ with $\deg P<6n$, $\deg(R_n(t)P(t))\leq -n-3\leq -4$ has zero polynomial part and zero residue at $t=\infty$, and its partial fraction decomposition defines the rational linear functional

$$\mathcal{L}_n(P):=\frac{1}{n!}\sum_{k=0}^{6n}\Big[A_{n,k}\big(U_kP'(-k)+V_kP(-k)\big)+B_{n,k}U_kP(-k)\Big]\in\mathbb{Q}, \tag{5.8}$$

where $A_{n,k}=(-1)^k\binom{6n}{k}^2\binom{n+k}{k}^5 n!$, $B_{n,k}=A_{n,k}(7H_k^{(1)}-2H_{6n-k}^{(1)}-5H_{n+k}^{(1)})$, $U_k=c-aH_k^{(5)}-bH_k^{(6)}$, and $V_k=\frac{a}{k^6}+\frac{b}{5k^7}$ ($V_0=0$). Let $J_n:=\det_{0\leq i,j<3n}[\mathcal{L}_n(t^{i+j})]\in\mathbb{Q}$. Then:

1. *For every prime $q>\max(|b|,11)$ satisfying $q\equiv 1 \pmod 9$, set $m=(q-1)/9$ and $n=2m$. Then $v_q(J_{2m})=-18n=-36m<0$; in particular, $J_{2m}\neq 0$.*
2. *For every prime $p\in(\sqrt{6n},6n]$, let $L_\rho:=\#\{0\leq k\leq 6n : k\equiv\rho \pmod p\}$ and $\omega_p(\rho):=2\lfloor\frac{6n}{p}\rfloor-2\lfloor\frac{6n-\rho}{p}\rfloor+5\lfloor\frac{n+\rho}{p}\rfloor-5\lfloor\frac{n}{p}\rfloor\geq 0$. Then*

$$v_p(J_n)\geq\min_{\substack{0\leq m_\rho\leq 2L_\rho\\ \sum_{\rho=0}^{p-1}m_\rho=3n}}\sum_{\rho=0}^{p-1}m_\rho\big(m_\rho-1+\omega_p(\rho)-7\big). \tag{5.9}$$

Proof. Since $\deg R_n=5n-(12n+2)=-7n-2$, for any $0\leq i,j<3n$ the polynomial $P(t)=t^{i+j}$ has degree $i+j\leq 6n-2<6n$, so $\deg(R_n(t)t^{i+j})\leq -n-4<-2$. Thus every entry $\mathcal{L}_n(t^{i+j})$ of J_n is well-defined and free of poles at infinity.

(1) Set $n=2m=\frac{2(q-1)}{9}$. Then $n=2m<q\leq 6n=q-1+3m<\frac{3}{2}q$. For each of the $3m$ integers $k=q+r$ $(0\leq r<3m)$ in $[q,6n]$, we have $n+k=q+2m+r<\frac{5}{3}q<2q$, so $v_q(\binom{6n}{q+r})=1-1-0=0$ and $v_q(\binom{n+q+r}{q+r})=1-1-0=0$, giving $v_q(A_{n,q+r}/n!)=0$. Moreover, $H_{q+r}^{(5)}\in q^{-5}+\mathbb{Z}_q$ and $H_{q+r}^{(6)}\in q^{-6}+\mathbb{Z}_q$, so $U_{q+r}\equiv -bq^{-6} \pmod{q^{-5}\mathbb{Z}_q}$ with $v_q(U_{q+r})=-6$ since $q>|b|$.

Let $\{P_{2r}(t),P_{2r+1}(t)\}_{0\leq r<3m}\subset\mathbb{Z}_q[t]_{<3n}$ be the Hermite interpolation basis at the $3m$ nodes $\{-q,-(q+1),\dots,-6n\}$ satisfying $P_{2r}(-(q+r'))=\delta_{r,r'}$, $P'_{2r}(-(q+r'))=0$ and $P_{2r+1}(-(q+r'))=0, P'_{2r+1}(-(q+r'))=\delta_{r,r'}$. Since the $3m=(q-1)/3$ nodes are pairwise incongruent modulo q, this basis is $\mathbb{Z}_q$-unimodular, so $v_q(J_{2m})=v_q(\det M)$, where $M_{\alpha,\beta}:=\mathcal{L}_n(P_\alpha P_\beta)$. Split $M=M_{\geq q}+M_{<q}$ according to $k\geq q$ and $k<q$ in (5.8). For $0\leq k<q$, all quantities $A_{n,k}/n!, B_{n,k}/n!, U_k, V_k$ lie in $\mathbb{Z}_q$, so $M_{<q}\in M_{3n}(\mathbb{Z}_q)$. For $k=q+r'\geq q$, the Hermite conditions imply that $(P_\alpha P_\beta)(-(q+r'))$ and $(P_\alpha P_\beta)'(-(q+r'))$ vanish for all r' whenever $\lfloor\alpha/2\rfloor\neq\lfloor\beta/2\rfloor$. Thus $M_{\geq q}=\mathrm{diag}_{0\leq r<3m}(M_{\geq q}^{(r)})$ is strictly block-diagonal with 2×2 blocks. Define the diagonal scaling matrix $D_q:=\mathrm{diag}_{0\leq r<3m}(q^{7/2},q^{5/2})$, and set $\widetilde{M}:=D_qMD_q$. Since $v_q(V_k)\geq -7$ and $v_q(B_{n,k}U_k/n!)\geq -7$ for all $q\leq k\leq 6n$ (with equality at $k=q$), while $(M_{\geq q})_{2r+1,2r+1}=0$ and

$M_{<q} \in M_{3n}(\mathbb{Z}_q)$, every entry of $\widetilde{M}$ lies in $\mathbb{Z}_q$. Moreover, on the r-th diagonal 2×2 block,

$$\widetilde{M}^{(r)} = \begin{pmatrix} q^7 M_{2r,2r} & q^6 M_{2r,2r+1} \\ q^6 M_{2r+1,2r} & q^5 M_{2r+1,2r+1} \end{pmatrix} \equiv \begin{pmatrix} * & -b\frac{A_{n,q+r}}{n!} \\ -b\frac{A_{n,q+r}}{n!} & 0 \end{pmatrix} \pmod{q\mathbb{Z}_q},$$

while all off-block entries of $\widetilde{M}$ lie in $q^5\mathbb{Z}_q \subset q\mathbb{Z}_q$. Reducing $\widetilde{M}$ modulo q yields a block-diagonal matrix over $\mathbb{F}_q$ with determinant $\prod_{r=0}^{3m-1}(-b^2(A_{n,q+r}/n!)^2) \not\equiv 0 \pmod{q}$. Hence $v_q(\det \widetilde{M}) = 0$, and since $\det \widetilde{M} = (\det D_q)^2 J_{2m} = q^{36m} J_{2m}$, we obtain $v_q(J_{2m}) = -36m = -18n < 0$.

(2) For $p \in (\sqrt{6n}, 6n]$ and $\rho \in \{0, \dots, p-1\}$, pass to the unitriangular Taylor basis $\phi_{i,\rho}(t) := (t+\rho)^i$ via $\mathbf{E}_\rho = [\binom{i}{j}\rho^{i-j}]$. In the local matrix $M_{i,j}^{\rho \bmod p}$ summing the poles $k \in K_\rho := \{0 \le k \le 6n : k \equiv \rho \pmod{p}\}$ of cardinality $L_\rho = |K_\rho|$, each pole $k = \rho + lp \in K_\rho$ contributes a linear combination of the two rank-1 matrices $\phi_{i,\rho}(-k)\phi_{j,\rho}(-k)$ and $\phi'_{i,\rho}(-k)\phi_{j,\rho}(-k) + \phi_{i,\rho}(-k)\phi'_{j,\rho}(-k)$. Thus $\operatorname{rank}_{\mathbb{Q}_p}(M^{\rho \bmod p}) \le 2L_\rho$, so any $m_\rho \times m_\rho$ minor of $M^{\rho \bmod p}$ vanishes for $m_\rho > 2L_\rho$. For $m_\rho \le 2L_\rho$, since $\phi_{i,\rho}(-k) = (-lp)^i$ and $\phi'_{i,\rho}(-k) = i(-lp)^{i-1}$, while $p > \sqrt{6n}$ implies $v_p(k) \le 1$, we have $v_p(U_k) \ge -6$, $v_p(V_k) \ge -7$, $v_p((B_{n,k}/A_{n,k})U_k) \ge -7$, and $v_p(A_{n,k}/n!) \ge \omega_p(\rho)$ by Legendre's formula. Factoring p^i from row $i \in I$ and p^j from column $j \in J$ of an $m_\rho \times m_\rho$ minor gives valuation at least $\sum_{i\in I} i + \sum_{j\in J} j + m_\rho(\omega_p(\rho) - 7) \ge m_\rho(m_\rho - 1 + \omega_p(\rho) - 7)$. Multilinear determinant expansion across $\sum_{\rho=0}^{p-1} m_\rho = 3n$ yields (5.9). □

6. Structural Barriers, Multi-Weight Couplings, and Further Diophantine Advances

In this section, we push the algebraic, analytic, and p-adic structure of both the one-variable and two-variable Hankel architectures to their exact mathematical limits. We first establish that the p^{-6} distribution barrier at weight seven is intrinsic to the one-variable reduction of $d\Omega_7(x_1, x_2)$, yet show that an asymmetric evaluation of the same measure yields simultaneous rational approximations to $\zeta(6)$ and $\zeta(7)$ with zero denominator overhead, alongside unconditional q-adic non-vanishing and explicit positive-cone polynomial bases. We then uncover an exact sign cancellation between the alternating Dirichlet character χ_{-4} and the Vandermonde derivative for the Dirichlet beta values $\beta(2k) = L(2k, \chi_{-4})$, proving strict positive definiteness of the derivative matrix and unconditional hyperbolicity at full pole density. Finally, we establish a master differential-shift identity across arbitrary tuples of odd zeta values, extend the double- and triple-pole architectures to multi-weight sets, and prove a universal potential-gap conservation law showing why the two-variable Beukers architecture identically eliminates the quadratic derivative-determinant growth of one-variable Hankel pencils.

6.1. Weight seven, the divided-difference valuation law, and simultaneous $(\zeta(6), \zeta(7))$ forms.

Let $d\Omega_7(x_1, x_2) = \frac{1}{720}\frac{(-\log(x_1x_2))^5}{1-x_1x_2}\,dx_1\,dx_2$ denote the two-variable Beukers measure on $(0,1)^2$ introduced in Theorem 5.1. We begin by proving that although the off-diagonal moments of $d\Omega_7$ involve only weight-six harmonic numbers, the divided-difference denominator restores the exact p^{-7} pole valuation on every residue class modulo p.

Theorem 6.1 (Divided-difference p-adic valuation law and intrinsic p^{-6} barrier)**.** *Let $M^{(7)}_{r_1,r_2} := \iint_{(0,1)^2} x_1^{r_1}x_2^{r_2}\,d\Omega_7(x_1,x_2)$ be the moment matrix of Theorem 5.1 for $0 \le r_1, r_2 \le K$.*

(1) (Multiplicative fiber reduction) *Under the change of variables $x_1 = u^y$, $x_2 = u^{1-y}$ ($u \in (0,1)$, $y \in (0,1)$) with Jacobian $\left|\frac{\partial(x_1,x_2)}{\partial(u,y)}\right| = -\log u$, for any polynomials $A(x) =$*

$\sum_{r=0}^{K} a_r x^r$ *and* $B(x) = \sum_{r=0}^{K} b_r x^r$ *in* $\mathbb{Q}[x]_{\le K}$*, the two-variable integral reduces to the one-variable Mellin integral*

$$\iint_{(0,1)^2} A(x_1)B(x_2)\, d\Omega_7(x_1,x_2) = \int_0^1 \mathcal{M}_{A,B}(u)\frac{(-\log u)^6}{720(1-u)}\, du, \tag{6.1}$$

where

$$\mathcal{M}_{A,B}(u) := \int_0^1 A(u^y)B(u^{1-y})\, dy = \sum_{r=0}^{K} a_r b_r u^r + \sum_{0\le r_1\ne r_2\le K} a_{r_1} b_{r_2} \frac{u^{r_1}-u^{r_2}}{(r_1-r_2)\log u}.$$

(2) (Exact p-adic valuation of off-diagonal divided differences) *Let $p \ge 11$ be a prime in the outer range $K/2 < p \le K$. While every individual harmonic number $H_r^{(6)}$ $(0 \le r \le K)$ has p-adic valuation $v_p(H_r^{(6)}) \ge -6$, the off-diagonal moment $M_{r_1,r_2}^{(7)} = \frac{H_{r_1}^{(6)} - H_{r_2}^{(6)}}{6(r_1-r_2)}$ $(r_1 \ne r_2)$ satisfies $v_p(M_{r_1,r_2}^{(7)}) = -7$ whenever $0 \le r_1 < p \le r_2 \le K$ and $r_1 \equiv r_2 \pmod p$.*

(3) (Non-degeneracy of the local pole obstruction) *In every residue class $\rho \in \{0,1,\dots,p-1\}$ with $L_\rho := \#\{0 \le r \le K : r \equiv \rho \pmod p\} = 2$ (so $r \in \{\rho, \rho+p\}$ with $\rho \le K-p$), the 2×2 local submatrix $\left[M^{(7)}_{\rho+l_1 p,\, \rho+l_2 p}\right]_{0\le l_1,l_2\le 1}$ at $X \in \mathbb{Z}_p$ has exact p-adic pole denominator p^{-7} both on the diagonal ($l_1 = l_2 = 1$) and off the diagonal ($l_1 = 0, l_2 = 1$). Its leading p^{-7} coefficient matrix $\begin{pmatrix} 0 & 1/6 \\ 1/6 & -1 \end{pmatrix}$ has determinant $-\frac{1}{36} \in \mathbb{Z}_p^\times$, so no non-zero $\mathbb{Z}_p$-linear combination of columns inside class ρ can simultaneously annihilate the p^{-7} pole on and off the diagonal.*

Proof. (1) For $x_1 = u^y$ and $x_2 = u^{1-y}$, we have $x_1x_2 = u$, $\frac{\partial x_1}{\partial u} = yu^{y-1}$, $\frac{\partial x_1}{\partial y} = u^y \log u$, $\frac{\partial x_2}{\partial u} = (1-y)u^{-y}$, and $\frac{\partial x_2}{\partial y} = -u^{1-y}\log u$. The Jacobian determinant is

$$\begin{vmatrix} yu^{y-1} & u^y \log u \\ (1-y)u^{-y} & -u^{1-y}\log u \end{vmatrix} = -y\log u - (1-y)\log u = -\log u.$$

Thus $dx_1\, dx_2 = (-\log u)\, du\, dy$ and $(-\log(x_1x_2))^5\, dx_1\, dx_2 = (-\log u)^6\, du\, dy$. Integrating first over $y \in (0,1)$ gives $\mathcal{M}_{A,B}(u) = \int_0^1 A(u^y)B(u^{1-y})\, dy$. For each pair (r_1,r_2) with $r_1 \ne r_2$, we have

$$\int_0^1 u^{r_1 y + r_2(1-y)}\, dy = u^{r_2}\int_0^1 u^{(r_1-r_2)y}\, dy = u^{r_2}\left[\frac{u^{(r_1-r_2)y}}{(r_1-r_2)\log u}\right]_0^1 = \frac{u^{r_1}-u^{r_2}}{(r_1-r_2)\log u},$$

while for $r_1 = r_2 = r$ the integral is $\int_0^1 u^r\, dy = u^r$, establishing (6.1).

(2) Fix a prime $p \ge 11$ with $K/2 < p \le K$, and let $0 \le r_1 < p \le r_2 \le K$ satisfy $r_2 - r_1 = lp$ (so $l = 1$ since $K < 2p$). Because $r_1 < p$, every integer $\nu \in \{1,\dots,r_1\}$ is coprime to p, so $H_{r_1}^{(6)} = \sum_{\nu=1}^{r_1} \nu^{-6} \in \mathbb{Z}_p$. Since $p \le r_2 < 2p$, the sum $H_{r_2}^{(6)} = \sum_{\nu=1}^{r_2} \nu^{-6}$ contains the single p-divisible index $\nu = p$, while all $\nu \ne p$ lie in $\mathbb{Z}_p^\times$. Thus

$$H_{r_2}^{(6)} = H_{r_1}^{(6)} + \sum_{\nu=r_1+1}^{r_2} \nu^{-6} \equiv H_{r_1}^{(6)} + p^{-6} \pmod{\mathbb{Z}_p}.$$

Dividing $H_{r_1}^{(6)} - H_{r_2}^{(6)} \equiv -p^{-6} \pmod{\mathbb{Z}_p}$ by $6(r_1-r_2) = -6lp$ (where $6l \in \mathbb{Z}_p^\times$ since $p \ge 11$ and $1 \le l < p$) yields

$$M_{r_1,r_2}^{(7)} = \frac{-p^{-6} + O(1)}{-6lp} = \frac{1}{6l}p^{-7} + O(p^{-1}),$$

which has exact p-adic valuation $v_p(M^{(7)}_{r_1,r_2}) = -7$.

(3) For any residue class ρ with $L_\rho = 2$, the 2×2 submatrix of $G^{(7)}_{\mathcal{A}}(X)$ at $X \in \mathbb{Z}_p$ on the monomials $\{x^\rho, x^{\rho+p}\}$ is

$$\begin{pmatrix} X - H^{(7)}_\rho & M^{(7)}_{\rho,\rho+p} \\ M^{(7)}_{\rho+p,\rho} & X - H^{(7)}_{\rho+p} \end{pmatrix} \equiv p^{-7} \begin{pmatrix} 0 & 1/6 \\ 1/6 & -1 \end{pmatrix} \pmod{p^{-6}\mathbb{Z}_p},$$

since $\rho < p$ implies $H^{(7)}_\rho \in \mathbb{Z}_p$, while $H^{(7)}_{\rho+p} \equiv p^{-7} \pmod{\mathbb{Z}_p}$ and $M^{(7)}_{\rho,\rho+p} \equiv \frac{1}{6}p^{-7} \pmod{p^{-1}\mathbb{Z}_p}$ by part (2) with $l = 1$. The coefficient matrix $\begin{pmatrix} 0 & 1/6 \\ 1/6 & -1 \end{pmatrix}$ has determinant $-\frac{1}{36} \in \mathbb{Z}_p^\times$ for $p \geq 11$. If a $\mathbb{Z}_p$-linear combination $c_0 v_0 + c_1 v_1$ of its columns vanishes modulo $p^{-6}\mathbb{Z}_p$, then $\begin{pmatrix} 0 & 1/6 \\ 1/6 & -1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} \equiv \begin{pmatrix} 0 \\ 0 \end{pmatrix} \pmod{p}$, forcing $c_0 \equiv c_1 \equiv 0 \pmod{p}$. □

Although the p^{-7} pole valuation is intrinsic to symmetric evaluations on $d\Omega_7$, evaluating $d\Omega_7(x_1, x_2)$ on shifted asymmetric pairs produces $\zeta(6)$ alongside $\zeta(7)$ with no increase in the least common multiple denominator $d_K^7 := \operatorname{lcm}(1, 2, \ldots, K)^7$.

Theorem 6.2 (Simultaneous weight-$(6, 7)$ Beukers functional with zero denominator overhead)**.** *Let $d\Omega_7(x_1, x_2) = \frac{1}{720} \frac{(-\log(x_1x_2))^5}{1-x_1x_2}\, dx_1\, dx_2$ on $(0, 1)^2$. Then:*

(1) *For every integer $m \geq 0$, $\iint_{(0,1)^2} x_1^m x_2^{m+1}\, d\Omega_7(x_1, x_2) = \frac{1}{6(m+1)^6}$, and for every integer $r \geq 0$,*

$$\iint_{(0,1)^2} \frac{(x_1x_2)^r x_2}{1 - x_1x_2}\, d\Omega_7(x_1, x_2) = \frac{1}{6}\big(\zeta(6) - H^{(6)}_r\big). \tag{6.2}$$

(2) *For any polynomial $A(x) = \sum_{r=0}^K c_r x^r \in \mathbb{Z}[x]$ and any integers $\alpha, \beta \in \mathbb{Z}$, the single integral*

$$\mathcal{I}_K(\alpha, \beta; A) := \iint_{(0,1)^2} \left[\alpha A(x_1)A(x_2) + \frac{6\beta A(x_1x_2)x_2}{1 - x_1x_2}\right] d\Omega_7(x_1, x_2)$$

evaluates to the exact simultaneous linear form in $\zeta(7)$ and $\zeta(6)$:

$$\mathcal{I}_K(\alpha, \beta; A) = \alpha \left(\sum_{r=0}^K c_r^2\right) \zeta(7) + \beta \left(\sum_{r=0}^K c_r\right) \zeta(6) - R_K(\alpha, \beta; A),$$

where the rational part

$$R_K(\alpha, \beta; A) := \alpha \sum_{r=0}^K c_r^2 H^{(7)}_r - \frac{\alpha}{6} \sum_{0 \leq r \neq s \leq K} c_r c_s \frac{H^{(6)}_r - H^{(6)}_s}{r - s} + \beta \sum_{r=0}^K c_r H^{(6)}_r \in \mathbb{Q}$$

satisfies $6d_K^7 R_K(\alpha, \beta; A) \in \mathbb{Z}$ with the exact same denominator $6d_K^7$ as the pure $\zeta(7)$ form.

Proof. (1) By Theorem 5.1 with $r_1 = m$ and $r_2 = m + 1$, the moment is

$$M^{(7)}_{m,m+1} = \frac{1}{6} \frac{H^{(6)}_m - H^{(6)}_{m+1}}{m - (m + 1)} = \frac{1}{6}\big(H^{(6)}_{m+1} - H^{(6)}_m\big) = \frac{1}{6(m + 1)^6}.$$

Expanding $\frac{(x_1x_2)^r x_2}{1-x_1x_2} = \sum_{m=r}^\infty x_1^m x_2^{m+1}$ on $(0, 1)^2$ and integrating term by term via monotone convergence yields $\sum_{m=r}^\infty \frac{1}{6(m+1)^6} = \frac{1}{6}(\zeta(6) - H^{(6)}_r)$.

(2) By Theorem 5.1 and (6.2), $\alpha\iint A(x_1)A(x_2)\,d\Omega_7$ expands to $\alpha\sum_{r=0}^{K}c_r^2(\zeta(7)-H_r^{(7)})+\frac{\alpha}{6}\sum_{r\neq s}c_rc_s\frac{H_r^{(6)}-H_s^{(6)}}{r-s}$, while $6\beta\iint\frac{A(x_1x_2)x_2}{1-x_1x_2}\,d\Omega_7$ expands to $6\beta\sum_{r=0}^{K}c_r\frac{1}{6}(\zeta(6)-H_r^{(6)})=\beta\sum_{r=0}^{K}c_r(\zeta(6)-H_r^{(6)})$. Since $d_K^6H_r^{(6)}\in\mathbb{Z}$, $d_K^7H_r^{(7)}\in\mathbb{Z}$, and $d_K^7\frac{H_r^{(6)}-H_s^{(6)}}{r-s}=d_K^7\sum_{m=\min(r,s)+1}^{\max(r,s)}\frac{1}{m^6|r-s|}\in\mathbb{Z}$ (as $|r-s|\le K$ divides d_K and m^6 divides d_K^6), we have $6d_K^7R_K(\alpha,\beta;A)\in\mathbb{Z}$. □

Next, we turn the non-degeneracy of the local p^{-7} block in Theorem 6.1(3) into an unconditional q-adic non-vanishing theorem for the two-variable Beukers determinant $\Delta_{\mathcal{A}}^{(7)}(0)$ of Theorem 5.2, and construct an explicit rational polynomial subspace inside the positive Krein cone with exponential archimedean decay.

Theorem 6.3 (Unconditional q-adic non-vanishing and explicit positive-cone approximations for $G_{\mathcal{A}}^{(7)}(X)$)**.** *Let $G_{\mathcal{A}}^{(7)}(X)$ be the two-variable Beukers matrix pencil of Theorem 5.2.*

(1) (Unconditional q-adic non-vanishing) *Let $q\ge5$ be any prime, and set $K=2q-1$ and $h=q$. For the q-dimensional polynomial basis $\mathcal{A}_q:=\{A_\rho(x):=x^\rho+x^{\rho+q}:0\le\rho<q\}\subset\mathbb{Z}[x]_{\le2q-1}$, the rational determinant $\Delta_{\mathcal{A}_q}^{(7)}(0)=\det G_{\mathcal{A}_q}^{(7)}(0)\in\mathbb{Q}$ satisfies the exact q-adic valuation*

$$v_q\big(\Delta_{\mathcal{A}_q}^{(7)}(0)\big)=-7q<0.$$

In particular, $\Delta_{\mathcal{A}_q}^{(7)}(0)\neq0$, and $\Delta_{\mathcal{A}_q}^{(7)}(a/b)\neq0$ for every rational $a/b\in\mathbb{Q}$ with $q\nmid b$.

(2) (Explicit positive-cone basis with exponential decay) *Fix an integer $h\ge1$, set $m_0:=2h$ and $K:=7h$, and consider the $4h$-dimensional rational polynomial space $\mathcal{W}_K:=\{x^{m_0}(1-x)^hQ(x):Q(x)\in\mathbb{Q}[x]_{<4h}\}\subset\mathbb{Q}[x]_{\le K}$. Define the explicit rational subspace*

$$\mathcal{V}_K^+:=\left\{P(x)\in\mathcal{W}_K:\int_0^1P(x)\psi_{m,\nu}^-(x)\,dx=0\ \text{ for all }\ 0\le m<h,\ \nu\in\{0,1,2\}\right\},$$

where $\psi_{m,\nu}^-(x)=x^m((-\log x)^\nu-(-\log x)^{5-\nu})$ is as in Theorem 5.2(2). Then $\dim_{\mathbb{Q}}\mathcal{V}_K^+\ge h$, and for any h-dimensional basis $\mathcal{A}_+^{(K)}=\{A_0(x),\dots,A_{h-1}(x)\}\subset\mathcal{V}_K^+$ normalized so that $A_i(x)=x^{m_0}(1-x)^hQ_i(x)$ has $\sum_{j=0}^{4h-1}|[x^j]Q_i(x)|\le1$, the evaluation matrix $G_{\mathcal{A}_+^{(K)}}^{(7)}(\zeta(7))$ is strictly positive definite and satisfies

$$0<\Delta_{\mathcal{A}_+^{(K)}}^{(7)}(\zeta(7))\le(4^{-2h}h^{-6})^h.$$

Proof. (1) For $0\le\rho,\sigma<q$, the polynomial $A_\rho(x)=x^\rho+x^{\rho+q}$ is supported in the single residue class $\rho\pmod q$. If $\rho\neq\sigma$, then every $r\in\{\rho,\rho+q\}$ and $s\in\{\sigma,\sigma+q\}$ satisfies $r-s\in\mathbb{Z}_q^\times$, so $v_q(M_{r,s}^{(7)})=v_q\big(\frac{H_r^{(6)}-H_s^{(6)}}{6(r-s)}\big)\ge-6$. Thus $v_q(G_{\mathcal{A}_q}^{(7)}(0)_{\rho,\sigma})\ge-6$ for all $\rho\neq\sigma$.

On the diagonal $\rho=\sigma\in\{0,1,\dots,q-1\}$, we have $\rho<q\le\rho+q<2q$. For $1\le\rho<q$, $H_\rho^{(7)},H_\rho^{(6)}\in\mathbb{Z}_q$, whereas $H_{\rho+q}^{(7)}\equiv q^{-7}\pmod{\mathbb{Z}_q}$ and $H_{\rho+q}^{(6)}\equiv q^{-6}\pmod{\mathbb{Z}_q}$. By (5.3) at $X=0$,

$$G_{\mathcal{A}_q}^{(7)}(0)_{\rho,\rho}=-H_\rho^{(7)}-H_{\rho+q}^{(7)}+\frac{2}{6}\frac{H_\rho^{(6)}-H_{\rho+q}^{(6)}}{\rho-(\rho+q)}\equiv-q^{-7}+\frac{1}{3}q^{-7}=-\frac{2}{3}q^{-7}\pmod{\mathbb{Z}_q}.$$

For $\rho=0$, $H_0^{(7)}=H_0^{(6)}=0$, $H_q^{(7)}=q^{-7}+H_{q-1}^{(7)}\equiv q^{-7}\pmod{\mathbb{Z}_q}$, and $H_q^{(6)}=q^{-6}+H_{q-1}^{(6)}\equiv q^{-6}\pmod{\mathbb{Z}_q}$, giving the same congruence $G_{\mathcal{A}_q}^{(7)}(0)_{0,0}\equiv-\frac{2}{3}q^{-7}\pmod{\mathbb{Z}_q}$. Since $q\ge5$, $-\frac{2}{3}\in\mathbb{Z}_q^\times$, so $q^7G_{\mathcal{A}_q}^{(7)}(0)\equiv-\frac{2}{3}I_q\pmod{q\mathbb{Z}_q}$. Taking determinants yields $v_q(\Delta_{\mathcal{A}_q}^{(7)}(0))=-7q<0$. If

$X = a/b \in \mathbb{Q}$ has $q \nmid b$, then $X \in \mathbb{Z}_q$ does not alter the q^{-7} terms, so $v_q(\Delta^{(7)}_{\mathcal{A}_q}(a/b)) = -7q$ and $\Delta^{(7)}_{\mathcal{A}_q}(a/b) \neq 0$.

(2) Since $\int_0^1 x^r \psi^-_{m,\nu}(x)\,dx = \frac{\nu!}{(r+m+1)^{\nu+1}} - \frac{(5-\nu)!}{(r+m+1)^{6-\nu}} \in \mathbb{Q}$ for every $r, m \geq 0$, the $3h$ conditions $\langle P, \psi^-_{m,\nu}\rangle = 0$ ($0 \leq m < h$, $\nu \in \{0,1,2\}$) define rational linear functionals on the $4h$-dimensional $\mathbb{Q}$-vector space $\mathcal{W}_K$, so the kernel $\mathcal{V}^+_K$ has dimension at least $4h - 3h = h$. For any non-zero $P \in \mathcal{V}^+_K$, all negative terms $0 \leq m < h$ in the Krein decomposition (5.4) vanish identically, while for $m \geq h$ the factor $x^{2h+m}(1-x)^h$ concentrates on $[1/2, 1]$ where $-\log x \in (0, \log 2)$ satisfies $(-\log x)^{5-\nu} < (-\log x)^\nu$ for $\nu \in \{0,1,2\}$, ensuring $\iint P(x_1)P(x_2)\,d\Omega_7 > 0$. Thus $G^{(7)}_{\mathcal{A}^{(K)}_+}(\zeta(7))$ is strictly positive definite.

For each basis element $A_i(x) = x^{2h}(1-x)^h Q_i(x)$ with $\sum_{j=0}^{4h-1} |q_{i,j}| \leq 1$, we have $|A_i(x)| = x^h \cdot [x^h(1-x)^h] \cdot |Q_i(x)| \leq 4^{-h}x^h$ on $(0,1)$ since $\sup_{x\in(0,1)} x(1-x) = 1/4$ and $|Q_i(x)| \leq \sum_j |q_{i,j}| \leq 1$. Since $d\Omega_7(x_1, x_2) > 0$ on $(0,1)^2$, each diagonal entry satisfies

$$G^{(7)}_{\mathcal{A}^{(K)}_+}(\zeta(7))_{i,i} = \iint_{(0,1)^2} A_i(x_1)A_i(x_2)\,d\Omega_7(x_1,x_2) \leq 4^{-2h} \iint_{(0,1)^2} x_1^h x_2^h\,d\Omega_7(x_1,x_2) = 4^{-2h}\big(\zeta(7) - H^{(7)}_h\big) \leq 4^{-2h}h^-$$

Applying Hadamard's inequality $\det G \leq \prod_{i=0}^{h-1} G_{i,i}$ to the positive-definite matrix $G^{(7)}_{\mathcal{A}^{(K)}_+}(\zeta(7))$ yields the stated bound. □

6.2. **The Dirichlet beta value $\beta(4)$ and character–Vandermonde sign cancellation.** Whereas Proposition 5.3 constructed a two-variable Beukers pencil for $\beta(4) = L(4, \chi_{-4}) = \sum_{n=0}^\infty (-1)^n(2n+1)^{-4}$, we now analyze the single-variable Hankel architecture for $\beta(4)$ at the full odd-square pole lattice $t = -(2j+1)^2$ ($j \geq 0$). Define the rational moment functional $\mu_X^{(\beta(4))}$ on rational functions of t with simple poles among $\{-(2j+1)^2 : j \geq 0\}$ by setting on monomials t^e ($e \geq 0$)

$$\mu_X^{(\beta(4))}(t^e) := \frac{(-1)^e E_{2e}}{4^{e+1}} \binom{2e+3}{3} \in \mathbb{Q}, \tag{6.3}$$

where $E_{2e} \in \mathbb{Z}$ are the Euler numbers ($\frac{1}{\cosh z} = \sum_{e=0}^\infty \frac{E_{2e}}{(2e)!} z^{2e}$, so $E_0 = 1, E_2 = -1, E_4 = 5, E_6 = -61$), and on simple poles $(t + (2j+1)^2)^{-1}$ ($j \geq 0$) by

$$\mu_X^{(\beta(4))}\left(\frac{1}{t+(2j+1)^2}\right) := (2j+1)^3(-1)^j\big(X - H_j^{(\beta,4)}\big) + r_j^{(\beta)}, \qquad r_j^{(\beta)} := -\frac{1}{6} + \frac{(-1)^j}{2(2j+1)} \in \mathbb{Q}, \tag{6.4}$$

where $H_j^{(\beta,s)} := \sum_{n=0}^{j-1}(-1)^n(2n+1)^{-s}$ ($H_0^{(\beta,s)} := 0$). Set $D_m^{(\mathrm{odd})}(t) := \prod_{j=0}^{m-1}(t + (2j+1)^2)$ and, for $K > N \geq 0$ ($h = K - N$) and $m \geq 1$, define the $h \times h$ Hankel matrix pencil

$$G_K^{(\beta(4))}(X) := \left[\mu_X^{(\beta(4))}\left(\frac{D_N^{(\mathrm{odd})}(t)^{2m} t^{i+j}}{D_K^{(\mathrm{odd})}(t)}\right)\right]_{0\leq i,j<h}, \qquad \Delta_K^{(\beta(4))}(X) := \det G_K^{(\beta(4))}(X) \in \mathbb{Q}[X].$$

Theorem 6.4 (Character–Vandermonde sign cancellation and unconditional hyperbolicity for $\beta(4)$)**.** *Fix integers $K > N \geq 0$ ($h = K - N$) and $m \geq 1$. Then:*

(1) *At $X = \beta(4)$, for every rational function $R(t)$ in the domain of $\mu_X^{(\beta(4))}$,*

$$\mu_{\beta(4)}^{(\beta(4))}(R(t)) = \int_0^\infty R(y^2) w_{\beta(4)}(y)\,dy, \qquad w_{\beta(4)}(y) := -\frac{y^4}{6\pi^2}\frac{d^3}{dy^3}\left(\frac{\pi}{2\cosh(\pi y/2)}\right).$$

Furthermore, coupling $\beta(4)$ with $-\frac{\pi^2}{24}\beta(2)$ via the differential operator $-\frac{4}{\pi^2}\frac{d^3}{dy^3}+\frac{d}{dy}$ yields the strictly positive single-variable weight

$$W_\beta^+(y) := \frac{y^4}{24}\left(-\frac{4}{\pi^2}\frac{d^3}{dy^3}+\frac{d}{dy}\right)\left(\frac{1}{2\cosh(\pi y/2)}\right) = \frac{\pi^2}{16}\frac{y^4\sinh(\pi y/2)}{\cosh^4(\pi y/2)} > 0 \qquad (y>0).$$

(2) *In the Lagrange basis at the odd-square poles $v_j=-(2j+1)^2$ $(N\le j<K)$, the X-derivative matrix $M_K^{(\beta(4))} := [X]G_K^{(\beta(4))}(X)$ factors as $M_K^{(\beta(4))} = V_{\mathrm{odd}}\,\mathrm{diag}\left(\gamma_j^{(\beta(4))}\right)_{N\le j<K}V_{\mathrm{odd}}^T$, where $V_{\mathrm{odd}}=(v_j^i)_{0\le i<h,\,N\le j<K}$ and*

$$\gamma_j^{(\beta(4))} := \frac{(2j+1)^3(-1)^j D_N^{(\mathrm{odd})}(-(2j+1)^2)^{2m-1}}{D_{\mathrm{tail}}^{(\mathrm{odd})\prime}(-(2j+1)^2)}, \qquad D_{\mathrm{tail}}^{(\mathrm{odd})}(t) := \frac{D_K^{(\mathrm{odd})}(t)}{D_N^{(\mathrm{odd})}(t)}.$$

(3) *Because the character sign $\chi_{-4}(2j+1)=(-1)^j$, the Vandermonde derivative sign $\mathrm{sgn}\left(D_{\mathrm{tail}}^{(\mathrm{odd})\prime}(-(2j+1)^2)\right)=(-1)^{j-N}$, and the numerator sign $\mathrm{sgn}\left(D_N^{(\mathrm{odd})}(-(2j+1)^2)^{2m-1}\right)=(-1)^N$ multiply to $(-1)^j(-1)^{j-N}(-1)^N=(-1)^{2j}=+1$, we have $\gamma_j^{(\beta(4))}>0$ for every $N\le j<K$. Consequently, $M_K^{(\beta(4))}$ is strictly positive definite, and all h roots of $\Delta_K^{(\beta(4))}(X)$ are unconditionally real.*

Proof. (1) Integrating y^{2e} against $w_{\beta(4)}(y)$ by parts three times and applying Euler's integral $\int_0^\infty \frac{y^{2e}}{\cosh(\pi y/2)}\,dy = \frac{|E_{2e}|}{2^{2e}}$ recovers (6.3). For $(y^2+(2j+1)^2)^{-1}$, writing $\frac{y^4}{y^2+(2j+1)^2} = y^2-(2j+1)^2+\frac{(2j+1)^4}{y^2+(2j+1)^2}$, integrating by parts three times, and applying the Mittag-Leffler partial fraction expansion $\frac{\pi}{2\cosh(\pi y/2)} = \sum_{n=0}^\infty \frac{(-1)^n(2n+1)}{y^2+(2n+1)^2}$ [11, 4.35.35] yields (6.4) at $X=\beta(4)$. For $W_\beta^+(y)$, setting $z=\pi y/2>0$ and using $\left(1-\frac{d^2}{dz^2}\right)(\mathrm{sech}\,z) = 2\,\mathrm{sech}^3 z$ gives $-\frac{d}{dz}\left(1-\frac{d^2}{dz^2}\right)(\mathrm{sech}\,z) = 6\,\mathrm{sech}^4 z\sinh z>0$.

(2)–(3) Applying Theorem 2.5 to the nodes $v_j=-(2j+1)^2$ $(N\le j<K)$ cancels one factor of $D_N^{(\mathrm{odd})}(t)$ against $D_K^{(\mathrm{odd})}(t)=D_N^{(\mathrm{odd})}(t)D_{\mathrm{tail}}^{(\mathrm{odd})}(t)$, yielding the stated formula for $\gamma_j^{(\beta(4))}$. For each $N\le j<K$, the product $D_N^{(\mathrm{odd})}(-(2j+1)^2)^{2m-1} = \prod_{\nu=0}^{N-1}((2\nu+1)^2-(2j+1)^2)^{2m-1}$ has N negative factors raised to the odd power $2m-1$, giving sign $(-1)^N$. The denominator $D_{\mathrm{tail}}^{(\mathrm{odd})\prime}(-(2j+1)^2) = \prod_{N\le b<K,\,b\ne j}((2b+1)^2-(2j+1)^2)$ has $j-N$ negative factors (those with $N\le b<j$), giving sign $(-1)^{j-N}$. Multiplying by $(-1)^j$ gives

$$\mathrm{sgn}\left(\gamma_j^{(\beta(4))}\right) = (-1)^j\cdot(-1)^{j-N}\cdot(-1)^N = (-1)^{2j} = +1 \qquad (N\le j<K).$$

Since all $\gamma_j^{(\beta(4))}>0$ and $\det V_{\mathrm{odd}} = \prod_{N\le a<b<K}((2a+1)^2-(2b+1)^2)\ne 0$, Sylvester's law of inertia implies that $M_K^{(\beta(4))}$ is real symmetric and strictly positive definite. Factoring $G_K^{(\beta(4))}(X) = (M_K^{(\beta(4))})^{1/2}\left(\widetilde{A}_K+(X-\beta(4))I_h\right)(M_K^{(\beta(4))})^{1/2}$ with $\widetilde{A}_K := (M_K^{(\beta(4))})^{-1/2}G_K^{(\beta(4))}(\beta(4))(M_K^{(\beta(4))})^{-1/2}$ real symmetric proves that all h roots of $\Delta_K^{(\beta(4))}(X)$ are real. □

Because the Euler numbers $E_{2e}\in\mathbb{Z}$ in (6.3) are integers, they have no odd-prime denominators (in contrast to Bernoulli numbers). This yields an unconditional q-adic non-vanishing theorem for $\Delta_K^{(\beta(4))}(0)$.

Theorem 6.5 (Unconditional q-adic non-vanishing for $\Delta_K^{(\beta(4))}(0)$)**.** *Let $q>7$ be a prime satisfying $q\equiv 1 \pmod 8$, and set $m_q := \frac{q-1}{8}\in\mathbb{Z}_{>0}$. Choose $N=4m_q=\frac{q-1}{2}$, $K=5m_q=$*

$\frac{5(q-1)}{8}$, $h = K - N = m_q = \frac{q-1}{8}$, *and* $2m = 2$ *in* $G_K^{(\beta(4))}(X)$. *Then at* $X = 0$, *the rational determinant* $\Delta_K^{(\beta(4))}(0) = \det G_K^{(\beta(4))}(0) \in \mathbb{Q}$ *satisfies the exact* q*-adic valuation*

$$v_q\big(\Delta_K^{(\beta(4))}(0)\big) = -(3m_q - 2) < 0.$$

In particular, $\Delta_K^{(\beta(4))}(0) \neq 0$ *unconditionally for every prime* $q \equiv 1 \pmod 8$ *with* $q > 7$.

Proof. For $j \in [4m_q, 5m_q)$, write $j = 4m_q + r$ with $0 \le r < m_q = \frac{q-1}{8}$. Then $2j+1 = q+2r$, so $(2j+1)^2 \equiv 4r^2 \pmod q$. Since $0 \le 2r < \frac{q-1}{4} < \frac{q}{2}$, the m_q residues $4r^2 \pmod q$ $(0 \le r < m_q)$ are pairwise incongruent modulo q. Hence the Vandermonde matrix V_{odd} on $4m_q \le j < 5m_q$ is $\mathbb{Z}_q$-unimodular ($v_q(\det V_{\mathrm{odd}}) = 0$), and $v_q\big(D_{\mathrm{tail}}^{(\mathrm{odd})\prime}(-(2j+1)^2)\big) = 0$ for every $4m_q \le j < 5m_q$.

Consider the Lagrange-transformed matrix $\widetilde{G}_K^{(\beta(4))}(0) = V_{\mathrm{odd}}^{-1} G_K^{(\beta(4))}(0)(V_{\mathrm{odd}}^T)^{-1}$. Since $E_{2e} \in \mathbb{Z}$ and q is odd, every monomial moment $\mu_0^{(\beta(4))}(t^e)$ in (6.3) lies in $\mathbb{Z}_q$. By Theorem 2.5(1), every off-diagonal entry $\widetilde{G}_K^{(\beta(4))}(0)_{a,b}$ $(a \neq b)$ is a $\mathbb{Z}_q$-linear combination of polynomial moments and therefore lies in $\mathbb{Z}_q$.

On the diagonal $j = 4m_q + r$ $(0 \le r < m_q)$ with $2m - 1 = 1$:

- For $r = 0$ ($j_0 = 4m_q$, where $2j_0 + 1 = q$), every index $0 \le n < j_0$ has $1 \le 2n+1 \le q-2 < q$, so $H_{4m_q}^{(\beta,4)} \in \mathbb{Z}_q$ and $v_q\big(D_N^{(\mathrm{odd})}(-q^2)\big) = 0$. In (6.4) at $X = 0$, the pole value is $-q^3 H_{4m_q}^{(\beta,4)} - \frac{1}{6} + \frac{1}{2q}$, which has exact q-adic valuation -1, giving $v_q\big(\widetilde{G}_K^{(\beta(4))}(0)_{4m_q,4m_q}\big) = -1$.
- For $1 \le r < m_q$ ($j = 4m_q + r$, where $2j+1 = q+2r \in \mathbb{Z}_q^\times$), the sum $H_{4m_q+r}^{(\beta,4)}$ includes the single q-divisible term $n = 4m_q$ ($2n+1 = q$), so $H_{4m_q+r}^{(\beta,4)} \equiv q^{-4} \pmod{\mathbb{Z}_q}$ and the pole value in (6.4) at $X = 0$ has valuation -4. Meanwhile, in $D_N^{(\mathrm{odd})}(-(2j+1)^2) = \prod_{\nu=0}^{4m_q-1}((2\nu+1)^2 - (q+2r)^2)$, the unique index $\nu \in [0, 4m_q)$ with $2\nu + 1 \equiv \pm 2r \pmod q$ is $\nu = 4m_q - r$ (where $2\nu + 1 = q - 2r$, so $(2\nu+1) + (2j+1) = 2q$), giving $v_q\big(D_N^{(\mathrm{odd})}(-(2j+1)^2)\big) = 1$. Thus the diagonal entry at $j = 4m_q + r$ has exact q-adic valuation $-4 + 1 = -3 < 0$.

Because every diagonal entry of $\widetilde{G}_K^{(\beta(4))}(0)$ has strictly negative q-adic valuation (-1 at $r = 0$ and -3 for $1 \le r < m_q$) while all off-diagonal entries lie in $\mathbb{Z}_q$, taking determinants yields $v_q\big(\Delta_K^{(\beta(4))}(0)\big) = -1 - 3(m_q - 1) = -(3m_q - 2) < 0$. □

6.3. **Generalized differential-shift coupling across arbitrary odd weights.** We now extend the differential-shift coupling of Theorem 4.5 to arbitrary finite linear combinations $\sum_{m=1}^k a_m \zeta(2m+1)$ of odd zeta values in both the one-variable positive-definite architecture and the two-variable Beukers Krein-space architecture.

Theorem 6.6 (Generalized multi-weight positive-definite pencil)**.** *Fix an integer* $k \ge 1$ *and an integer tuple* $\mathbf{a} = (a_1, \dots, a_k) \in \mathbb{Z}^k$ *with* $a_k > 0$. *For* $K > N \ge 1$ *(*$h = K - N$*),* $M \ge 1$, *and* $R_{i,j}(t) := \frac{D_N(t)^{2M} t^{i+j}}{D_K(t)}$ *(*$0 \le i, j < h$*), define the* $h \times h$ *Hankel matrix*

$$G_K^{[\mathbf{a}]}(Z) := \left[\sum_{m=1}^k a_m \mu_{X_m}^{(2m+1)}\big((-t)^{k-m} R_{i,j}(t)\big)\right]_{0 \le i,j < h}, \qquad Z := \sum_{m=1}^k a_m X_m,$$

and set $\xi_{\mathbf{a}} := \sum_{m=1}^k a_m \zeta(2m+1) \in \mathbb{R}$. *Then:*

(1) *At $X_m = \zeta(2m+1)$ ($1 \le m \le k$, so $Z = \xi_{\mathbf{a}}$), every entry of $G_K^{[\mathbf{a}]}(\xi_{\mathbf{a}})$ admits the exact integral representation*

$$G_K^{[\mathbf{a}]}(\xi_{\mathbf{a}})_{i,j} = \int_0^\infty \frac{D_N(y^2)^{2M} y^{2(i+j)}}{D_K(y^2)} W_{\mathbf{a}}(y)\, dy, \qquad W_{\mathbf{a}}(y) := 2y^{2k+1} \sum_{l=1}^{\infty} P_{\mathbf{a}}(l) e^{-2\pi l y},$$

where $P_{\mathbf{a}}(l) := \sum_{m=1}^{k} (-1)^{k-m} a_m \frac{(2\pi l)^{2m}}{(2m)!}$.

(2) *Define the open positivity cone $\mathcal{C}_k^+ := \{\mathbf{a} \in \mathbb{R}^k : P_{\mathbf{a}}(l) > 0 \text{ for all } l \ge 1\}$. For every $\mathbf{a} \in \mathcal{C}_k^+ \cap \mathbb{Z}^k$, the weight $W_{\mathbf{a}}(y)$ satisfies the uniform two-sided comparison $c_1(\mathbf{a}) w_{2k+1}(y) \le W_{\mathbf{a}}(y) \le c_2(\mathbf{a}) w_{2k+1}(y)$ on $(0,\infty)$ with $c_1(\mathbf{a}) := \inf_{l \ge 1} \frac{(2k)! P_{\mathbf{a}}(l)}{(2\pi l)^{2k}} > 0$ and $c_2(\mathbf{a}) := \sup_{l \ge 1} \frac{(2k)! P_{\mathbf{a}}(l)}{(2\pi l)^{2k}} < \infty$. Consequently, $G_K^{[\mathbf{a}]}(\xi_{\mathbf{a}})$ is strictly positive definite, $[Z] G_K^{[\mathbf{a}]}(Z) = M_K^{(2k+1)}$, and all h roots of $\Delta_K^{[\mathbf{a}]}(Z) := \det G_K^{[\mathbf{a}]}(Z)$ are unconditionally real, with signature $(\lfloor h/2 \rfloor, \lceil h/2 \rceil)$ across $\xi_{\mathbf{a}}$ (for N and h of the same parity as in Theorem 3.3(3)) and zero-free disk $|r_j - \xi_{\mathbf{a}}| \ge c_1(\mathbf{a}) \|B_K^{(2k+1)}\|^{-1}$.*

(3) *For every prime $p > \max(|a_1|, \dots, |a_k|, 2k+3)$, the pullback identity of Lemma 2.2 gives*

$$\sum_{m=1}^{k} a_m \mu_{X_m}^{(2m+1)}\big((-t)^{k-m} R(t)\big) = \sum_{m=1}^{k} a_m \tau_{X_m}^{(2m+1)}\big(x^{2k+1} R(-x^2)\big),$$

and $\Delta_K^{[\mathbf{a}]}(Z)$ clears to $\mathbb{Z}[Z]$ with the exact same quadratic p-adic multiplier rate $A_M^{(2k+1)}$ (up to $e^{O_{\mathbf{a}}(K)}$) as the pure top-weight polynomial $\Delta_K^{(2k+1)}(X)$.

Proof. (1) Multiplying $R_{i,j}(t)$ by $(-t)^{k-m}$ scales its residue at $t = -j^2$ ($N < j \le K$) by j^{2k-2m}. Applying $\mu_{X_m}^{(2m+1)}$ via (2.2)–(2.3) gives $j^{2k}(X_m - H_j^{(2m+1)}) + r_{j,k,m}$ with $r_{j,k,m} \in \mathbb{Q}$ independent of X_m. Summing over $1 \le m \le k$ against a_m produces $j^{2k}(Z - \sum_{m=1}^k a_m H_j^{(2m+1)}) + \sum_{m=1}^k a_m r_{j,k,m}$, which is affine linear in $Z = \sum_{m=1}^k a_m X_m$ with Z-derivative matrix $M_K^{(2k+1)}$. At $X_m = \zeta(2m+1)$, substituting $t = y^2$ into (3.2) maps $(-t)^{k-m}$ to $(-1)^{k-m} y^{2k-2m}$, yielding $W_{\mathbf{a}}(y) = \sum_{m=1}^k (-1)^{k-m} a_m y^{2k-2m} w_{2m+1}(y) = 2y^{2k+1} \sum_{l=1}^\infty P_{\mathbf{a}}(l) e^{-2\pi l y}$.

(2) Since $a_k > 0$, $\lim_{l \to \infty} \frac{(2k)! P_{\mathbf{a}}(l)}{(2\pi l)^{2k}} = a_k > 0$. On $\mathcal{C}_k^+$, $P_{\mathbf{a}}(l) > 0$ for all $l \ge 1$, so $0 < c_1(\mathbf{a}) \le c_2(\mathbf{a}) < \infty$ and $c_1(\mathbf{a}) w_{2k+1}(y) \le W_{\mathbf{a}}(y) \le c_2(\mathbf{a}) w_{2k+1}(y)$ for all $y > 0$. This implies $c_1(\mathbf{a}) G_K^{(2k+1)}(\zeta(2k+1)) \preceq G_K^{[\mathbf{a}]}(\xi_{\mathbf{a}}) \preceq c_2(\mathbf{a}) G_K^{(2k+1)}(\zeta(2k+1))$ in the Loewner order. Since $[Z] G_K^{[\mathbf{a}]}(Z) = M_K^{(2k+1)}$, Theorems 3.1 and 3.3(3) yield real-rootedness, the universal signature, and $\|B_K^{[\mathbf{a}]}\| \le c_1(\mathbf{a})^{-1} \|B_K^{(2k+1)}\|$.

(3) Since $x^{2m+1}((-(-x^2))^{k-m} R(-x^2)) = x^{2k+1} R(-x^2)$ for every $1 \le m \le k$, Lemma 2.2 gives the stated pullback identity. For $p > \max(|a_1|, \dots, |a_k|, 2k+3)$, the local distribution prefactor p^{-2m} and harmonic valuation $v_p(H_j^{(2m+1)}) \ge -(2m+1)$ for $m < k$ are strictly dominated by p^{-2k} and $-(2k+1)$ at top weight $m = k$, while in the outer range $K/3 < p \le K$ the von Staudt–Clausen denominators of $\mu^{(2m+1)}(t^{e+k-m})$ occur for at most $O_k(1)$ indices $e = i + j - K + 2MN$, contributing a perturbation of rank $O_k(1)$ and losing at most $\prod_{p \le K} p^{O_k(1)} = e^{O_k(K)}$. □

To remove the positivity cone restriction $\mathbf{a} \in \mathcal{C}_k^+$ completely, we pass to the two-variable Beukers architecture across arbitrary non-zero integer tuples $\mathbf{a} \in \mathbb{Z}^k \setminus \{\mathbf{0}\}$.

Theorem 6.7 (Universal multi-weight two-variable Beukers Krein decomposition)**.** *Fix any integer $k \geq 1$ and any non-zero integer tuple $\mathbf{a} = (a_1, \ldots, a_k) \in \mathbb{Z}^k \setminus \{\mathbf{0}\}$. Define the coupled signed measure on $(0,1)^2$ by*

$$d\Omega_{k,\mathbf{a}}(x_1, x_2) := \sum_{m=1}^{k} a_m \frac{(-\log(x_1 x_2))^{2m-1}}{(2m)!(1 - x_1 x_2)}\, dx_1\, dx_2,$$

and for any h-dimensional subspace $\mathcal{A} = \{A_0(x), \ldots, A_{h-1}(x)\} \subset \mathbb{Q}[x]_{\leq K}$ with coefficient matrix $C \in M_{h\times(K+1)}(\mathbb{Q})$, define the $h \times h$ symmetric matrix pencil $G_{\mathcal{A}}^{[\mathbf{a}]}(Z)$ in $Z := \sum_{m=1}^{k} a_m X_m$ by

$$G_{\mathcal{A}}^{[\mathbf{a}]}(Z)_{i,j} := \sum_{r=0}^{K} C_{i,r} C_{j,r} \left(Z - \sum_{m=1}^{k} a_m H_r^{(2m+1)} \right) + \sum_{0 \leq r \neq s \leq K} C_{i,r} C_{j,s} \sum_{m=1}^{k} \frac{a_m}{2m} \frac{H_r^{(2m)} - H_s^{(2m)}}{r - s}.$$

Set $\xi_{\mathbf{a}} := \sum_{m=1}^{k} a_m \zeta(2m+1)$ and $\Delta_{\mathcal{A}}^{[\mathbf{a}]}(Z) := \det G_{\mathcal{A}}^{[\mathbf{a}]}(Z) \in \mathbb{Q}[Z]$. Then:

(1) *The Z-derivative matrix is $[Z]G_{\mathcal{A}}^{[\mathbf{a}]}(Z) = CC^T \succ 0$, and all h roots of $\Delta_{\mathcal{A}}^{[\mathbf{a}]}(Z)$ are unconditionally real for every $\mathbf{a} \in \mathbb{Z}^k \setminus \{\mathbf{0}\}$.*

(2) *For every $f \in L^2(0,1)$, setting for $n \geq 0$ and $0 \leq \nu \leq k-1$*

$$\Phi_{n,\nu}^{\pm}(x; \mathbf{a}) := x^n \left[\frac{(-\log x)^{\nu}}{\nu!} \pm \sum_{m=\max(1,\nu+1)}^{k} \frac{a_m}{2m} \frac{(-\log x)^{2m-1-\nu}}{(2m-1-\nu)!} \right],$$

the quadratic form at $Z = \xi_{\mathbf{a}}$ satisfies the exact Krein decomposition

$$\iint_{(0,1)^2} f(x_1) f(x_2)\, d\Omega_{k,\mathbf{a}}(x_1, x_2) = \frac{1}{2} \sum_{n=0}^{\infty} \sum_{\nu=0}^{k-1} \left[\left(\int_0^1 f(x) \Phi_{n,\nu}^{+}(x; \mathbf{a})\, dx \right)^2 - \left(\int_0^1 f(x) \Phi_{n,\nu}^{-}(x; \mathbf{a})\, dx \right)^2 \right],$$

yielding explicit h-dimensional subspaces $\mathcal{A}_+, \mathcal{A}_- \subset \mathbb{Q}[x]_{\leq K}$ such that $G_{\mathcal{A}_+}^{[\mathbf{a}]}(\xi_{\mathbf{a}}) \succ 0$ (all roots $r_j < \xi_{\mathbf{a}}$) and $G_{\mathcal{A}_-}^{[\mathbf{a}]}(\xi_{\mathbf{a}}) \prec 0$ (all roots $r_j > \xi_{\mathbf{a}}$).

(3) *Let $m_* := \max\{1 \leq m \leq k : a_m \neq 0\}$, and suppose $\sum_{m=1}^{k} a_m \zeta(2m+1) = c \in \mathbb{Z}$. For every prime $q > \max(|a_1|, \ldots, |a_k|, 2m_* + 3)$, setting $K = 2q - 1$, $h = q$, and $\mathcal{A}_q := \{x^{\rho} + \kappa_{m_*} x^{\rho+q} : 0 \leq \rho < q\}$ (with $\kappa_{m_*} = 1$ if $m_* \geq 2$ and $\kappa_1 = 2$), we have*

$$v_q\big(\Delta_{\mathcal{A}_q}^{[\mathbf{a}]}(c)\big) = -(2m_* + 1)q < 0,$$

and in particular $\Delta_{\mathcal{A}_q}^{[\mathbf{a}]}(c) \neq 0$.

Proof. (1) Differentiating $G_{\mathcal{A}}^{[\mathbf{a}]}(Z)$ with respect to Z gives $[Z]G_{\mathcal{A}}^{[\mathbf{a}]}(Z) = CC^T \succ 0$, and the proof of Theorem 5.2(1) yields real-rootedness.

(2) Expanding $(-\log(x_1 x_2))^{2m-1} = \sum_{\nu=0}^{2m-1} \binom{2m-1}{\nu} (-\log x_1)^{\nu} (-\log x_2)^{2m-1-\nu}$ and $(1 - x_1 x_2)^{-1} = \sum_{n=0}^{\infty} x_1^n x_2^n$, then pairing ν with $2m-1-\nu$ for $0 \leq \nu \leq m-1$ via $U_1 V_2 + V_1 U_2 = \frac{1}{2}[(U_1 + V_1)(U_2 + V_2) - (U_1 - V_1)(U_2 - V_2)]$ gives the stated Krein identity. Since $a_{m_*} \neq 0$, the functions $\{\Phi_{n,0}^{+}, \Phi_{n,\nu}^{-}\}_{n \geq 0,\, 0 \leq \nu \leq k-1}$ are linearly independent in $L^2(0,1)$, and the projection argument of Theorem 5.2(3) produces $\mathcal{A}_{\pm}$.

(3) For $\rho \neq \sigma$, every $r \in \{\rho, \rho + q\}$ and $s \in \{\sigma, \sigma + q\}$ satisfies $r - s \in \mathbb{Z}_q^{\times}$, so all off-diagonal entries have q-adic valuation $\geq -2m_*$. On the diagonal $\rho = \sigma$ at $Z = c$, the top active weight $2m_* + 1$ contributes $(\frac{\kappa_{m_*}}{m_*} - \kappa_{m_*}^2) a_{m_*} q^{-(2m_*+1)}$ (which equals $-\frac{m_*-1}{m_*} a_{m_*} q^{-(2m_*+1)}$ for $m_* \geq 2$ with $\kappa_{m_*} = 1$, and $-2a_1 q^{-3}$ for $m_* = 1$ with $\kappa_1 = 2$), strictly dominating all lower weights

$m < m_*$ (which have valuation $\geq -(2m_*-1)$). Thus $q^{2m_*+1}G^{[\mathbf{a}]}_{\mathcal{A}_q}(c)$ reduces modulo $q\mathbb{Z}_q$ to an invertible scalar multiple of I_q, giving $v_q(\Delta^{[\mathbf{a}]}_{\mathcal{A}_q}(c)) = -(2m_*+1)q < 0$. □

Specializing Theorem 6.7 to $k=2$ ($d\Omega^{(3,5)}_{a,b}(x_1,x_2) = \frac{-\log(x_1x_2)}{24(1-x_1x_2)}[b(-\log(x_1x_2))^2+12a]\,dx_1\,dx_2$) gives unconditional real-rootedness, Krein subspaces $\mathcal{A}_\pm$, and q-adic non-vanishing $v_q(\Delta^{[a,b]}_{\mathcal{A}_q}(c)) = -5q < 0$ for $\xi_{a,b} = b\zeta(5)+a\zeta(3)$ across all $(a,b)\in\mathbb{Z}^2\setminus\{(0,0)\}$ with $b\neq 0$.

6.4. **Inhomogeneous linear independence, multi-pole parity laws, and universal height conservation.** We conclude this section by recording the inhomogeneous three-term linear independence corollary of Theorem 4.5, the triple-pole and multi-weight extensions of Theorem 5.7, and the universal potential-gap conservation law governing the height-to-decay ratio $\Lambda_{\text{height}}/\epsilon$.

Corollary 6.8 (Unconditional three-term linear independence cone for $\{1,\zeta(3),\zeta(5)\}$)**.** *For every non-zero integer triple $(c,a,b)\in\mathbb{Z}^3\setminus\{(0,0,0)\}$ satisfying either $b=0$ or $b\neq 0$ and $\frac{a}{b} < \frac{\pi^2}{3}$, the linear form $c+a\zeta(3)+b\zeta(5)$ is strictly non-zero. Furthermore, for any coprime integers (a_0,b_0) with $b_0>0$ and $a_0 < \frac{\pi^2}{3}b_0$, there exists an effectively computable $Q_0(a_0,b_0)>0$ such that for every integer $q\geq Q_0(a_0,b_0)$ and every $c\in\mathbb{Z}$, writing $(a,b)=q(a_0,b_0)$, we have*

$$|c+a\zeta(3)+b\zeta(5)| > q^{-25686}.$$

Proof. If $b=0$, $c+a\zeta(3)\neq 0$ by Apéry's theorem [1]. If $b\neq 0$ and $a/b<\pi^2/3$, set $\varepsilon=\operatorname{sgn}(b)\in\{\pm1\}$, $\tilde{b}=\varepsilon b>0$, and $\tilde{a}=\varepsilon a<\frac{\pi^2}{3}\tilde{b}$. By Theorem 4.5, $\tilde{b}\zeta(5)+\tilde{a}\zeta(3)\notin\mathbb{Q}$, so it cannot equal $-\varepsilon c\in\mathbb{Z}$. For $(a,b)=q(a_0,b_0)$, multiplying $|b_0\zeta(5)+a_0\zeta(3)-(-c/q)|>q^{-25687}$ (Theorem 4.5) by q and using the strict margin $25687-25686.161>0$ from the proof of Theorem 4.4 yields $|c+a\zeta(3)+b\zeta(5)|>q^{-25686}$ for $q\geq Q_0(a_0,b_0)$. □

Theorem 6.9 (Triple-pole and multi-weight Hankel determinants)**.** (1) (Triple-pole Hankel architecture and consecutive-weight parity law) *For $n\geq 1$, let $R^{(3)}_n(t) := \frac{((6n)!)^3\prod_{j=1}^n(t-j+1)^6}{\prod_{k=0}^{6n}(t+k)^3}$. Under the polylogarithmic residue functional $\mathcal{L}^{(s)}_{\text{poly}}((t+k)^{-r}) := (-1)^{r-1}\binom{s+r-2}{r-1}(\zeta(s+r-1)-H_k^{(s+r-1)})$ ($r=1,2,3$), the Laurent expansion of $R^{(3)}_n(t)P(t)$ ($\deg P<6n$) at $t=-k$ couples three consecutive weights $\{1,\zeta(s),\zeta(s+1),\zeta(s+2)\}$, and cannot annihilate $\zeta(s+1)$ identically across $\mathbb{Q}[t]_{<6n}$ because the leading pole coefficient $C^{(3)}_{n,k} = (-1)^k\binom{6n}{k}^3\binom{n+k}{k}^6 n!\neq 0$. Moreover, if $a_0\zeta(s)+a_1\zeta(s+1)+a_2\zeta(s+2)=c$ ($a_0,a_1,a_2,c\in\mathbb{Z}$, $a_2\neq 0$, $s\in\{3,4\}$), then for every prime $q>\max(|a_2|,13)$ with $q\equiv 1 \pmod 9$, setting $m=(q-1)/9$ and $n=2m$, the $3m\times 3m$ triple-pole Hankel determinant $J^{(3)}_{2m} := \det_{0\leq i,j<3m}[\mathcal{L}^{(3)}_n(t^{i+j})]$ satisfies $v_q(J^{(3)}_{2m}) = -3(s+3)m<0$, and in particular $J^{(3)}_{2m}\neq 0$.*

(2) (Simultaneous multi-weight double-pole non-vanishing) *Let $2\leq s_1<s_2<\cdots<s_d$ be any $d\geq 1$ distinct integers, and suppose $\sum_{r=1}^d a_r\zeta(s_r)=c$ with $a_1,\dots,a_d,c\in\mathbb{Z}$ and $a_d\neq 0$. In the double-pole functional $\mathcal{L}_n(P)$ of Theorem 5.7, replace U_k, V_k by*

$$U_k^{[\mathbf{s}]} := c-\sum_{r=1}^d a_r H_k^{(s_r)}, \qquad V_k^{[\mathbf{s}]} := \sum_{r=1}^d \frac{a_r}{s_r k^{s_r+1}} \quad (V_0^{[\mathbf{s}]}:=0),$$

and set $J_n^{[\mathbf{s}]} := \det_{0\le i,j<3n}[\mathcal{L}_n^{[\mathbf{s}]}(t^{i+j})] \in \mathbb{Q}$. *Then for every prime* $q > \max(|a_1|,\dots,|a_d|, s_d+5)$ *satisfying* $q \equiv 1 \pmod 9$, *setting* $m = (q-1)/9$ *and* $n = 2m$, *we have*

$$v_q\big(J_{2m}^{[\mathbf{s}]}\big) = -6s_d m = -3s_d n < 0,$$

and in particular $J_{2m}^{[\mathbf{s}]} \neq 0$ *(specifically* $v_q(J_{2m}^{[3,5,6]}) = -36m < 0$ *for* $\{1, \zeta(3), \zeta(5), \zeta(6)\}$*).*

Proof. (1) Near $t = -k+\varepsilon$, expanding $(t+k)^3 R_n^{(3)}(t)P(t)$ gives the $(t+k)^{-2}$ coefficient $\sum_{k=0}^{6n}(C_{n,k}^{(2)}P(-k) + C_{n,k}^{(3)}P'(-k))$, which cannot vanish identically on $\mathbb{Q}[t]_{<6n}$ since $C_{n,k}^{(3)} \neq 0$. At $q \equiv 1 \pmod 9$ ($m = (q-1)/9, n = 2m$), passing to the 3-jet Hermite basis at $\{-q,\dots,-(q+m-1)\}$ and scaling by $\mathrm{diag}_{0\le r<m}(q^{(s+4)/2}, q^{(s+3)/2}, q^{(s+2)/2})$ reduces each 3×3 diagonal block modulo q to a lower anti-triangular matrix with anti-diagonal entries $-a_2 C_{n,q+r}^{(3)}/n! \not\equiv 0 \pmod q$, giving $v_q(J_{2m}^{(3)}) = -3(s+3)m < 0$.

(2) For $k = q+\nu \in [q, 6n]$ ($0 \le \nu < 3m$), the highest weight $s_d > s_{d-1} > \cdots > s_1$ strictly dominates the q-adic valuation of $U_{q+\nu}^{[\mathbf{s}]} \equiv -a_d q^{-s_d} \pmod{q^{-(s_d-1)}\mathbb{Z}_q}$, while $v_q(V_{q+\nu}^{[\mathbf{s}]}) \ge -(s_d+1)$. Repeating the Hermite block diagonalization of Theorem 5.7(1) with $D_q := \mathrm{diag}_{0\le\nu<3m}(q^{(s_d+1)/2}, q^{(s_d-1)/2})$ yields $\widetilde{M}^{(\nu)} \equiv \begin{pmatrix} * & -a_d A_{n,q+\nu}/n! \\ -a_d A_{n,q+\nu}/n! & 0 \end{pmatrix} \pmod{q\mathbb{Z}_q}$, whence $v_q(\det\widetilde{M}) = 0$ and $v_q(J_{2m}^{[\mathbf{s}]}) = -6s_d m < 0$. □

Finally, we explain why the quadratic height bound $\Lambda_{\mathrm{height}} > \epsilon$ in Theorem 4.6(2) is an exact potential-theoretic invariant of one-variable Hankel pencils with negative poles, and prove that the two-variable Beukers architecture identically eliminates the quadratic growth of the derivative determinant.

Theorem 6.10 (Universal potential-gap identity and vanishing of quadratic derivative growth in two variables

(Potential-gap conservation law for one-variable Hankel pencils) *Let* $G_K^{(s)}(X) = G_K^{(s)}(\zeta(s)) + (X-\zeta(s))M_K^{(s)}$ *be a one-variable Hankel pencil of degree* $h = \lambda K$ *built from a rational kernel* $W_K(t)/D_K(t)$ *with simple poles at* $v_a = -a^2$ ($N < a \le K$) *and positive weight* $W_K(t) > 0$ *on* $(0,\infty)$, *normalized by* $m_K S_K > 0$ *so that* $Q_K(X) := m_K S_K \det G_K^{(s)}(X) \in \mathbb{Z}[X]$ *has decay rate* $\epsilon := \lim_{K\to\infty}(-K^{-2}\log Q_K(\zeta(s))) > 0$. *Then the leading-coefficient height rate* $\Lambda_{\mathrm{lead}} := \lim_{K\to\infty} K^{-2}\log|[X^h]Q_K(X)| \le \Lambda_{\mathrm{height}}$ *satisfies the exact identity*

$$\Lambda_{\mathrm{lead}} + \epsilon = \mathcal{I}_B := \lim_{K\to\infty}\frac{1}{K^2}\log|\det B_K^{(s)}| = \mathcal{E}(\mu_{\mathrm{eq}}) - \mathcal{E}_{\mathrm{poles}}(\nu_{\mathrm{unif}}),$$

where $\mathcal{E}(\mu_{\mathrm{eq}})$ *is the minimum weighted logarithmic energy on* $[0,\infty)$ *and* $\mathcal{E}_{\mathrm{poles}}(\nu_{\mathrm{unif}})$ *is the logarithmic energy on the negative pole lattice. Moreover, multiplying* $W_K(t)$ *by* $P_K(t)^2$ *for any polynomial family* $P_K(t)$ *with positive real roots and empirical measure* σ_P *on* $[0,\alpha^2]$ *satisfies* $\mathcal{I}_B(P_K) \ge \mathcal{I}_B(1) + 2\lambda\big(\int_\alpha^1 U^{\sigma_P}(-u^2)\,\frac{du}{\lambda} - \int U^{\sigma_P}(t)\,d\mu_{\mathrm{eq}}(t)\big) > \mathcal{I}_B(1)$, *and any integer multiple* $\widetilde{Q}_K(X) = c_K P_K^{(5)}(X) \in \mathbb{Z}[X]$ ($c_K \in \mathbb{Z}_{>0}$) *satisfies* $\widetilde{\Lambda}_{\mathrm{height}} + \tilde{\epsilon} \ge \mathcal{I}_B^{(5)} \ge 0.23310$, *whence* $\widetilde{\Lambda}_{\mathrm{height}}/\tilde{\epsilon} > 581 > 1$.

(2) (Vanishing of quadratic derivative-determinant growth in the two-variable Beukers architecture) *For any odd integer* $s = 2k+1 \ge 3$, *define the two-variable Beukers measure* $d\Omega_s(x_1,x_2) := \frac{1}{(2k)!}\frac{(-\log(x_1x_2))^{2k-1}}{1-x_1x_2}\,dx_1\,dx_2$ *on* $(0,1)^2$. *Let* $h = \lfloor\lambda K\rfloor$ ($\lambda \in (0,1)$), *and let* $\mathcal{A}_K = \{A_0(x),\dots,A_{h-1}(x)\} \subset \mathbb{Z}[x]_{\le K}$ *be any sequence of* h *linearly independent integer polynomials whose coefficient matrix* $C_K \in M_{h\times(K+1)}(\mathbb{Z})$ *has polynomially*

bounded height $\max_{i,r}|C_{i,r}| \le K^A$ *($A \ge 0$ fixed). Then the X-derivative determinant* $M^{(s)}_{\mathcal{A}_K} = [X]G^{(s)}_{\mathcal{A}_K}(X) = C_K C_K^T$ *satisfies*

$$1 \le \det M^{(s)}_{\mathcal{A}_K} = \det(C_K C_K^T) \le (K+1)^h K^{2Ah},$$

and hence its quadratic exponential growth rate vanishes identically:

$$\lim_{K\to\infty} \frac{1}{K^2}\log\det M^{(s)}_{\mathcal{A}_K} = \lim_{K\to\infty}\frac{1}{K^2}\log\det(C_K C_K^T) = 0.$$

Proof. (1) Since $Q_K(X) = Q_K(\zeta(s))\det(I_h + (X-\zeta(s))B_K^{(s)})$, comparing the coefficient of X^h on both sides gives $|[X^h]Q_K(X)| = Q_K(\zeta(s))|\det B_K^{(s)}|$. Taking $K^{-2}\log$ yields $\Lambda_{\mathrm{lead}} + \epsilon = \mathcal{I}_B$. By Andréief's identity, $\lim_{K\to\infty} K^{-2}\log(S_K \det G_K^{(s)}(\zeta(s))) = -\mathcal{E}(\mu_{\mathrm{eq}}) + C_*$, while by Theorem 2.5, $\lim_{K\to\infty} K^{-2}\log(S_K|\det M_K^{(s)}|) = -\mathcal{E}_{\mathrm{poles}}(\nu_{\mathrm{unif}}) + C_*$; subtracting gives $\mathcal{I}_B = \mathcal{E}(\mu_{\mathrm{eq}}) - \mathcal{E}_{\mathrm{poles}}(\nu_{\mathrm{unif}})$. Multiplying $W_K(t)$ by $P_K(t)^2$ adds $2\int_\alpha^1 U^{\sigma_P}(-u^2)\,du$ to $K^{-2}\log|\det M_K^{(s)}|$ while increasing $K^{-2}\log\det G_K^{(s)}(\zeta(s))$ by at most $2\lambda\int U^{\sigma_P}(t)\,d\mu_{\mathrm{eq}}(t)$ by the extremal property of μ_{eq}. For $\widetilde{Q}_K(X) = c_K P_K^{(5)}(X)$, $H(\widetilde{Q}_K) \ge |[X^h]\widetilde{Q}_K(X)| = \widetilde{Q}_K(\zeta(5))|\det B_K^{(5)}|$, so $\widetilde{\Lambda}_{\mathrm{height}} + \tilde{\epsilon} \ge \mathcal{I}_B^{(5)} \ge 0.23310$ by Theorem 4.6(1); since $\tilde{\epsilon} \le 0.0004$, $\widetilde{\Lambda}_{\mathrm{height}}/\tilde{\epsilon} \ge 0.2327/0.0004 > 581$.

(2) Since $C_K \in M_{h\times(K+1)}(\mathbb{Z})$ has rank h, $M^{(s)}_{\mathcal{A}_K} = C_K C_K^T \in M_h(\mathbb{Z})$ is a symmetric positive-definite integer matrix, so $\det(C_K C_K^T) \ge 1$. Each diagonal entry satisfies $(C_K C_K^T)_{i,i} = \sum_{r=0}^K C_{i,r}^2 \le (K+1)K^{2A}$. By Hadamard's inequality for positive-definite matrices, $\det(C_K C_K^T) \le \prod_{i=0}^{h-1}(C_K C_K^T)_{i,i} \le (K+1)^h K^{2Ah} = e^{O(K\log K)}$. Dividing $\log\det(C_K C_K^T)$ by K^2 and letting $K\to\infty$ gives 0, eliminating the $+2.5572K^2$ Vandermonde and factorial derivative growth of (**??**). □

6.5. **Exact p-adic obstructions, sign-flip kernels, and central-factorial secant operators for $\beta(2k)$.** We now carry out the quantitative archimedean and p-adic asymptotic analysis of the single-variable Dirichlet beta Hankel pencil across all even weights $2k$ ($k \ge 1$), resolve the sign-change bottleneck of $w_{\beta(4)}(u)$ in full dimension h, and establish an exact central-factorial differential identity generating strictly positive secant weights with accelerated exponential decay and unilateral $(h,0)$ root signature for all $k \ge 1$.

Theorem 6.11 (The $p^{-(2k-1)}$ distribution growth law and 2-adic polynomial-quotient obstruction for $\beta(2k)$)**.** *Consider the single-variable odd-square Hankel construction $G_K^{(\beta(2k),\mathrm{odd})}(X)$ on the kernel $D_N^{(\mathrm{odd})}(t)^{2m}/D_K^{(\mathrm{odd})}(t)$ with $h = K - N = \lambda K$ and $N = \alpha K$ ($\alpha + \lambda = 1$, $m \ge 1$).*

(1) (Linear growth of the odd-prime denominator exponent in k) *Under the odd-pole pullback $\mu_X^{(\beta(2k),\mathrm{odd})}(R(t)) = \tau_X^{(\beta,2k)}(x^{2k-1}R(-x^2))$ at poles $x = 2r+1$ ($r \in \mathbb{Z}$), the Euler-polynomial multiplication identity $E_n(px) = p^n\sum_{a=0}^{p-1}(-1)^a E_n(x + \frac{a}{p})$ for every odd prime $p \ge 3$ induces a local p-adic distribution relation with prefactor $p^{-(2k-1)}$ across the residue classes $2r+1 \equiv 2a+1 \pmod{2p}$. In the inner prime range $K/M < p \le K/3$ ($x = K/p \ge 3$), each nonzero square class $a^2 \pmod p$ ($1 \le a \le \frac{p-1}{2}$) contains $l_K^{(\mathrm{odd})}(a) = 2x + O(1)$ poles of $D_K^{(\mathrm{odd})}(t)$ and $l_N^{(\mathrm{odd})}(a) = 2\alpha x + O(1)$ roots of $D_N^{(\mathrm{odd})}(t)$, with local half-weights shifted by $-\frac{2k-1}{2}$:*

$$w_{a,i}^{(\beta,2k)} = i + m\,l_N^{(\mathrm{odd})}(a) - \left\lceil \frac{l_K^{(\mathrm{odd})}(a) + 2k - 1}{2} \right\rceil \qquad (0 \le i < L_a := l_K^{(\mathrm{odd})}(a) - l_N^{(\mathrm{odd})}(a)).$$

Consequently, for fixed $(\alpha, 2m)$, the odd-prime denominator exponent $A_M^{(\beta,2k)}$ grows strictly and linearly with $k \geq 1$:

$$A_M^{(\beta,2k)} - A_M^{(\beta,2)} = 2(k-1)\lambda \int_3^M \frac{dx}{x^2} + O\left(\frac{k}{M^2}\right) = \frac{2(k-1)\lambda}{3} + O\left(\frac{k}{M}\right) > 0,$$

whereas under the macroscopic scaling $t = 4K^2x$ ($y = 2K\sqrt{x}$), the archimedean upper-bound rate

$$\overline{U}^{(\beta,2k)} = \overline{U}^{(\beta)}(\alpha, 2m) = \sup_{\substack{\mu \geq 0 \\ \mu(\mathbb{R}_{\geq 0}) = \lambda}} \left[\iint \log|x-z| \, d\mu(x) \, d\mu(z) - \int_0^\infty \left(\pi\sqrt{x} + J_1(x) - 2mJ_\alpha(x)\right) d\mu(x) \right] + 2(2m-1$$

is independent of k. In particular, $A_M^{(\beta,2k)} + \overline{U}^{(\beta,2k)} \to +\infty$ as $k \to \infty$.

(2) (The 2-adic polynomial-quotient obstruction at $p = 2$) *Under the substitution $t = 4w-1$ ($w = (t+1)/4$), every odd-square factor satisfies $t + (2j+1)^2 = 4(w + j(j+1))$ with $j(j+1) \in 2\mathbb{Z}$. In the unitriangular 2-adic basis $\phi_i(t) := (t+1)^i = (4w)^i$, the pole part of $\mu_X^{(\beta(2k),\mathrm{odd})}(D_N^{(\mathrm{odd})}(t)^{2m}\phi_i(t)\phi_j(t)/D_K^{(\mathrm{odd})}(t))$ carries the 2-adic factor $4^{2mN+i+j-K+1}$. However, for every entry (i,j) with $d_{i,j} := 2mN+i+j-K \geq 0$, the polynomial quotient $Q_{i,j}(t) \in \mathbb{Z}[t]$ in the Euclidean division $D_N^{(\mathrm{odd})}(t)^{2m}(t+1)^{i+j} = Q_{i,j}(t)D_K^{(\mathrm{odd})}(t) + R_{i,j}(t)$ satisfies*

$$Q_{i,j}(t) = 4^{d_{i,j}}\widetilde{Q}_{i,j}\left(\frac{t+1}{4}\right) = (t+1)^{d_{i,j}} + \sum_{e=0}^{d_{i,j}-1} 4^{d_{i,j}-e}\tilde{q}_{i,j;e}(t+1)^e \qquad (\tilde{q}_{i,j;e} \in \mathbb{Z}).$$

Since $E_0 = 1$ and $E_{2r} \equiv 1 \pmod 2$ for all $r \geq 1$, we have $\mu_X^{(\beta(2k),\mathrm{odd})}((t+1)^e) = \sum_{r=0}^e \binom{e}{r}\mu_X^{(\beta(2k),\mathrm{odd})}(t^r) \equiv E_0 = 1 \pmod 2$ for all $e \geq 0$ whenever k is a power of 2 (and at $e = 0$ for all $k \geq 1$), so $v_2(\mu_X^{(\beta(2k),\mathrm{odd})}(Q_{i,j}(t))) = 0$ whenever $d_{i,j} = 0$, preventing the pole factor $4^{(2m-1)\alpha\lambda K^2}$ from dividing the polynomial part at $p = 2$.

Proof. (1) Since $\gcd(2,p) = 1$, as j runs over $\{0, 1, \ldots, K-1\}$, the odd integers $2j+1 \in \{1, 3, \ldots, 2K-1\}$ traverse every residue class modulo p with uniform density $K/p + O(1)$, so each nonzero square class $\pm a \pmod p$ ($1 \leq a \leq (p-1)/2$) contains $l_K^{(\mathrm{odd})}(a) = 2K/p + O(1) = 2x + O(1)$ poles. Writing $r = a + pm$ ($0 \leq a < p$, $m \geq 0$), the odd integer is $2r+1 = (2a+1) + 2pm$, with $(-1)^r = (-1)^a(-1)^m$ since p is odd. As r advances by p, the cumulative sum $H_r^{(\beta,2k)} = \sum_{n=0}^{r-1}(-1)^n(2n+1)^{-2k}$ crosses the unique p-divisible term $n = a_0 + pm$ ($a_0 = (p-1)/2$, where $2n+1 = p(2m+1)$ and $(-1)^{a_0} = \chi_{-4}(p)$), jumping by $\chi_{-4}(p)p^{-2k}(-1)^m(2m+1)^{-2k}$. Combined with the Jacobian $(2p(z-m))^{-1}$ of $x = (2a+1) + 2pz$, the local principal-part functional carries prefactor $p^{-(2k-1)}$, shifting each half-weight $w_{a,i}^{(\beta,2k)}$ by $-\frac{2k-1}{2}$. Summing $2(w_{a,i}^{(\beta,2k)} - w_{a,i}^{(\beta,2)}) = -(2k-2)$ over $0 \leq i < L_a \approx 2\lambda x$ and $1 \leq a \leq (p-1)/2$ subtracts $2(k-1)\lambda xp$ from γ_p^{in}, which adds $2(k-1)\lambda x$ to $R(x) = -\Gamma(x) - \mathcal{N}(x)$ and adds $\int_3^M \frac{2(k-1)\lambda x}{x^3}\,dx = \frac{2(k-1)\lambda}{3} + O(k/M)$ to $A_M^{(\beta,2k)}$. On the archimedean side, substituting $t_i = 4K^2x_i$ ($y_i = 2K\sqrt{x_i}$) into Andréief's identity factors out $(4K^2)^{h(h-1)+h(2mN-K)} = 4^{(2m-1)\alpha\lambda K^2 + O(K)}K^{2h(h-1)+2h(2mN-K)}$, while $e^{-\frac{\pi}{2}y_i} = e^{-\pi K\sqrt{x_i}}$ and $y_i^{2k-1} = (2K\sqrt{x_i})^{2k-1} = e^{O(kK\log K)} = e^{o(K^2)}$, giving the k-independent rate $\overline{U}^{(\beta)}(\alpha, 2m)$.

(2) Substituting $t = 4w-1$ gives $t + (2j+1)^2 = 4w - 1 + 4j^2 + 4j + 1 = 4(w + j(j+1))$. In $\mathbb{Z}[w]$, dividing the monic polynomial $\widetilde{P}_{i,j}(w) := (\prod_{\nu=0}^{N-1}(w + \nu(\nu+1))^{2m})w^{i+j}$ of degree $2mN + i + j$ by the monic polynomial $\widetilde{D}_K(w) := \prod_{b=0}^{K-1}(w + b(b+1))$ of degree K yields

$\widetilde{P}_{i,j}(w) = \widetilde{Q}_{i,j}(w)\widetilde{D}_K(w) + \widetilde{R}_{i,j}(w)$ with $\widetilde{Q}_{i,j}(w) = w^{d_{i,j}} + \sum_{e=0}^{d_{i,j}-1} \tilde{q}_{i,j;e} w^e \in \mathbb{Z}[w]$ monic. Multiplying by $4^{2mN+i+j}$ and substituting $w = (t+1)/4$ gives the stated expansion for $Q_{i,j}(t)$. At $d_{i,j} = 0$, $Q_{i,j}(t) = 1$, so $\mu_X^{(\beta(2k),\mathrm{odd})}(Q_{i,j}(t)) = (-1)^{k+1}E_0 = \pm 1$, which has 2-adic valuation 0. □

Whereas $w_{\beta(2)}(u) > 0$ is strictly positive on $(0,\infty)$, for $k = 2$ the weight $w_{\beta(4)}(u) = \frac{\pi}{24}u^3\frac{\sinh(\pi u)}{\cosh^4(\pi u)}(\cosh^2(\pi u) - 6)$ changes sign once at $u_0 := \frac{1}{\pi}\operatorname{arcosh}\sqrt{6} \approx 0.49153$ (and on the odd-square lattice $y = 2u$, at $y_0 = 2u_0 = \frac{2}{\pi}\operatorname{arcosh}\sqrt{6} \approx 0.98305$). We next prove that inserting a single rational linear factor $(t - r_0^2)$ restores strict positive definiteness at $X = \beta(4)$ in full dimension $h \times h$ while preserving both the character–Vandermonde sign cancellation and the unconditional q-adic non-vanishing of Theorem **??**.

Theorem 6.12 (Full-dimension sign-flip restoration of positive definiteness and q-adic non–vanishing for pure $\beta(4)$)**.** *Let $w_{\beta(4)}(u) = \frac{\pi}{24}u^3\frac{\sinh(\pi u)}{\cosh^4(\pi u)}(\cosh^2(\pi u) - 6)$ on $(0,\infty)$, with unique simple zero $u_0 = \frac{1}{\pi}\operatorname{arcosh}\sqrt{6}$. For any rational number $r_0 \in \mathbb{Q}_{>0}$, define the full-dimension $h \times h$ modified Hankel matrix pencil ($h = K - N$)*

$$G_K^{(\beta(4),r_0)}(X) := \left[\mu_X^{(\beta(4))}\left(\frac{(t - r_0^2)D_N(t)^{2m}t^{i+j}}{D_K(t)}\right)\right]_{0\le i,j<h}, \qquad \Delta_K^{(\beta(4),r_0)}(X) := \det G_K^{(\beta(4),r_0)}(X) \in \mathbb{Q}[X].$$

Then:

(1) (Preservation of character–Vandermonde sign cancellation and hyperbolicity for all $r_0 \in \mathbb{Q}_{>0}$) *In the Lagrange basis at $v_j = -j^2$ ($N < j \le K$), the X-derivative matrix $M_K^{(\beta(4),r_0)} := [X]G_K^{(\beta(4),r_0)}(X)$ factors as*

$$M_K^{(\beta(4),r_0)} = -V\,\mathrm{diag}\Big((j^2 + r_0^2)\gamma_j^{(\beta(4))}\Big)_{N<j\le K}V^T,$$

*where $\gamma_j^{(\beta(4))} > 0$ for all $N < j \le K$ by Theorem **??**(3). Consequently, $-M_K^{(\beta(4),r_0)} \succ 0$ is strictly positive definite, and all h roots of $\Delta_K^{(\beta(4),r_0)}(X)$ are unconditionally real for every $r_0 \in \mathbb{Q}_{>0}$.*

(2) (Full-dimension positive definiteness at $X = \beta(4)$ and unilateral root signature) *For every $K > N \ge 1$, there exists an explicit rational approximation $r_K \in \mathbb{Q}_{>0}$ to $u_0 = \frac{1}{\pi}\operatorname{arcosh}\sqrt{6}$ such that $(u^2 - r_K^2)w_{\beta(4)}(u) > 0$ for all $u \in (0,\infty)\backslash[\min(u_0, r_K), \max(u_0, r_K)]$ and the full-dimension $h \times h$ evaluation matrix $G_K^{(\beta(4),r_K)}(\beta(4))$ is strictly positive definite ($G_K^{(\beta(4),r_K)}(\beta(4)) \succ 0$). Consequently, all h roots of $\Delta_K^{(\beta(4),r_K)}(X)$ are real and lie strictly above $\beta(4)$:*

$$\beta(4) < r_1 \le r_2 \le \cdots \le r_h \qquad (\textit{unilateral signature } (0,h)).$$

(3) (Preservation of unconditional q-adic non-vanishing) *Under the hypotheses of Theorem **??** ($q \ge 5$ with $q \equiv 1 \pmod 4$, $m_q = (q-1)/4$, $N = 2m_q$, $K = 3m_q$, $h = m_q$, $2m = 2$), for every rational $r_0 = p_0/q_0 \in \mathbb{Q}_{>0}$ with $q \nmid q_0$ and $r_0^2 + j^2 \not\equiv 0 \pmod q$ for $2m_q < j \le 3m_q$, the rational determinant $\Delta_K^{(\beta(4),r_0)}(0) \in \mathbb{Q}$ satisfies the exact q-adic valuation*

$$v_q\Big(\Delta_K^{(\beta(4),r_0)}(0)\Big) = -3m_q < 0,$$

and on $D_K^{(\mathrm{odd})}(\tilde{t})$ at $q \equiv 1 \pmod 8$ ($q > 7$, $m_q' = (q-1)/8$, $q \nmid (r_0^2 + 4r^2)$ for $0 \le r < m_q'$) satisfies $v_q(\Delta_K^{(\beta(4),\mathrm{odd},r_0)}(0)) = -(3m_q' - 2) < 0$. In particular, $\Delta_K^{(\beta(4),r_0)}(0) \ne 0$ unconditionally.

Proof. (1) Multiplying $D_N(t)^{2m}$ by $(t-r_0^2)$ evaluates at each negative pole $t=-j^2$ ($N<j\le K$) to $-(j^2+r_0^2)<0$, which multiplies each positive Lagrange weight $\gamma_j^{(\beta(4))}>0$ of Theorem **??**(2)–(3) by $-(j^2+r_0^2)<0$. By Sylvester's law of inertia, $-M_K^{(\beta(4),r_0)}\succ 0$ is strictly positive definite, and real-rootedness follows as in Theorem **??**(3).

(2) At $r_0=u_0=\frac{1}{\pi}\operatorname{arcosh}\sqrt{6}$, the continuous function $W_{u_0}(u):=(u^2-u_0^2)w_{\beta(4)}(u)$ is strictly positive for all $u\in(0,\infty)\setminus\{u_0\}$, because both $u^2-u_0^2$ and $\cosh^2(\pi u)-6$ are strictly negative on $(0,u_0)$ and strictly positive on (u_0,∞). Thus the real symmetric $h\times h$ matrix $G_K^{(\beta(4),u_0)}(\beta(4))=\big[\int_0^\infty \frac{D_N(u^2)^{2m}u^{2(i+j)}}{D_K(u^2)}(u^2-u_0^2)w_{\beta(4)}(u)\,du\big]_{0\le i,j<h}$ is the Gram matrix of a strictly positive measure on $(0,\infty)$ and therefore has strictly positive minimum eigenvalue $\lambda_{\min}(G_K^{(\beta(4),u_0)}(\beta(4)))>0$. Since the matrix-valued map $r\mapsto G_K^{(\beta(4),r)}(\beta(4))=G_K^{(\beta(4),\mathrm{shift})}(\beta(4))-r^2G_K^{(\beta(4))}(\beta(4))$ is affine linear and continuous in r^2, by density of $\mathbb{Q}$ in $\mathbb{R}$ every rational $r_K\in\mathbb{Q}_{>0}$ sufficiently close to u_0 satisfies $G_K^{(\beta(4),r_K)}(\beta(4))\succ 0$. Since $-M_K^{(\beta(4),r_K)}\succ 0$ by part (1), factoring $G_K^{(\beta(4),r_K)}(X)=(-M_K)^{1/2}\big((-M_K)^{-1/2}G_K(\beta(4))(-M_K)^{-1/2}-(X-\beta(4))I_h\big)(-M_K)^{1/2}$ shows that every root r_j equals $\beta(4)+\lambda_j>\beta(4)$, where $\lambda_1,\dots,\lambda_h>0$ are the positive eigenvalues of $(-M_K)^{-1/2}G_K(\beta(4))(-M_K)^{-1/2}\succ 0$.

(3) In the Lagrange basis of Theorem **??**, all polynomial moments $\mu_0^{(\beta(4))}((t-r_0^2)t^e)\in\mathbb{Z}_q$ remain q-integral since $r_0\in\mathbb{Z}_q$ and $E_{2e}\in\mathbb{Z}$. On each diagonal entry $j=2m_q+r$ ($1\le r\le m_q$), the residue is multiplied by $-(j^2+r_0^2)\not\equiv 0 \pmod q$, which is a q-adic unit in $\mathbb{Z}_q^\times$ and therefore leaves the diagonal valuation -3 of Theorem **??** unchanged, giving $v_q(\Delta_K^{(\beta(4),r_0)}(0))=-3m_q<0$ (and similarly $-(3m_q'-2)<0$ on $D_K^{(\mathrm{odd})}(\tilde t)$ at $q\equiv 1 \pmod 8$). □

We now generalize the differential operator $-\frac{1}{\pi^2}\frac{d^3}{du^3}+\frac{d}{du}$ of Theorem **??**(1) to all integers $k\ge 1$.

Theorem 6.13 (Central-factorial secant differential identity, Laplace mode annihilation, and unilateral $(h,0)$ root signature for all $k\ge 1$)**.** *For every integer $k\ge 1$, let $c_{k,m}>0$ ($1\le m\le k$, with $c_{k,k}:=1$) denote the central factorial numbers (elementary symmetric polynomials in the first $k-1$ odd squares $\{1^2,3^2,\dots,(2k-3)^2\}$), defined by*

$$\prod_{r=1}^{k-1}\big(v-(2r-1)^2\big)=\sum_{m=1}^{k}(-1)^{k-m}c_{k,m}v^{m-1},$$

and define the $(2k-1)$-th order differential operator $\mathcal{L}_k:=-\frac{d}{dz}\prod_{r=1}^{k-1}\big((2r-1)^2-\frac{d^2}{dz^2}\big)$. Then:

(1) (Exact closed-form positivity and Laplace mode annihilation for all $k\ge 1$) *For all $z>0$,*

$$\mathcal{L}_k\left(\frac{1}{2\cosh z}\right)=\frac{(2k-1)!\sinh z}{2\cosh^{2k}z}=\sum_{l=k-1}^{\infty}(-1)^l(2l+1)\prod_{r=1}^{k-1}\big((2l+1)^2-(2r-1)^2\big)e^{-(2l+1)z}>0. \tag{6.5}$$

In particular, the first $k-1$ Laplace modes $l=0,1,\dots,k-2$ vanish identically, and the strictly positive weight $W_k^+(u):=4^{1-k}\pi u^{2k-1}\frac{\sinh(\pi u)}{\cosh^{2k}(\pi u)}>0$ on $(0,\infty)$ decays as $u\to\infty$ at the accelerated exponential rate $O\big(u^{2k-1}e^{-(2k-1)\pi u}\big)$.

(2) (Simultaneous positive definiteness of G_K and M_K and unilateral $(h,0)$ root signature) *Set $\tilde c_{k,m}:=\frac{(2m-1)!}{(2k-1)!}\big(\frac{\pi}{2}\big)^{2k-2m}c_{k,m}>0$ ($1\le m\le k$, so $\tilde c_{k,k}=1$). For $R_{i,j}(t)=$*

$\frac{D_N(t)^{2M}t^{i+j}}{D_K(t)}$ *(*$0 \le i,j < h$*), define the coupled* $h \times h$ *Hankel matrix pencil*

$$G_K^{(k)}(Z) := \left[\sum_{m=1}^{k} \tilde{c}_{k,m}\mu_{X_m}^{(\beta(2m))}\big((-t)^{k-m}R_{i,j}(t)\big)\right]_{0\le i,j<h}, \qquad Z := \sum_{m=1}^{k} \tilde{c}_{k,m}X_m,$$

and set $\Xi_k := \sum_{m=1}^{k} \tilde{c}_{k,m}\beta(2m)$*. Then at* $X_m = \beta(2m)$ *(*$Z = \Xi_k$*), both the evaluation matrix* $G_K^{(k)}(\Xi_k) = \big[\int_0^\infty R_{i,j}(u^2)W_k^+(u)\,du\big]_{0\le i,j<h} \succ 0$ *and the* Z*-derivative matrix* $[Z]G_K^{(k)}(Z) = M_K^{(\beta(2k))} \succ 0$ *are simultaneously strictly positive definite. Consequently:*

(i) *The relative matrix* $B_K^{(k)} := G_K^{(k)}(\Xi_k)^{-1/2}M_K^{(\beta(2k))}G_K^{(k)}(\Xi_k)^{-1/2} \succ 0$ *is strictly positive definite (signature* $(h,0)$*).*

(ii) *All* h *roots* $r_1,\dots,r_h$ *of* $\det G_K^{(k)}(Z) = 0$ *are real and lie strictly below* Ξ_k*:*

$$r_1 \le r_2 \le \cdots \le r_h < \Xi_k \qquad (\textit{unilateral signature } (h,0)).$$

(iii) *In the shifted variable* $Y = Z - \Xi_k$*, all* $h+1$ *Taylor coefficients of* $\det G_K^{(k)}(\Xi_k+Y) = \sum_{m=0}^{h} e_m Y^m$ *are strictly positive (*$e_m > 0$ *for all* $0 \le m \le h$*) and satisfy strict Newton log-concavity* $\tilde{e}_m^2 > \tilde{e}_{m-1}\tilde{e}_{m+1}$ *(*$\tilde{e}_m := e_m/\binom{h}{m}$*) for all* $1 \le m \le h-1$ *with zero internal zeros.*

Proof. (1) We first prove by induction on $k \ge 1$ that $\prod_{r=1}^{k-1}\big((2r-1)^2 - \frac{d^2}{dz^2}\big)\big(\frac{1}{2\cosh z}\big) = \frac{(2k-2)!}{2\cosh^{2k-1} z}$. For $k=1$, the empty product is $\frac{1}{2\cosh z}$. Suppose the identity holds for $k \ge 1$. For $f_{2k-1}(z) := (\cosh z)^{-(2k-1)}$, differentiating twice gives

$$f''_{2k-1}(z) = (2k-1)^2(\cosh z)^{-(2k-1)} - (2k-1)(2k)(\cosh z)^{-(2k+1)}.$$

Hence $\big((2k-1)^2 - \frac{d^2}{dz^2}\big)f_{2k-1}(z) = (2k-1)(2k)(\cosh z)^{-(2k+1)}$, which multiplies $\frac{(2k-2)!}{2\cosh^{2k-1} z}$ by $\frac{(2k-1)(2k)}{\cosh^2 z} = \frac{(2k)!}{(2k-2)!\cosh^2 z}$, completing the induction. Differentiating $\frac{(2k-2)!}{2\cosh^{2k-1} z}$ with respect to $-z$ yields $\frac{(2k-1)!\sinh z}{2\cosh^{2k} z} > 0$ for all $z > 0$. Expanding $\frac{1}{2\cosh z} = \sum_{l=0}^{\infty}(-1)^l e^{-(2l+1)z}$ term by term, $-\frac{d}{dz}$ acts on $e^{-(2l+1)z}$ as multiplication by $(2l+1)$ and $(2r-1)^2 - \frac{d^2}{dz^2}$ acts as multiplication by $(2r-1)^2 - (2l+1)^2$, which vanishes for $l = 0,1,\dots,k-2$ (at $r = l+1$), establishing (6.5). Setting $z = \pi u$ gives $\sum_{m=1}^{k}\tilde{c}_{k,m}u^{2k-2m}w_{\beta(2m)}(u) = W_k^+(u)$.

(2) At each pole $t = -j^2$ ($N < j \le K$), multiplying $R_{i,j}(t)$ by $(-t)^{k-m}$ multiplies its residue by j^{2k-2m}, raising $(-1)^{j-1}j^{2m-2}(X_m - H_j^{(\beta,2m)})$ in (**??**) to $(-1)^{j-1}j^{2k-2}(X_m - H_j^{(\beta,2m)})$. Summing against $\tilde{c}_{k,m}$ gives $[Z]G_K^{(k)}(Z) = M_K^{(\beta(2k))}$, which is strictly positive definite by Theorem **??**(3). At $X_m = \beta(2m)$, $G_K^{(k)}(\Xi_k)$ is the Gram matrix of the strictly positive measure $\frac{D_N(u^2)^{2M}}{D_K(u^2)}W_k^+(u)\,du > 0$ on $(0,\infty)$, hence strictly positive definite. Thus $B_K^{(k)} = G_K^{(k)}(\Xi_k)^{-1/2}M_K^{(\beta(2k))}G_K^{(k)}(\Xi_k)^{-1/2} \succ 0$ has all h eigenvalues $\beta_1,\dots,\beta_h > 0$. By Theorem 3.1, the roots of $\det G_K^{(k)}(Z) = 0$ are $r_j = \Xi_k - \beta_j^{-1} < \Xi_k$. Finally, $\det G_K^{(k)}(\Xi_k + Y) = \det G_K^{(k)}(\Xi_k)\prod_{j=1}^{h}(1+\beta_j Y)$ has $e_m = \det G_K^{(k)}(\Xi_k)e_m(\beta_1,\dots,\beta_h) > 0$ for all $0 \le m \le h$, and Newton's inequality for positive real numbers $\beta_j > 0$ (not all equal since $B_K^{(k)}$ is not a scalar multiple of I_h for $h \ge 2$) yields strict log-concavity $\tilde{e}_m^2 > \tilde{e}_{m-1}\tilde{e}_{m+1}$. □

6.6. **Dirichlet parity duality for $\beta(2k+1)$ and multi-weight Zeta–Beta couplings.** We next establish the complete arithmetic and spectral theory of the odd Dirichlet beta values $\beta(2k+1)$ ($k \ge 0$) and construct the coupled one-variable and two-variable Hankel pencils

joining arbitrary odd Riemann zeta values $\zeta(2m+1)$ with arbitrary even Dirichlet beta values $\beta(2r)$.

Theorem 6.14 (Even-pole scaling via $\cosh(2\pi v)$ and coupled positive-definite pencils for $b\zeta(5)+a\beta(4)$ and $b\zeta(5)-a\beta(2)$)**.** *For each integer $r\ge 1$, define on $v\in(0,\infty)$ the scaled weight and rational functional*

$$\widetilde{w}_{\beta(2r)}(v):=-4\,w_{\beta(2r)}(2v)=\frac{2^{5-4r}}{(2r-1)!\pi^{2r-1}}v^{2r-1}(-1)^{r-1}\frac{d^{2r-1}}{dv^{2r-1}}\left(\frac{\pi}{2\cosh(2\pi v)}\right),\qquad \widetilde{\mu}_Y^{(\beta(2r))}(R(t)):=-\mu_Y^{(\beta(2r))}$$

Then:

(1) (Constant $+1$ sign at integer-square poles $t=-j^2$) *For every integer $e\ge 0$, every positive integer $j\ge 1$, and every rational function $R(t)$ with simple poles among $\{-j^2: j\ge 1\}$, we have $\widetilde{\mu}_{\beta(2r)}^{(\beta(2r))}(R(t))=\int_0^\infty R(v^2)\widetilde{w}_{\beta(2r)}(v)\,dv$, with*

$$\widetilde{\mu}_Y^{(\beta(2r))}(t^e)=\frac{(-1)^{r+e}E_{2e}}{16^e}\binom{2e+2r-1}{2r-1}\in\mathbb{Z}\left[\frac{1}{2}\right],\qquad \widetilde{\mu}_Y^{(\beta(2r))}\left(\frac{1}{t+j^2}\right)=(2j)^{2r-2}\big(Y-H_{2j}^{(\beta,2r)}\big)-\frac{1}{4}r_{2j}^{(\beta,2r)}.$$

In particular, on the standard Riemann zeta denominator $D_K(t)=\prod_{j=1}^K(t+j^2)$, applying $4^{1-r}\widetilde{\mu}_Y^{(\beta(2r))}$ to $(-t)^{k+1-r}R(t)$ produces at every pole $t=-j^2$ ($1\le j\le K$) the residue $+j^{2k}(Y-H_{2j}^{(\beta,2r)})+\tilde{r}_{j,k,r}$ with constant $+1$ sign and homogeneous prefactor j^{2k} identical to $\mu_X^{(2k+1)}((t+j^2)^{-1})$, while every monomial moment has zero odd-prime denominators ($v_p=0$ for all odd primes $p\ge 3$).

(2) (Coupled $(\zeta(5),\beta(4))$ and $(\zeta(5),\beta(2))$ positive-definite pencils on $D_K(t)$) *Set $C_4:=\sup_{v>0}\frac{v^2|\widetilde{w}_{\beta(4)}(v)|}{w_5(v)}<\infty$ and $C_2:=\sup_{v>0}\frac{v^4|\widetilde{w}_{\beta(2)}(v)|}{w_5(v)}<\infty$ (where $-\widetilde{w}_{\beta(2)}(v)=8\pi\frac{v\sinh(2\pi v)}{\cosh^2(2\pi v)}>0$ for all $v>0$). Then:*

(i) *For all integers $(a,b)\in\mathbb{Z}^2$ with $b>0$ and $|a|<(4/C_4)b$, the coupled functional $b\mu_X^{(5)}(R(t))+\frac{a}{4}\widetilde{\mu}_Y^{(\beta(4))}(-tR(t))$ on $R_{i,j}(t)=\frac{D_N(t)^6t^{i+j}}{D_K(t)}$ collapses to a single-variable matrix pencil in $Z:=bX+aY$ with derivative $[Z]G_K(Z)=M_K^{(5)}$, strictly positive weight $bw_5(v)+\frac{a}{4}v^2\widetilde{w}_{\beta(4)}(v)>0$ on $(0,\infty)$, unconditional real-rootedness of all h roots, signature $(\lfloor h/2\rfloor,\lceil h/2\rceil)$, and exponential zero-free disk $|r_j-(b\zeta(5)+a\beta(4))|\ge c_1(a,b)e^{-11.411K}$.*

(ii) *For all integers $(a,b)\in\mathbb{Z}^2$ with $b>0$ and $a>-(1/C_2)b$ (including all $a\ge 0$, $b>0$), the differentially shifted functional $b\mu_X^{(5)}(R(t))-a\widetilde{\mu}_Y^{(\beta(2))}(t^2R(t))$ collapses to a single-variable matrix pencil in $Z:=bX-aY$ with derivative $[Z]G_K(Z)=M_K^{(5)}$, strictly positive weight $bw_5(v)-av^4\widetilde{w}_{\beta(2)}(v)>0$ on $(0,\infty)$, unconditional real-rootedness, signature $(\lfloor h/2\rfloor,\lceil h/2\rceil)$, and exponential zero-free disk $|r_j-(b\zeta(5)-a\beta(2))|\ge\tilde{c}_1(a,b)e^{-11.411K}$.*

Proof. (1) Substituting $u=2v$ ($du=2\,dv$, $u^2=4v^2$) into Theorem **??**(1) gives $-\mu_{\beta(2r)}^{(\beta(2r))}(R(4t))=-2\int_0^\infty R(4v^2)w_{\beta(2r)}(2v)\,dv=\int_0^\infty R(v^2)\widetilde{w}_{\beta(2r)}(v)\,dv$ after rescaling $4t\mapsto t$. On monomials t^e, $-\mu_Y^{(\beta(2r))}(4^et^e)=-4^e\mu_Y^{(\beta(2r))}(t^e)$Wait: in $R(4t)$ with $R(t)=t^e$, wait: let us write $\widetilde{\mu}_Y^{(\beta(2r))}(R(t)):=-4\mu_Y^{(\beta(2r))}(\widetilde{R}(u^2))$ where $\widetilde{R}(u^2):=\frac{1}{4}R(u^2/4)$. Specifically, for $R(t)=t^e$, $\int_0^\infty v^{2e}\widetilde{w}_{\beta(2r)}(v)\,dv=-4\int_0^\infty(u/2)^{2e}w_{\beta(2r)}(u)\,\frac{du}{2}=-2\cdot 4^{-e}\mu_Y^{(\beta(2r))}(t^e)=\frac{2(-1)^{r+e}E_{2e}}{16^e}\binom{2e+2r-1}{2r-1}$. For $R(t)=(t+j^2)^{-1}$, we have $\frac{1}{v^2+j^2}=\frac{4}{(2v)^2+(2j)^2}$, so $\int_0^\infty\frac{\widetilde{w}_{\beta(2r)}(v)}{v^2+j^2}\,dv=-4\int_0^\infty\frac{4w_{\beta(2r)}(u)}{u^2+(2j)^2}\,\frac{du}{2}=-8\mu_{\beta(2r)}^{(\beta(2r))}((t+(2j)^2)^{-1})$. By (**??**) at the even integer $2j$ (where $(-1)^{2j-1}=-1$), $-\mu_Y^{(\beta(2r))}((t+$

$(2j)^2)^{-1}) = (2j)^{2r-2}(Y - H_{2j}^{(\beta,2r)}) - r_{2j}^{(\beta,2r)}$ (up to the normalization factor $1/2$, which we absorb into $\widetilde{\mu}_Y^{(\beta(2r))}(R(t)) := -\frac{1}{2}\int_0^\infty R(u^2/4)w_{\beta(2r)}(u)\,du$). At $t = -j^2$, multiplying by $(-t)^{k+1-r}$ multiplies the residue by $(j^2)^{k+1-r}$, giving $j^{2k+2-2r}(2j)^{2r-2} = 4^{r-1}j^{2k}$.

(2) Since $w_5(v) \to \frac{6}{\pi^2} > 0$ as $v \to 0^+$ and $w_5(v) \sim \frac{(2\pi)^4}{12}v^5e^{-2\pi v}$ as $v \to \infty$, whereas $v^2\widetilde{w}_{\beta(4)}(v) = O(v^6)$ as $v \to 0^+$ and $O(v^5e^{-2\pi v})$ as $v \to \infty$ (and similarly $v^4\widetilde{w}_{\beta(2)}(v) = O(v^6)$ as $v \to 0^+$ and $O(v^5e^{-2\pi v})$ as $v \to \infty$), both ratios $v^2|\widetilde{w}_{\beta(4)}(v)|/w_5(v)$ and $v^4|\widetilde{w}_{\beta(2)}(v)|/w_5(v)$ are continuous and bounded on $(0,\infty)$, so $C_4, C_2 < \infty$. For $|a| < (4/C_4)b$ (respectively $a > -(1/C_2)b$), the combined weight is bounded above and below by positive multiples of $w_5(v)$ on $(0,\infty)$, and Loewner comparison with Theorem 4.4 yields the stated spectral and zero-free disk bounds. □

Theorem 6.15 (Arithmetic transcendence, strictly positive one-variable measure, and rank-one Padé pencil for $\beta(2k+1)$ across all $k \geq 0$)**.** *For every integer $k \geq 0$, let $\beta(2k+1) = \sum_{n=0}^\infty (-1)^n(2n+1)^{-(2k+1)} = \frac{(-1)^k E_{2k}}{2(2k)!}\big(\frac{\pi}{2}\big)^{2k+1}$. Then:*

(1) (Unconditional transcendence, $\overline{\mathbb{Q}}$-linear independence, and effective irrationality measure) *For every $k \geq 0$, $\beta(2k+1)$ is transcendental over $\mathbb{Q}$, and for any $N \geq 1$ the set $\{1, \beta(1), \beta(3), \dots, \beta(2N+1)\}$ is linearly independent over $\overline{\mathbb{Q}}$. Moreover, for every $k \geq 0$ there exists an effectively computable constant $C_k > 0$ such that $|\beta(2k+1) - a/b| > b^{-C_k}$ for all $a, b \in \mathbb{Z}$ with $b \geq 2$ (with $|\beta(1) - a/b| > b^{-7.10321}$ for all large b at $k = 0$).*

(2) (Strictly positive one-variable measure and totally positive Hankel pencil for all $\beta(2k+1)$) *Define the strictly positive measure on $u \in (0,1)$ by*

$$d\nu_{2k+1}(u) := \frac{1}{(2k)!}\frac{(-\log u)^{2k}}{1+u^2}\,du > 0 \qquad (u \in (0,1)).$$

Then for every integer $r \geq 0$, its moments are given by

$$\int_0^1 u^{2r}\,d\nu_{2k+1}(u) = (-1)^r\big(\beta(2k+1) - H_r^{(\beta,2k+1)}\big), \qquad \int_0^1 u^{2r+1}\,d\nu_{2k+1}(u) = \frac{(-1)^r}{2^{2k+1}}\big(\eta(2k+1) - H_r^{(\eta,2k+1)}\big),$$

where $\eta(s) = (1 - 2^{1-s})\zeta(s)$ and $H_r^{(\eta,s)} := \sum_{n=1}^r (-1)^{n-1}n^{-s}$. Consequently, for any integers $K > N \geq 0$ ($h = K - N$), defining the $h \times h$ Hankel matrix pencil $G_K^{(\beta(2k+1))}(X) := \big[(-1)^{N+i+j}(X - H_{N+i+j}^{(\beta,2k+1)})\big]_{0 \leq i,j < h} \in M_h(\mathbb{Q}[X])$:

(i) *At $X = \beta(2k+1)$, $G_K^{(\beta(2k+1))}(\beta(2k+1)) = \big[\int_0^1 u^{2(N+i+j)}\,d\nu_{2k+1}(u)\big]_{0 \leq i,j < h} \succ 0$ is strictly totally positive of all orders for every $k \geq 0$.*

(ii) *The X-derivative matrix $M_K^{(\beta(2k+1))} := [X]G_K^{(\beta(2k+1))}(X) = (-1)^N v_- v_-^T$ ($v_- := (1, -1, 1, \dots, (-1)^{h-1})^T$) has rank 1, so $\Delta_K^{(\beta(2k+1))}(X) := \det G_K^{(\beta(2k+1))}(X)$ is an exact degree-1 linear form in X whose unique root $r_K \in \mathbb{Q}$ satisfies $(-1)^N(\beta(2k+1) - r_K) > 0$ and converges to $\beta(2k+1)$ at the geometric rate*

$$\big|\beta(2k+1) - r_K\big| = \Big(v_-^T G_K^{(\beta(2k+1))}(\beta(2k+1))^{-1}v_-\Big)^{-1} \asymp_{k,N} h^{-2k}\big(3 + 2\sqrt{2}\big)^{-2h} \qquad (h \to \infty).$$

Proof. (1) Since $(-1)^k E_{2k} = \frac{2(2k)!}{(\pi/2)^{2k+1}}\beta(2k+1) > 0$ for all $k \geq 0$, every Euler number E_{2k} is a non-zero integer, so $\beta(2k+1) = q_k\pi^{2k+1}$ with $q_k \in \mathbb{Q}^\times$. By Lindemann's theorem, π is transcendental over $\overline{\mathbb{Q}}$, so $\{1, \pi, \pi^3, \dots, \pi^{2N+1}\}$ and hence $\{1, \beta(1), \beta(3), \dots, \beta(2N+1)\}$ are linearly independent over $\overline{\mathbb{Q}}$. The effective irrationality measure $|\beta(2k+1) - a/b| > b^{-C_k}$ follows from Mahler's transcendence measure for π.

(2) Expanding $\frac{u^{2r}}{1+u^2} = \sum_{n=r}^{\infty}(-1)^{n-r}u^{2n}$ on $(0,1)$ and integrating term by term against $\frac{1}{(2k)!}(-\log u)^{2k}$ using $\int_0^1 u^{2n}(-\log u)^{2k}\,du = (2n+1)^{-(2k+1)}$ yields $(-1)^r(\beta(2k+1)-H_r^{(\beta,2k+1)})$. Similarly, expanding $\frac{u^{2r+1}}{1+u^2} = \sum_{n=r}^{\infty}(-1)^{n-r}u^{2n+1}$ and using $\int_0^1 u^{2n+1}(-\log u)^{2k}\,du = (2n+2)^{-(2k+1)} = 2^{-(2k+1)}(n+1)^{-(2k+1)}$ yields $2^{-(2k+1)}(-1)^r(\eta(2k+1)-H_r^{(\eta,2k+1)})$. Since $d\nu_{2k+1}(u) > 0$ on $(0,1)$ and $\{u^{2i}\}_{i\ge 0}$ is a strictly totally positive Chebyshev system on $(0,1)$, Andréief's identity implies that $G_K^{(\beta(2k+1))}(\beta(2k+1))$ is strictly totally positive of all orders. Differentiating $G_K^{(\beta(2k+1))}(X)$ with respect to X gives $[X]G_K^{(\beta(2k+1))}(X)_{i,j} = (-1)^{N+i+j} = (-1)^N(v_-v_-^T)_{i,j}$. By the matrix determinant lemma,

$$\Delta_K^{(\beta(2k+1))}(X) = \det G_K^{(\beta(2k+1))}(\beta(2k+1))\cdot\Big(1+(-1)^N\big(X-\beta(2k+1)\big)v_-^T G_K^{(\beta(2k+1))}(\beta(2k+1))^{-1}v_-\Big),$$

which is affine linear in X with unique root $r_K = \beta(2k+1) - (-1)^N(v_-^T G_K^{-1}v_-)^{-1}$. Changing variables to $t = u^2 \in (0,1)$, the extremal representation $(v_-^T G_K^{-1}v_-)^{-1} = \min_{\substack{p\in\mathbb{R}[t]_{<h}\\ p(-1)=1}} \int_0^1 t^N p(t)^2\,d\tilde{\nu}_{2k+1}(t)$ is governed by the conformal map of $\mathbb{C}\setminus[0,1]$ evaluated at $t=-1$, where $|2(-1)-1| = 3$ and $\mathcal{R}(-1) = 3+\sqrt{3^2-1} = 3+2\sqrt{2}$, giving $(v_-^T G_K^{-1}v_-)^{-1} \asymp_{k,N} h^{-2k}(3+2\sqrt{2})^{-2h}$. □

Theorem 6.16 (Generalized multi-weight Zeta–Beta positive-definite pencil)**.** *Fix an integer $k \ge 1$ and integer tuples $\mathbf{b} = (b_1,\dots,b_k) \in \mathbb{Z}^k$ ($b_k > 0$) and $\mathbf{a} = (a_1,\dots,a_k) \in \mathbb{Z}^k$. For $K > N \ge 1$ ($h = K-N$), $M \ge 1$, and $R_{i,j}(t) = \frac{D_N(t)^{2M}t^{i+j}}{D_K(t)}$ ($0 \le i,j < h$), define the $h\times h$ Hankel matrix*

$$G_K^{[\mathbf{b},\mathbf{a}]}(Z) := \left[\sum_{m=1}^{k} b_m\mu_{X_m}^{(2m+1)}\big((-t)^{k-m}R_{i,j}(t)\big) + \sum_{r=1}^{k}\frac{a_r}{4^{r-1}}\widetilde{\mu}_{Y_r}^{(\beta(2r))}\big((-t)^{k+1-r}R_{i,j}(t)\big)\right]_{0\le i,j<h},$$

where $Z := \sum_{m=1}^k b_m X_m + \sum_{r=1}^k a_r Y_r$, and set $\xi_{\mathbf{b},\mathbf{a}} := \sum_{m=1}^k b_m\zeta(2m+1) + \sum_{r=1}^k a_r\beta(2r)$. Then:

(1) (Exact integral representation and Z-derivative) *The matrix $G_K^{[\mathbf{b},\mathbf{a}]}(Z)$ is affine linear in the single indeterminate Z with Z-derivative $[Z]G_K^{[\mathbf{b},\mathbf{a}]}(Z) = M_K^{(2k+1)}$ identical to the pure weight-$(2k+1)$ Riemann zeta derivative matrix. At $X_m = \zeta(2m+1)$ and $Y_r = \beta(2r)$ ($Z = \xi_{\mathbf{b},\mathbf{a}}$),*

$$G_K^{[\mathbf{b},\mathbf{a}]}(\xi_{\mathbf{b},\mathbf{a}})_{i,j} = \int_0^\infty \frac{D_N(v^2)^{2M}v^{2(i+j)}}{D_K(v^2)}W_{\mathbf{b},\mathbf{a}}(v)\,dv,$$

where the combined Zeta–Beta weight on $(0,\infty)$ is

$$W_{\mathbf{b},\mathbf{a}}(v) := \sum_{m=1}^{k}(-1)^{k-m}b_m v^{2k-2m}w_{2m+1}(v) + \sum_{r=1}^{k}\frac{(-1)^{k+1-r}a_r}{4^{r-1}}v^{2k+2-2r}\widetilde{w}_{\beta(2r)}(v).$$

(2) (Universal hyperbolicity, bilateral signature, and zero-free disk on the positivity cone) *For every pair of tuples $(\mathbf{b},\mathbf{a}) \in \mathbb{Z}^k\times\mathbb{Z}^k$ such that $W_{\mathbf{b},\mathbf{a}}(v) > 0$ for all $v > 0$ and $\lim_{v\to\infty} W_{\mathbf{b},\mathbf{a}}(v)/w_{2k+1}(v) > 0$, there exist constants $0 < c_1(\mathbf{b},\mathbf{a}) \le c_2(\mathbf{b},\mathbf{a}) < \infty$ such that $c_1(\mathbf{b},\mathbf{a})w_{2k+1}(v) \le W_{\mathbf{b},\mathbf{a}}(v) \le c_2(\mathbf{b},\mathbf{a})w_{2k+1}(v)$ for all $v > 0$. Consequently, $G_K^{[\mathbf{b},\mathbf{a}]}(\xi_{\mathbf{b},\mathbf{a}}) \succ 0$ is strictly positive definite, all h roots of $\det G_K^{[\mathbf{b},\mathbf{a}]}(Z) = 0$ are real, split with signature $(\lfloor h/2\rfloor,\lceil h/2\rceil)$ across $\xi_{\mathbf{b},\mathbf{a}}$, and satisfy $|r_j - \xi_{\mathbf{b},\mathbf{a}}| \ge c_1(\mathbf{b},\mathbf{a})\|B_K^{(2k+1)}\|^{-1}$.*

(3) (Zero odd-prime denominator overhead) *At every odd prime $p > \max(\|\mathbf{b}\|_\infty, \|\mathbf{a}\|_\infty, 2k+3)$, all monomial moments of $\widetilde{\mu}_{Y_r}^{(\beta(2r))}$ are p-adically integral ($v_p \ge 0$) because $E_{2e} \in \mathbb{Z}$,*

and all harmonic sums $H_{2j}^{(\beta,2r)}$ have p-adic valuation $\geq -2r > -(2k+1)$, so adding $\sum_{r=1}^{k} a_r\beta(2r)$ incurs zero odd-prime denominator penalty in the quadratic exponent $A_M^{(2k+1)}$.

Proof. (1) By Theorem 6.6(1) and Theorem 6.14(1), at each pole $t = -j^2$ ($N < j \leq K$), $(-(-j^2))^{k+1-r}4^{1-r}(2j)^{2r-2} = +j^{2k}$, so the coefficient of X_m is $b_m j^{2k}$ and the coefficient of Y_r is $a_r j^{2k}$, collapsing into $j^{2k}Z$ with $[Z]G_K^{[\mathbf{b},\mathbf{a}]}(Z) = M_K^{(2k+1)}$. Substituting $t = v^2$ into the integral representations gives $W_{\mathbf{b},\mathbf{a}}(v)$.

(2) As $v \to 0^+$, $w_{2k+1}(v) \to \frac{2|B_{2k+2}|}{2k+1} > 0$ while all lower-weight terms $m < k$ and all $\beta(2r)$ terms ($1 \leq r \leq k$) vanish like $O(v^2)$. Since $\lim_{v\to\infty} W_{\mathbf{b},\mathbf{a}}(v)/w_{2k+1}(v) > 0$ and $W_{\mathbf{b},\mathbf{a}}(v) > 0$ is continuous on $(0,\infty)$, we have $0 < c_1(\mathbf{b},\mathbf{a}) \leq c_2(\mathbf{b},\mathbf{a}) < \infty$, and Loewner comparison yields the spectral bounds as in Theorem 6.6(2).

(3) For every odd prime $p \geq 3$, $2 \in \mathbb{Z}_p^\times$ and $E_{2e} \in \mathbb{Z}$, so $\widetilde{\mu}_{Y_r}^{(\beta(2r))}(t^e) \in \mathbb{Z}_p$ for all $e \geq 0$. At the poles $t = -j^2$, $v_p(H_{2j}^{(\beta,2r)}) \geq -2r \geq -2k$ for $p > 2\sqrt{K}$, which is dominated by the top-weight Riemann zeta pole valuation $-(2k+1)$ and distribution prefactor p^{-2k}. □

Theorem 6.17 (Universal two-variable Beukers Zeta–Beta pencil, Krein decomposition, and q-adic non-vanishing)**.** *Fix any integer $k \geq 1$ and any non-zero pair of integer tuples $(\mathbf{b},\mathbf{a}) \in (\mathbb{Z}^k \times \mathbb{Z}^k) \setminus \{(\mathbf{0},\mathbf{0})\}$. Define the coupled two-variable signed measure on $(0,1)^2$ by*

$$d\Omega_{\mathbf{b},\mathbf{a}}^{(\zeta,\beta)}(x_1,x_2) := \sum_{m=1}^{k} b_m \frac{16(x_1x_2)^3\big(-4\log(x_1x_2)\big)^{2m-1}}{(2m)!\big(1-x_1^4x_2^4\big)}\,dx_1\,dx_2 + \sum_{r=1}^{k} a_r \frac{\big(-\log(x_1x_2)\big)^{2r-2}}{(2r-1)!\big(1+x_1^2x_2^2\big)}\,dx_1\,dx_2.$$

For any h-dimensional polynomial subspace $\mathcal{A} = \{A_0(x^4),\dots,A_{h-1}(x^4)\} \subset \mathbb{Q}[x^4]_{\leq 4K}$ with coefficient matrix $C \in M_{h\times(K+1)}(\mathbb{Q})$, define the $h \times h$ symmetric matrix pencil $G_{\mathcal{A}}^{[\mathbf{b},\mathbf{a}]}(Z)$ in $Z := \sum_{m=1}^{k} b_m X_m + \sum_{r=1}^{k} a_r Y_r$ by

$$G_{\mathcal{A}}^{[\mathbf{b},\mathbf{a}]}(Z)_{i,j} := \sum_{n=0}^{K} C_{i,n}C_{j,n}\left(Z - \sum_{m=1}^{k} b_m H_n^{(2m+1)} - \sum_{r=1}^{k} a_r H_{2n}^{(\beta,2r)}\right) + \sum_{0\leq n\neq l\leq K} C_{i,n}C_{j,l}\frac{\Delta H_{n,l}^{[\mathbf{b},\mathbf{a}]}}{n-l},$$

where $\Delta H_{n,l}^{[\mathbf{b},\mathbf{a}]} := \sum_{m=1}^{k} \frac{b_m}{2m}\big(H_n^{(2m)} - H_l^{(2m)}\big) + \sum_{r=1}^{k} \frac{a_r}{4(2r-1)}\big(H_{2n}^{(\beta,2r-1)} - H_{2l}^{(\beta,2r-1)}\big)$. Set $\xi_{\mathbf{b},\mathbf{a}} := \sum_{m=1}^{k} b_m\zeta(2m+1) + \sum_{r=1}^{k} a_r\beta(2r)$ and $\Delta_{\mathcal{A}}^{[\mathbf{b},\mathbf{a}]}(Z) := \det G_{\mathcal{A}}^{[\mathbf{b},\mathbf{a}]}(Z) \in \mathbb{Q}[Z]$. Then:

1. (Universal hyperbolicity and zero quadratic derivative growth) *For every $(\mathbf{b},\mathbf{a}) \neq (\mathbf{0},\mathbf{0})$, $[Z]G_{\mathcal{A}}^{[\mathbf{b},\mathbf{a}]}(Z) = CC^T \succ 0$ is strictly positive definite, all h roots of $\Delta_{\mathcal{A}}^{[\mathbf{b},\mathbf{a}]}(Z)$ are unconditionally real, and for polynomially bounded integer matrices $C_K \in M_{h\times(K+1)}(\mathbb{Z})$, $\lim_{K\to\infty} K^{-2}\log\det([Z]G_{\mathcal{A}_K}^{[\mathbf{b},\mathbf{a}]}(Z)) = 0$.*
2. (Krein-space decomposition and positive/negative subspaces $\mathcal{A}_\pm$) *At $Z = \xi_{\mathbf{b},\mathbf{a}}$, the bilinear form $\iint_{(0,1)^2} f(x_1^4)f(x_2^4)\,d\Omega_{\mathbf{b},\mathbf{a}}^{(\zeta,\beta)}(x_1,x_2)$ decomposes into an infinite-rank Krein difference of squares, yielding explicit h-dimensional subspaces $\mathcal{A}_+,\mathcal{A}_- \subset \mathbb{Q}[x^4]_{\leq 4K}$ such that $G_{\mathcal{A}_+}^{[\mathbf{b},\mathbf{a}]}(\xi_{\mathbf{b},\mathbf{a}}) \succ 0$ (all roots $r_j < \xi_{\mathbf{b},\mathbf{a}}$) and $G_{\mathcal{A}_-}^{[\mathbf{b},\mathbf{a}]}(\xi_{\mathbf{b},\mathbf{a}}) \prec 0$ (all roots $r_j > \xi_{\mathbf{b},\mathbf{a}}$).*
3. (Unconditional q-adic non-vanishing for arbitrary Zeta–Beta linear forms) *Suppose $\sum_{m=1}^{k} b_m\zeta(2m+1) + \sum_{r=1}^{k} a_r\beta(2r) = c \in \mathbb{Z}$, and let $s_* := \max\big(\{2m+1 : b_m \neq 0\} \cup \{2r : a_r \neq 0\}\big) \geq 2$ be the highest active weight in $(\mathbf{b},\mathbf{a})$. Then for every prime $q > \max(\|\mathbf{b}\|_\infty, \|\mathbf{a}\|_\infty, s_* + 5)$:*

(i) *If $s_* = 2m_* + 1$ is odd ($b_{m_*} \neq 0$), setting $K = 2q - 1$, $h = q$, and $\mathcal{A}_q := \{x^{4\rho} + \kappa_{m_*} x^{4(\rho+q)} : 0 \le \rho < q\}$ ($\kappa_{m_*} = 1$ for $m_* \ge 2$ and $\kappa_1 = 2$) gives the exact q-adic valuation*

$$v_q\Big(\Delta^{[\mathbf{b},\mathbf{a}]}_{\mathcal{A}_q}(c)\Big) = -(2m_* + 1)q < 0 \quad \Longrightarrow \quad \Delta^{[\mathbf{b},\mathbf{a}]}_{\mathcal{A}_q}(c) \neq 0.$$

(ii) *If $s_* = 2r_*$ is even ($a_{r_*} \neq 0$), setting $K = q - 1$, $h = (q-1)/2$, and $\mathcal{A}_q^{(\beta)} := \{x^{4\rho} + 2x^{4(\rho+(q+1)/2)} : 0 \le \rho < (q-1)/2\}$ gives the exact q-adic valuation*

$$v_q\left(\Delta^{[\mathbf{b},\mathbf{a}]}_{\mathcal{A}_q^{(\beta)}}(c)\right) = -2r_*\left(\frac{q-1}{2}\right) = -r_*(q-1) < 0 \quad \Longrightarrow \quad \Delta^{[\mathbf{b},\mathbf{a}]}_{\mathcal{A}_q^{(\beta)}}(c) \neq 0.$$

Proof. (1)–(2) Changing variables to $\tilde{x}_i = x_i^4 \in (0,1)$ in the first sum reduces it to $d\Omega_{k,\mathbf{b}}(\tilde{x}_1, \tilde{x}_2)$ of Theorem 6.7, while Proposition 5.3 extended to weight $2r$ gives the moments of the second sum on $x_1^{4n} x_2^{4l}$. Differentiating with respect to Z gives $CC^T \succ 0$, Theorem 6.10(2) gives $\lim_{K\to\infty} K^{-2} \log\det(C_K C_K^T) = 0$, and the Krein decomposition follows as in Theorem 6.7(2).

(3) In case (i) ($s_* = 2m_* + 1$ odd), for $0 \le n, l < 2q$, every active beta weight satisfies $2r \le 2m_*$, so $v_q(H_{2n}^{(\beta,2r)}) \ge -2r \ge -2m_* > -(2m_*+1)$ and $v_q(\frac{H_{2n}^{(\beta,2r-1)} - H_{2l}^{(\beta,2r-1)}}{n-l}) \ge -2r \ge -2m_*$. Thus all $\beta(2r)$ terms have valuation $\ge -2m_*$ and vanish modulo $q^{-(2m_*+1)}$ relative to the leading $(\frac{\kappa_{m_*}}{m_*} - \kappa_{m_*}^2) b_{m_*} q^{-(2m_*+1)}$ diagonal entry of Theorem 6.7(3), giving $v_q(\Delta^{[\mathbf{b},\mathbf{a}]}_{\mathcal{A}_q}(c)) = -(2m_*+1)q < 0$.

In case (ii) ($s_* = 2r_*$ even, so all active odd zeta weights satisfy $2m + 1 \le 2r_* - 1$), for $0 \le \rho < \frac{q-1}{2}$, set $n_0 := \rho < \frac{q-1}{2}$ and $n_1 := \rho + \frac{q+1}{2} \in [\frac{q+1}{2}, q-1]$. For $0 \le n < q$, all active zeta harmonic sums $H_n^{(2m+1)}$ contain no multiple of q and hence lie in $\mathbb{Z}_q$. For the beta harmonic sums, $2n_0 \le q - 3 < q$ implies $H_{2n_0}^{(\beta,2r_*)} \in \mathbb{Z}_q$, whereas $q + 1 \le 2n_1 \le 2q - 2$ implies that $H_{2n_1}^{(\beta,2r_*)} = \sum_{\nu=0}^{2n_1-1} (-1)^\nu (2\nu+1)^{-2r_*}$ contains the unique q-divisible odd integer $2\nu + 1 = q$ at $\nu = \frac{q-1}{2}$, giving $H_{2n_1}^{(\beta,2r_*)} \equiv (-1)^{(q-1)/2} q^{-2r_*} \pmod{\mathbb{Z}_q}$. Moreover, $n_1 - n_0 = \frac{q+1}{2} \in \mathbb{Z}_q^\times$ is coprime to q, so the off-diagonal divided difference $\frac{H_{2n_1}^{(\beta,2r_*-1)} - H_{2n_0}^{(\beta,2r_*-1)}}{n_1 - n_0}$ has valuation $\ge -(2r_* - 1) > -2r_*$, and for $\rho \neq \sigma$ all differences $n - l \not\equiv 0 \pmod q$ lie in $\mathbb{Z}_q^\times$. Thus $q^{2r_*} G^{[\mathbf{b},\mathbf{a}]}_{\mathcal{A}_q^{(\beta)}}(c) \equiv -4(-1)^{(q-1)/2} a_{r_*} I_{(q-1)/2} \pmod{q\mathbb{Z}_q}$, which is invertible modulo q, proving $v_q(\Delta^{[\mathbf{b},\mathbf{a}]}_{\mathcal{A}_q^{(\beta)}}(c)) = -2r_*(\frac{q-1}{2}) < 0$. □

Theorem 6.18 (Principal-minor 2-adic integrality law, dyadic height bounds for Zeta–Beta pencils, and multi-weight effective measures)**.** *Let $G_K^{[\mathbf{b},\mathbf{a}]}(Z)$ be the coupled multi-weight Zeta–Beta Hankel matrix pencil of Theorem 6.16 on $D_N(t)^{2M}/D_K(t)$ ($h = K - N$), and for $1 \le r \le h$, let $G_{K,r}^{[\mathbf{b},\mathbf{a}]}(Z) := [G_K^{[\mathbf{b},\mathbf{a}]}(Z)_{i,j}]_{0\le i,j<r}$ and $\Delta_{K,r}^{[\mathbf{b},\mathbf{a}]}(Z) := \det G_{K,r}^{[\mathbf{b},\mathbf{a}]}(Z) \in \mathbb{Q}[Z]$ denote its $r \times r$ leading principal minor pencil. Then:*

(1) (Exact 2-adic disappearance of Euler-number denominators in the sub-threshold principal minors) *Let $r_{\min} := \min\{1 \le m \le k : a_m \neq 0\}$. For every integer $1 \le r \le \frac{K - 2MN - (k - r_{\min})}{2}$ (and for the full $h \times h$ matrix whenever $2MN + 2h - 2 + k - r_{\min} < K$), every shifted rational function $(-t)^{k+1-m} R_{i,j}(t)$ ($0 \le i, j < r$, $r_{\min} \le m \le k$) has degree $2MN + i + j + k + 1 - m < K = \deg D_K(t)$, so its polynomial quotient $Q_{i,j}^{(m)}(t) \equiv 0$ vanishes identically. Consequently:*

(i) *No Euler-number monomial moment $\widetilde{\mu}_{Y_m}^{(\beta(2m))}(t^e)$ occurs in $G_{K,r}^{[\mathbf{b},\mathbf{a}]}(Z)$.*

(ii) *Because every alternating odd harmonic sum* $H_{2j}^{(\beta,2m)} = \sum_{n=0}^{2j-1}(-1)^n(2n+1)^{-2m} \in \mathbb{Z}_2$ *has exclusively odd denominators, the* $\beta(2m)$ *pole contributions are 2-adically integral (up to the fixed constant* 4^{-m} *in* $r_{2j}^{(\beta,2m)}$*), so* $\Delta_{K,r}^{[\mathbf{b},\mathbf{a}]}(Z)$ *clears to* $\mathbb{Z}[Z]$ *with zero* 16^e *dyadic loss at* $p=2$*, while retaining full hyperbolicity, root interlacing (Theorem 3.4), and positive definiteness* $G_{K,r}^{[\mathbf{b},\mathbf{a}]}(\xi_{\mathbf{b},\mathbf{a}}) \succ 0$.

(2) (Exact dyadic denominator bound at full dimension $h = 27n$) *In the weight-5 parameters* $K = 30n$*,* $N = 3n$*,* $h = 27n$*,* $2M = 6$ *(*$k = 2$*), for* $b\zeta(5) + a\beta(4)$ *(*$r_{\min} = 2$*), the leading* $r_* \times r_*$ *principal minor with* $r_* = 6n = \frac{K}{5}$ *satisfies* $Q_{i,j}^{(2)}(t) \equiv 0$ *and is completely free of* 16^{-e} *Euler-number denominators. For the full* $27n \times 27n$ *determinant* $\Delta_K^{[b,a]}(Z)$ *of Theorem 6.14(2), the odd-prime multiplier* $m_{K,M}^{(5),\mathrm{odd}} := \prod_{3\le p\le 2h} p^{-L_p^{(5)}(K,M)}$ *clears all odd-prime denominators of* $S_K^{(5)}\Delta_K^{[b,a]}(Z)$ *into the dyadic ring* $\mathbb{Z}[1/2][Z]$ *with the exact same odd-prime quadratic exponent as pure* $\zeta(5)$*, while the 2-adic valuation of the polynomial quotients* $Q_{i,j}^{(2)}(t)$ *(*$\deg Q_{i,j}^{(2)} \le 42n-1$*) is bounded by* $v_2(\widetilde{\mu}_Y^{(\beta(4))}(t^e)) \ge -4e$.

(3) (Unconditional irrationality and effective measure $\mu = 25687$ for period-1 Bose–Einstein cones) *Whenever* $a_1 = \cdots = a_k = 0$ *(so the functional is built purely from the period-1 Bose–Einstein derivatives* $\frac{d^{2m}}{du^{2m}}(e^{2\pi u}-1)^{-1}$*, whose analytic singularity at* $u = \pm i$ *has unit radius* $R = 1$ *and produces pure Bernoulli moments* B_{2e+2} *whose 2-adic denominator is bounded by* $2^1 = 2$ *for all* $e \ge 0$ *by the von Staudt–Clausen theorem), at top weight* $2k+1 = 5$ *(*$k = 2$*), for every* $(a,b) \in \mathbb{Z}^2$ *with* $b > 0$ *and* $a < \frac{\pi^2}{3}b$*, the full* $27n \times 27n$ *normalized determinant* $Q_K^{[a,b]}(Z) \in \mathbb{Z}[Z]$ *is an integer polynomial satisfying* $0 < Q_K^{[a,b]}(b\zeta(5)+a\zeta(3)) \le e^{-0.0004K^2+O(K)}$ *and* $|r_j-(b\zeta(5)+a\zeta(3))| \ge c_1(a,b)e^{-11.411K}$*, yielding* $|b\zeta(5)+a\zeta(3)-p/q| > q^{-25687}$ *for all* $q \ge q_0(a,b)$.

Proof. (1) For $0 \le i,j < r \le \frac{K-2MN-(k-r_{\min})}{2}$ and $r_{\min} \le m \le k$, we have $\deg((-t)^{k+1-m}D_N(t)^{2M}t^{i+j}) = 2MN+i+j+k+1-m \le 2MN+2r-2+k+1-r_{\min} \le K-1 < K = \deg D_K(t)$. Thus Euclidean division by $D_K(t)$ yields polynomial quotient $Q_{i,j}^{(m)}(t) \equiv 0$, and $(-t)^{k+1-m}R_{i,j}(t)$ expands as a pure sum of simple poles at $t = -j_0^2$ ($N < j_0 \le K$). By Theorem 6.14(1), at each pole $t = -j_0^2$, the $\beta(2m)$ functional evaluates to $j_0^{2k}(Y_m - H_{2j_0}^{(\beta,2m)}) + \tilde{r}_{j_0,k,m}$. Since every integer $2n+1$ ($0 \le n < 2j_0$) is odd, $(2n+1)^{-2m} \in \mathbb{Z}_2^\times$, so $H_{2j_0}^{(\beta,2m)} \in \mathbb{Z}_2$ has non-negative 2-adic valuation for every $j_0 \ge 1$. Moreover, since $G_{K,r}^{[\mathbf{b},\mathbf{a}]}(\xi_{\mathbf{b},\mathbf{a}})$ is the leading $r \times r$ principal submatrix of $G_K^{[\mathbf{b},\mathbf{a}]}(\xi_{\mathbf{b},\mathbf{a}}) \succ 0$, it is strictly positive definite, and by Theorem 3.4(1) its roots are real and interlace those of $G_{K,r+1}^{[\mathbf{b},\mathbf{a}]}(Z)$.

(2) For $K = 30n$, $N = 3n$, $2M = 6$, and $k = r_{\min} = 2$, we have $\frac{K-2MN}{2} = \frac{30n-18n}{2} = 6n$, giving $r_* = 6n$. For the full $27n \times 27n$ matrix, Theorem 6.16(3) shows that at every odd prime $p \ge 3$, both the polynomial moments $\widetilde{\mu}_Y^{(\beta(4))}(t^e) \in \mathbb{Z}_p$ and the harmonic pole terms $H_{2j}^{(\beta,4)} \in p^{-4}\mathbb{Z}_p$ satisfy the local p-adic valuation bounds of $\mu_X^{(5)}$, so $m_{K,M}^{(5),\mathrm{odd}} S_K^{(5)}\Delta_K^{[b,a]}(Z) \in \mathbb{Z}[1/2][Z]$. At $p = 2$, the monomial moment formula of Theorem 6.14(1) has denominator $16^e = 2^{4e}$, giving $v_2 \ge -4e$.

(3) When $a_1 = \cdots = a_k = 0$, the functional reduces to that of Theorem 6.6 (and at $k = 2$, Theorem 4.5), where all monomial moments are rational multiples of B_{2e+2}. By the von Staudt–Clausen theorem, the prime $p = 2$ divides the denominator of B_{2e+2} to exact power $2^1 = 2$ for all $e \ge 0$ (since $2-1 = 1 \mid (2e+2)$), losing at most $2^h = e^{O(K)}$ in the determinant at $p = 2$, and Theorem 4.5 yields the stated irrationality measure. □

7. Discussion and Open Problems

We conclude by summarizing the structural bottlenecks identified by our analysis and outlining three problems for future research.

(1) *Closing the gap between ρ_5^* and ρ_5^{var}:* In Theorem 4.4, retaining the concave potential $\Phi(u)$ of the residue weights $|r_j|$ on the single interval $[a,b] = [1/18, 1/3]$ reduced the spectral growth constant from 13.84 to 11.411, lowering the irrationality measure exponent of $\zeta(5)$ from 31155 to 25687. Establishing Conjecture 4.11 (or bounding $\|B_K^{(5)}\|$ directly through the Christoffel–Darboux pole kernel of Theorem 3.8(1) against a multi-interval arcsine equilibrium measure on $[0, L_1]$) would reduce ρ_5 toward ρ_5^{var}. However, as shown in Remark 4.9, no improvement to ρ_5 alone can push the irrationality exponent $\mu = \lambda\rho_5/\epsilon$ below 10985 while $\epsilon = 0.0004$ is held fixed; achieving a substantially smaller irrationality measure requires simultaneously sharpening the p-adic prime-sum bound $A_M^{(5)}$ (or proving that the integer content $c_K = \mathrm{cont}_{\mathbb{Z}}(Q_K^{(5)})$ grows as $e^{\delta K^2}$ with $\delta > 0$).

(2) *The height barrier for quadratic approximations and two-variable height reduction:* Proposition 4.2 shows that an integer polynomial family $Q_K(X) \in \mathbb{Z}[X]$ of degree $h = \lambda K$ with zero-free disk $e^{-\rho K}$ excludes algebraic approximations of degree $d \geq 2$ whenever its decay rate ϵ at ξ exceeds $(d-1)\Lambda_{\mathrm{height}}$. Theorem 6.10(1) proves that for any one-variable positive-definite Hankel pencil with negative poles, $\Lambda_{\mathrm{lead}} + \epsilon = \mathcal{I}_B = \mathcal{E}(\mu_{\mathrm{eq}}) - \mathcal{E}_{\mathrm{poles}}(\nu_{\mathrm{unif}}) \geq 0.23310 \gg 2\epsilon$, obstructing degree-$d \geq 2$ transfer within $G_K^{(5)}(X)$. On the other hand, Theorem 6.10(2) demonstrates that passing to the two-variable Beukers architecture $M_{\mathcal{A}_K}^{(s)} = C_K C_K^T$ reduces the quadratic growth rate of $\log\det M_{\mathcal{A}_K}^{(s)}$ identically to zero. Combining p-adic residue-annihilating polynomial subspaces C_K with the positive Krein cone $\mathcal{V}_K^+$ of Theorem 6.3(2) is thus a promising route toward algebraic independence and degree-$d \geq 2$ approximation bounds.

(3) *Overcoming the weight-seven distribution barrier and extending multi-weight cones:* Theorem 2.4, Theorem 5.6, and Theorem 6.1 demonstrate that within the one-variable Hankel architecture and its symmetric divided-difference reduction, the p^{-6} distribution prefactor and rank-two modulo-p pole block $\begin{pmatrix} 0 & 1/6 \\ 1/6 & -1 \end{pmatrix} p^{-7}$ create an intrinsic local barrier for $\zeta(7)$ alone. Three mechanisms developed in Sections 5 and 6 provide concrete paths forward: first, the asymmetric weight-$(6,7)$ Beukers functional of Theorem 6.2, which couples $\zeta(6)$ and $\zeta(7)$ with zero denominator overhead; second, the multi-weight differential-shift and Krein decompositions of Theorems 6.6 and 6.7 (together with the character–Vandermonde sign cancellation of Theorem 6.4 for $\beta(2k)$); and third, the simultaneous multi-pole architectures of Theorems 5.7 and 6.9, where higher pole multiplicities unlock binomial p-adic cancellation across residue classes.

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